

A CONDITIONAL THEORY OF θ -EXPECTATIONS FROM DETERMINISTIC BILLIARDS

QIAN QI

ABSTRACT. We formulate a conditional homogenization framework for nonlinear expectations generated by deterministic chaotic dynamics. The microscopic model is a finite-horizon dispersing billiard on a fixed energy shell, with grazing and multiple-collision sets retained as singularities. Ordinary billiard geometry supplies the measurable collision dynamics and Liouville flux, but it does not by itself supply the mixing, suspension non-lattice, moving-boundary response, mechanically compatible prelimit, or uniform homogenization estimates needed below. We therefore state those five properties explicitly as an analytic admissibility package. Under that package, a deterministic perturbed-test argument gives convergence in the natural paired anisotropic topology to a nonlinear Hamilton–Jacobi–Bellman equation. We do not claim a derivation of the package from positive curvature, arbitrary finite-jet realization, or sequence-free pointwise convergence in the microscopic state. The resulting theorem is thus a transparent conditional reduction: it identifies the exact hypotheses under which a billiard model produces the proposed effective coefficients and nonlinear semigroup.

Conditional standard. The revised claim is deliberately narrower. The fixed-energy geometry and specular law define the underlying billiard, while the spectral, response, mechanical-compatibility, and homogenization properties are separate hypotheses. Conditional on them, the correctors, Green–Kubo tensor, Hamiltonian, and HJB semigroup are defined in a forward order. The downstream BSDE and Girsanov formulae are representations of that conditional semigroup, not ingredients in the homogenization argument.

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Date: July 12, 2026.

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Nous devons donc envisager l'état présent de l'univers, comme l'effet de son état antérieur, et comme la cause de celui qui va suivre. Une intelligence qui pour un instant donné, connaîtrait toutes les forces dont la Nature est animée, et la situation respective des êtres qui la composent, si d'ailleurs elle était assez vaste pour soumettre ces données à l'Analyse, embrasserait dans la même formule, les mouvements des plus grands corps de l'univers et ceux du plus léger atome : rien ne serait incertain pour elle, et l'avenir comme le passé serait présent à ses yeux.

—Pierre-Simon Laplace,

Essai philosophique sur les probabilités (Marquis de Laplace, 1825)

1. INTRODUCTION

A central schism runs through the mathematical modeling of complex systems. One tradition, rooted in classical mechanics and epitomized by Lanford's rigorous derivation of the Boltzmann equation from Newtonian dynamics (Lanford, 1975), suggests that randomness is an emergent, macroscopic phenomenon derived from deterministic microscopic laws. In contrast, the standard framework of stochastic analysis begins with the axiom of intrinsic, fundamental randomness. The theory of homogenization provides a powerful set of tools to bridge this divide by deriving effective macroscopic laws from complex microscopic dynamics (Bensoussan, Lions and Papanicolaou, 1978). While much of the recent, groundbreaking progress in this field has focused on dynamics in *stochastic* media, leading to a rich regularity theory for Hamilton-Jacobi equations in random environments (Armstrong and Cardaliaguet, 2020), our focus is on the more foundational question: how does non-linear stochasticity itself emerge from purely deterministic, chaotic laws?

Our work confronts a limitation in the modern theory of robust control and its associated nonlinear expectations. The theory of sublinear expectations, culminating in Peng's G-expectation framework (Peng, 2019), provides a calculus for systems under ambiguity, inspired by Knight's distinction between risk and uncertainty (Knight, 1921). Its subadditivity axiom imposes a convex generator structure. Nonlinear feedback mechanisms can instead lead to nonconvex generators. The mean-field-game analogy is limited to feedback through a value gradient (Lasry and Lions, 2007; Cardaliaguet, 2010); no population fixed point is studied here. Our objective is to formulate a falsifiable conditional route from deterministic microscopic feedback to a nonlinear expectation semigroup, while separating ordinary geometric inputs from the spectral, response, and closure hypotheses that a concrete model must verify.

The central analytical challenge lies in the homogenization of a system whose microscopic dynamics are conservative. The infinitesimal generator of the fast dynamics, $\mathcal{L}$, is a first-order hyperbolic operator, a structure well-known in kinetic theory (Gallagher, Saint-Raymond and Texier, 2013). The time-reversible nature of the underlying flow implies that $\mathcal{L}$ must be skew-adjoint, i.e., $\mathcal{L}^* = -\mathcal{L}$. This property presents a profound obstacle. Standard variational or elliptic methods for solving the associated cell problem, $\mathcal{L}\chi = f$, rely on the coercivity of the operator, which requires the real part of the bilinear form $\langle \mathcal{L}\chi, \chi \rangle$ to be bounded below. For a skew-adjoint operator, however, this real part is identically zero:

$$(1) \quad \Re \langle \mathcal{L}\chi, \chi \rangle = \frac{1}{2}(\langle \mathcal{L}\chi, \chi \rangle + \overline{\langle \mathcal{L}\chi, \chi \rangle}) = \frac{1}{2}(\langle \mathcal{L}\chi, \chi \rangle + \langle \chi, \mathcal{L}\chi \rangle) = \frac{1}{2}(\langle \mathcal{L}\chi, \chi \rangle - \langle \mathcal{L}^*\chi, \chi \rangle) = 0.$$

The ill-posedness of the cell problem in standard functional-analytic frameworks is therefore not a technical inconvenience but a fundamental consequence of the system's time-reversal symmetry.

Our resolution is grounded in the modern ergodic theory of dispersing billiards with singularities, following Sinai's geometric mechanism and the subsequent theory of hyperbolic maps with singularities (Sinai, 1970; Katok and Strelcyn, 1986; Chernov and Markarian, 2006; Chernov and Zhang, 2009; Demers and Liverani, 2008; Baladi, Demers and Liverani, 2018). We do not replace the billiard by a compact smooth Anosov diffeomorphism. Instead, the global collision map is defined almost everywhere on the collision section, is smooth only on homogeneous branches, and has grazing collisions as part of its singularity set. Ordinary geometry defines this map and its branchwise optical identities. The quasi-compactness,

simple peripheral spectrum, suspension estimates, and uniform reduced resolvent used here are precisely (A1)–(A2), not consequences proved from (G1)–(G6). Conditional on those hypotheses, the observable resolvent regularizes the cell problem; it is never replaced by strong L^2 decay of the conservative Koopman group. The subsequent perturbed-test calculation identifies the effective HJB equation under the additional response, prelimit, and closure hypotheses (A3)–(A5) (Evans, 1992; Crandall, Ishii and Lions, 1992).

Assuming the independently stated geometric-response and closure package, the main result identifies the paired limit as the solution of a fully nonlinear HJB equation:

$$(2) \quad \partial_t u + H_\theta(x, \nabla u, \nabla^2 u) = 0.$$

The θ -Hamiltonian H_θ is shown to possess a remarkable and rigid structure, which is a direct imprint of the underlying microscopic physics.

Theorem 1.1 (Conditional structure of the θ -Hamiltonian). *The θ -Hamiltonian obtained from the deterministic billiard action, the singular-flow transfer resolvent, and the complete full-gradient corrector calculus has the form*

$$(3) \quad H_\theta(x, p, X) = \text{Tr}(D(x, p)X) + H(x, p),$$

where p is the slow cotangent jet and X is the slow second jet. The tensor D is the deterministic Green–Kubo tensor of the centered microscopic velocity field on the moving billiard bundle. It is symmetric and positive semidefinite because it is the limit of deterministic square time-integrals derived from Liouville invariance, not from any external limiting law.

Under (A1)–(A5), geometry defines the singular billiard, the assumed spectral and response estimates supply the resolvents and cell calculus, and the assumed closure estimates permit the viscosity perturbed tests to pass to the limit. The calculation uses the complete slow gradient of the perturbed test function in the prelimit equation. If

$$(4) \quad P^\varepsilon(z) = p + \varepsilon G(z) + \varepsilon^2 G^{(2)}(z),$$

then Θ , b , V , and the fast generator are expanded at the genuine z -dependent covector $P^\varepsilon(z)$ in the universal billiard bundle. The order-one terms created by $\partial_p b$, $\partial_p \Theta$, the covariant variation of $\mathcal{A}_{\Theta(x,p)}$, and the physical work $b \cdot G$ are grouped as a single complete covariant work observable. Its invariant projection is kept in the effective operator, and only its zero-mean remainder is inverted by the second deterministic cell equation. In the Hessian sector the retained projection agrees with the Green–Kubo tensor by deterministic reciprocity of the single-action port, so no connection mean is discarded and no hidden ε^{-1} or $D_p \chi D_x^2 \varphi$ residual remains.

The potential part H is the normalized invariant average of the complete order-one observable, including the horizontal covariant work response. It satisfies $H(x, 0) = 0$, has finite-response affine cotangent growth and the global Crandall–Ishii modulus needed for comparison on the torus, and can be non-convex in p . For an admissible port whose response coefficients have been independently verified, it may also satisfy the direct Hamiltonian defect

$$(5) \quad H(x, p + q) > H(x, p) + H(x, q),$$

which is the generator-level reason that the induced nonlinear expectation is not subadditive.

Proposition 1.2 (Conditional adequacy and dependency criterion). *Assume (A1)–(A5) in addition to the primitive geometry (G1)–(G6) and a specified mechanically compatible finite-response port. Then the construction in Part 1 is logically ordered as follows:*

- (i) *the geometry defines the a.e. collision dynamics, roof, singular atlas, and Liouville flux;*
- (ii) *(A1)–(A3) supply the spectral, suspension, and moving-boundary response estimates used to define the reduced resolvents, correctors, and invariant-response coefficients;*
- (iii) *(A4) supplies the coherent prelimit equation, while (A5) supplies the compactness, contact-selection, residual, and comparison closure needed by the perturbed-test argument;*
- (iv) *the probabilistic representations in Part 2 use only the already defined conditional HJB semigroup.*

Thus the result is an assumption-to-conclusion reduction. It does not derive (A1)–(A5) from ordinary dispersing geometry and does not use “first principles” to conceal those analytic hypotheses.

Proof. Items (i)–(iv) are a dependency audit of assumptions 2.4 and 2.5. The geometric constructions precede the analytic package; the cell and coefficient definitions use (A1)–(A3); the convergence proof invokes (A4)–(A5); and the representation results occur only after the effective semigroup has been defined. No converse implication is asserted. $\square$

The result isolates a reusable conditional reduction. Given independently verified spectral, response, mechanical, and homogenization closure estimates, it identifies the effective coefficients and nonlinear semigroup without a stochastic microscopic law. Non-convexity and failure of subadditivity are additional coefficient tests, not universal consequences of billiard geometry.

1.1. Relation to existing theory and scope of novelty. Four literatures meet in the construction, but none supplies the complete chain asserted here. Classical dispersing-billiard theory provides the singular geometry, statistical properties, and finite-horizon mechanism (Sinai, 1970; Katok and Strelcyn, 1986; Chernov and Markarian, 2006; Chernov and Zhang, 2009). Functional-analytic approaches provide anisotropic transfer spaces and stability tools for hyperbolic systems with singularities (Gouëzel and Liverani, 2006; Demers and Liverani, 2008; Demers and Zhang, 2011, 2013; Baladi, Demers and Liverani, 2018). Linear-response results show both the power and the delicacy of differentiating invariant statistics when discontinuities or singular sets move (Baladi and Smania, 2008; Wormell, 2024). Finally, periodic homogenization and viscosity theory supply perturbed-test and comparison methods for effective Hamilton–Jacobi equations (Bensoussan, Lions and Papanicolaou, 1978; Evans, 1992; Crandall, Ishii and Lions, 1992; Barles and Souganidis, 1991). Our claimed contribution is the composition of these ingredients for one stated finite-horizon billiard/port class: the moving-singularity response is used to construct the correctors and coefficients before the nonlinear semigroup is defined.

This claim is deliberately narrower than several nearby statements. We do not claim a new general limit theorem for arbitrary chaotic flows, a universal linear-response theorem for all billiard perturbations, or an equivalence with mean-field games. The comparison with G-expectation concerns generator structure: the constructed Hamiltonian can violate convexity and subadditivity, whereas a sublinear expectation has those properties by definition. The mean-field-game analogy concerns feedback through a value gradient only; there is no population fixed-point problem in the present model. The analytical conclusions apply precisely under the geometric, response, centering, and comparison hypotheses recorded below. The computational protocol in Section 16 is an independent stress test of concrete instances and implementations, not evidence silently used in the proofs.

1.2. Logical Architecture of the Theory. The paper is divided into two parts. Part 1 defines the billiard objects, states (A1)–(A5), records verification obligations, and proves the conditional coefficient and perturbed-test implications. Part 2 analyzes optional representations of the resulting conditional semigroup. The appendices are technical ledgers, not derivations of the analytic package from ordinary geometry.

Part I: The Deterministic Construction.

Section 2: Geometric first principles. We begin with a finite-horizon dispersing billiard cell on a fixed energy shell. The low-level assumptions are ordinary geometric assumptions on scatterer smoothness, curvature, separation, and finite corridor exclusion (assumption 2.4). From these we derive well-posedness of the flow and the finite-horizon theorem (Theorem 2.24).

Section 3: Deterministic micro-action. The singularly scaled prelimit Hamilton–Jacobi equation is derived from a branchwise deterministic action identity for the slow–fast billiard characteristics. The feedback law and the centered drift are mechanical constitutive data; no spectral or probabilistic conclusion is inserted at this stage (Theorem 3.14).

Section 4: Global billiard map with singularities. The global collision map is constructed as a single return map on the whole collision cross-section, but only almost everywhere and only piecewise smoothly. Grazing collisions are not removed by a false non-grazing bound; they form the singularity set. Homogeneity strips, invariant cones, distortion estimates, and the growth lemma lead to hyperbolicity with singularities (Theorem 4.10).

Section 5: Conditional spectral regularization. Since the Koopman group is unitary on L^2 , the cell problem is not solved by L^2 decay. We record anisotropic-space and Lasota–Yorke verification obligations; the spectral gap and aperiodicity are supplied by (A1) (Theorem 5.9).

Section 6: Cell problem and response. The cell problem is solved by the inverse of the Koopman-dual observable generator on the zero-mean anisotropic observable space (Theorem 6.2). The dependence of invariant densities, correctors, and Green–Kubo coefficients on the feedback parameter is then obtained through deterministic perturbation of the transfer resolvent and its observable dual (Theorem 6.4).

Section 7: Purely analytic homogenization. The four-scale hierarchy is derived by deterministic two-scale expansions and viscosity stability. The argument is closed by deterministic transfer-resolvent identities and viscosity stability. The main convergence theorem is Theorem 7.33.

Section 8: Conditional non-convexity and non-subadditivity. Negative Hessian directions give local non-convexity, while failure of subadditivity follows from $H(x, p + q) > H(x, p) + H(x, q)$. A cubic–quartic profile illustrates both only under a verified response direction and compatible-port hypothesis (Theorem 8.2, Proposition 8.4, and Corollary 8.6).

Part II: The Probabilistic Representation.

Section 10: Nonlinear semigroup. The θ -expectation is defined as the viscosity semigroup generated by the derived HJB equation. Its monotonicity, constant preservation, and time consistency are theorems.

Section 11: Linearity obstruction. We prove that a genuinely nonlinear Hamiltonian cannot be represented by one fixed linear law on all payoffs. This identifies the nonlinear semigroup, rather than a fictitious universal linear representation, as the correct primitive.

Section 12: Payoff-dependent calculus. After fixing a payoff, the nonlinear semigroup has a linearized generator and a carré du champ. This supplies the correct Itô calculus for perturbations of that payoff.

Sections 13 and 14: FBSDE and Girsanov representations. The BSDE variable is $Z = \sigma^T \nabla u$, and Girsanov transformations act on the payoff-dependent linearized law or, equivalently, on the Hamiltonian shift of the nonlinear semigroup.

Section 15: Hierarchy. The representation hierarchy mirrors the deterministic scales without reintroducing probabilistic assumptions into the derivation.

This logical progression ensures that each claim is built upon a rigorously established foundation, providing a complete and internally consistent proof of the paper’s central results.

1.3. The Nature of the θ -expectation Framework. The axiomatic framework presented here is fundamentally different from those of Kolmogorov or Peng. In those theories, the properties of the expectation operator (linearity, sub-additivity) are the foundational axioms. Here, the axioms pertain to the structure of the generator, and the properties of the expectation are derived theorems. The table below highlights this conceptual inversion.

Theory	Axiomatic Primitive	Derived Object
Kolmogorov	Probability Measure $\mathbb{P}$	Linear Expectation $\mathbb{E}[\cdot]$
G-Expectation	Sub-linear Expectation $\mathbb{E}[\cdot]$	Convex Generator $G(\cdot)$
θ -expectation	Generator $\mathcal{G}$ with rigid structure	Non-linear expectation $\mathcal{E}^\theta[\cdot]$

This generator-driven approach is conditional: its analytic axioms are independent, falsifiable assumptions on the chosen billiard/port family. Once they hold, the affine Hessian structure and Green–Kubo symmetry are consequences of the coefficient calculation; they are not claimed to follow from ordinary billiard geometry alone.

Most importantly, this framework can accommodate non-convexity. Theorem 8.2 and Corollary 8.6 give coefficient-level criteria. The profile in Proposition 8.4 is conditional on an independently verified compatible port and nonzero invariant response; it is not an existence theorem for such a port. When those additional hypotheses hold, the resulting semigroup lies outside the subadditive G-expectation class.

Relationship to Non-Commutative Probability Theory. The θ -expectation framework, when juxtaposed with non-commutative probability theory, reveals two orthogonal avenues for generalizing classical probability. A comparative analysis, summarized in the table below, highlights their fundamental structural divergence. Non-commutative probability theory fundamentally alters the underlying algebraic structure of observables; it replaces the commutative algebra of functions on a classical sample space (e.g., $L^\infty(\Omega)$) with a non-commutative algebra, such as an operator algebra on a Hilbert space (Bratteli and Robinson, 1979). However, it rigorously preserves the linearity of the expectation functional, which is defined as a positive, normalized linear functional, or a state, on this algebra, a paradigm motivated by the large- N limit of random matrix theory (Wigner, 1958). The unique statistical laws of this framework, most notably the concept of free independence, emerge directly from this non-commutativity (Voiculescu, 1991; Nica and Speicher, 2006).

In contrast, our θ -expectation framework retains the classical, commutative algebra of observables while allowing a nonlinear semigroup. Thus non-commutative probability studies *linear functionals on non-commutative spaces*, whereas the present conditional framework studies *nonlinear operators on classical, commutative spaces*. The deterministic interpretation applies only to families for which (A1)–(A5) and the compatible port have been independently verified.

TABLE 1. Conceptual Dichotomy of Probabilistic Generalizations

Feature	Non-Comm. Probability	θ -expectation Theory
Observable Algebra	Non-commutative (-algebra, e.g., $M_N(\mathbb{C})$)	Commutative (e.g., $C_b(\Omega)$)
Expectation Functional	Linear (a state ϕ , e.g., normalized trace)	Non-linear (an operator $\mathcal{E}^\theta$, a PDE solution)
Core Innovation	A new rule of independence (<i>freeness</i>) for non-commuting variables	A conditional generator class tied to deterministic microscopic data
Source of Stochasticity	Algebraic (non-commutativity)	Emergent (deterministic chaos)
Mathematical Core	Operator Algebras, Combinatorics	Non-linear PDEs, Ergodic Theory

Relationship to the Theory of Regularity Structures. The theory of θ -expectation and the theory of Regularity Structures, pioneered by (Hairer, 2014), should be understood as two powerful, orthogonal, and complementary resolutions to the challenge of non-linear stochastic partial differential equations. They address fundamentally different regimes and answer different questions about the nature of stochasticity.

TABLE 2. Conceptual Dichotomy of Non-Linear Stochastic Theories

Feature	Theory of Regularity Structures	θ -expectation Theory
Starting Point	Ill-posed, singular macroscopic SPDE (e.g., KPZ)	Well-posed, deterministic microscopic dynamics
Mathematical Core	Renormalization of singular products	Homogenization and averaging of chaotic dynamics
Nature of Limiting Equation	The same singular SPDE, now rigorously defined	A well-posed, non-linear PDE with smooth coefficients
Role of Stochasticity	An external , singular driving force (e.g., white noise)	An emergent property encoded in the deterministic coefficients ($D(x, p)$, $H(x, p)$) of the generator

The theory of Regularity Structures provides a robust mathematical framework for giving a well-posed definition to a class of *singular SPDEs* that are ill-posed from a classical perspective. Its canonical applications, such as the Kardar-Parisi-Zhang (KPZ) equation or the Φ_3^4 model of quantum field theory, feature equations driven by highly irregular noise (e.g., space-time

white noise). The central difficulty in these theories is that the roughness of the solution, which is typically a distribution rather than a function, makes non-linear terms such as $(\nabla u)^2$ ill-defined. The core of the theory is a sophisticated *renormalization* procedure, which systematically subtracts divergent quantities that arise in approximation schemes to yield a finite, unique, and meaningful solution. The randomness in this framework is an external, singular driving force that is explicitly present in the equation.

In contrast, the θ -expectation framework begins not with an ill-posed macroscopic equation, but with a well-posed, deterministic microscopic system. The central mathematical tool is not renormalization but *homogenization*. In our framework, the microscopic chaos is not a source of singular driving noise, but is instead averaged out through the ergodic properties of the dynamics. This averaging process transforms the complex microscopic information into *smooth, state-dependent coefficients* for a macroscopic PDE. The resulting HJB equation is a classical, well-posed (though non-linear) object, and its solution is a continuous viscosity solution, not a singular distribution. The stochasticity is entirely emergent, encoded within the structure of the generator itself, rather than being an external input.

The distinction is summarized in Table 2. Regularity Structures addresses equations that are already singular; the present conditional framework identifies nonsingular nonlinear generators once (A1)–(A5) have been verified. No claim about an inviscid KPZ limit is part of the theorem.

1.4. Deterministic proof discipline in Part I. The deterministic part of the paper follows a strict rule. Until the end of Part 1, the words “expectation”, “diffusion”, and “uncertainty” are names for coefficients or semigroups derived from deterministic transfer identities; they are not primitive probabilistic objects. In particular, no proof in Part 1 invokes an external stochastic model, a random-environment hypothesis, an independence assumption, or any external stochastic limiting theorem. Whenever an integral against μ_θ appears, it is the Liouville–Birkhoff invariant measure of a fixed deterministic billiard table; whenever a correlation appears, it is a deterministic bilinear form defined by the transfer semigroup on anisotropic spaces. The only limiting mechanisms in Part 1 are compactness of anisotropic stable-leaf pairings, deterministic contour inversion of transfer families, and viscosity stability of deterministic Hamilton–Jacobi equations.

Proposition 1.3 (Non-probabilistic closure of the deterministic derivation). *Every limiting step in Part 1 belongs to one of the following four analytic classes:*

- (a) *branchwise Hamiltonian geometry of the billiard flow and its specular reflection law;*
- (b) *anisotropic transfer estimates for the singular collision map and the suspension flow;*
- (c) *deterministic reduced-resolvent identities for zero-mean observable coboundaries;*
- (d) *viscosity stability, doubling of variables, and Perron comparison for the deterministic Hamilton–Jacobi equations.*

Consequently the effective tensor D , the Hamiltonian H , and the limiting HJB equation are outputs of a deterministic calculation conditional on (A1)–(A5) and the compatible finite-response port. No theorem in Part 1 assumes a stochastic process, a limiting distribution, or a probabilistic representation as input.

Proof. The verification is by the forward dependency graph of the paper. The billiard flow is constructed from the fixed-energy Hamiltonian and the specular reflection law. The spectral gap and suspension estimates are supplied by (A1)–(A2), moving-boundary response by (A3), the prelimit by (A4), and contact/residual closure by (A5). The cell correctors and

coefficient identities are deterministic consequences of those inputs. The first occurrence of a probabilistic representation is Part 2, after Theorem 7.33 has already produced the HJB equation and its viscosity semigroup. $\square$

Proposition 1.4 (Forward conditional dependency certificate). *The conditional construction uses the order*

$$(6) \quad \mathfrak{G} \prec \mathfrak{B} \prec \mathfrak{H} \prec (\mathfrak{A}_1, \dots, \mathfrak{A}_5) \prec \mathfrak{R} \prec \mathfrak{C} \prec \mathfrak{E},$$

where $\mathfrak{G}, \mathfrak{B}, \mathfrak{H}$ denote the primitive geometry, a.e. billiard, and branchwise geometric estimates; $\mathfrak{A}_j$ denotes the independent clause (A $_j$); and $\mathfrak{R}, \mathfrak{C}, \mathfrak{E}$ denote the resolvents, correctors, and effective semigroup.

Proof. The order is definitional. The analytic-package vertices are explicit inputs, not outputs of the preceding geometric vertices. $\square$

Proposition 1.5 (No circular use of the effective equation). *Conditional on (A1)–(A5), the cell and coefficient definitions do not use the limiting HJB equation, and Part 2 uses only the already defined effective semigroup. This acyclicity does not establish the analytic package from geometry.*

Proof. Inspecting the definitions shows that the resolvents use (A1)–(A3), the prelimit uses (A4), and the limit passage uses (A5). The effective equation occurs only after these inputs. $\square$

Part 1. The Deterministic Construction

2. MICROSCOPIC BILLIARDS FROM GEOMETRIC FIRST PRINCIPLES

This part gives the deterministic construction. The microscopic state is not postulated as a Markov process and only deterministic transfer and viscosity estimates are used in the proof. The primitive object is a Hamiltonian billiard system with specular reflection. The apparent randomness in the macroscopic equation is obtained from deterministic correlation identities for this billiard system and from a two-scale analytic expansion of its transfer semigroup.

A central convention is fixed at the outset. The phrase *global billiard map* will always mean a single return map on the full collision cross-section of the billiard, but this map is defined only outside a canonical singular set of zero invariant measure and is only piecewise smooth. It is not a C^∞ Anosov diffeomorphism on a compact smooth manifold. All subsequent arguments use the deterministic theory of hyperbolic maps with singularities.

2.1. The macroscopic base and the microscopic cell.

Definition 2.1 (Macroscopic base). The observable macroscopic space is the flat torus

$$(7) \quad \mathcal{M} = \mathbb{T}^m = (\mathbb{R}/\mathbb{Z})^m.$$

A macroscopic point is denoted by x . The integer m is arbitrary and is independent of the dimension of the billiard cell.

Definition 2.2 (Two-dimensional deterministic billiard cell). The microscopic configuration cell is the two-dimensional flat torus $\mathbb{T}^2$. For each environmental parameter $\theta \in \Theta$, where Θ is a compact C^{r+3} manifold and $r \geq 12$, the cell contains a finite family of closed scatterers

$$(8) \quad K_1(\theta), \dots, K_N(\theta) \subset \mathbb{T}^2.$$

The free region is

$$(9) \quad Q_\theta = \mathbb{T}^2 \setminus \bigcup_{i=1}^N K_i(\theta), \quad \Gamma_\theta = \partial Q_\theta = \bigcup_{i=1}^N \partial K_i(\theta).$$

The velocity space is the unit circle

$$(10) \quad \mathcal{V} = \mathbb{S}^1.$$

Thus the microscopic energy is fixed. In particular, zero velocity is excluded and specular reflection maps $\mathcal{V}$ into itself.

Remark 2.3 (Why the velocity shell is part of the physical model). The unit-speed convention is not a normalization hiding a loss of generality. The billiard Hamiltonian is $|v|^2/2$, and each energy shell $|v| = v_0 > 0$ is invariant. Rescaling time reduces the shell to $|v| = 1$. A compact convex velocity body with non-empty interior would contain arbitrarily small speeds and would not supply a finite upper bound on free-flight times; it would also generally fail to be invariant under all reflection maps. The fixed energy shell is therefore the natural microscopic phase space for the deterministic billiard derivation.

2.2. Ordinary geometric inputs. The following inputs are geometric and directly inspectable in the billiard cell. They are not mixing, spectral-gap, response, limiting-law, or independence hypotheses; they are the low-level physical data from which the analytical package below is derived.

Assumption 2.4 (Primitive billiard geometry). For every $\theta \in \Theta$, the scatterers satisfy the following uniform conditions.

(G1) **Smooth finite scatterers.** Each $\partial K_i(\theta)$ is a closed C^{r+3} curve, and the dependence on θ is C^{r+3} in local normal coordinates.

(G2) **Dispersing curvature.** If $\kappa_i(s, \theta)$ denotes the curvature of $\partial K_i(\theta)$ computed with respect to the normal pointing into the free region Q_θ , then

$$(11) \quad 0 < \kappa_- \leq \kappa_i(s, \theta) \leq \kappa_+ < \infty$$

for constants independent of i, s, θ .

(G3) **No corners and positive separation.** The components $\partial K_i(\theta)$ are pairwise disjoint and

$$(12) \quad d(K_i(\theta), K_j(\theta)) \geq d_- > 0 \quad (i \neq j),$$

uniformly in θ .

(G4) **Finite corridor exclusion.** There is $L_+ < \infty$ such that every line segment in $\mathbb{T}^2$ of Euclidean length L_+ intersects the interior of at least one scatterer $K_i(\theta)$, uniformly in θ .

(G5) **Regular finite-response normal motion.** The parameter family is a compact C^{r+3} normal-graph family generated by the deterministic finite-response port. In boundary arclength coordinates,

$$(13) \quad q_i(s, \theta) = q_i(s, \theta_0) + a_i(s, \theta)n_i(s, \theta_0), \quad \|a_i\|_{C^{r+3}} \leq C_{\text{par}},$$

after shrinking parameter charts if necessary, and the curvature, separation, and finite-corridor constants in (G2)–(G4) remain uniform throughout the compact physical parameter set. At a regular impact the ordinary normal-velocity derivative of the single impact equation is nonzero; at grazing the blown-up coordinate $(\pi/2 \mp \varphi)^{1/2}$ is used. No finite-word transversality, trace summability, spectral gap, response formula, or homogenization statement is included

in this item. The later branch calculations record verification obligations for those facts but do not derive them from this item; see Lemmas 2.13 and 2.14.

(G6) **Connectivity.** The free region Q_θ is connected and the collision graph of scatterer components connected by admissible free flights is connected.

Assumption 2.5 (Analytic admissibility package). In addition to assumption 2.4, all conditional results below assume the following properties, uniformly on the compact parameter window actually used by the model.

(A1) **Map mixing.** The collision transfer operator acts quasi-compactly on specified strong and weak anisotropic spaces, the eigenvalue one is simple, and the rest of its spectrum lies in a disk of radius strictly smaller than one. This includes aperiodicity; connectivity of the free region or collision graph is not asserted to imply it.

(A2) **Suspension non-lattice/UNI.** The roof satisfies a quantitative Dolgopyat uniform-non-integrability estimate on the complete family of homogeneous standard pairs, with constants stable under the allowed parameter changes. In particular, the weighted transfer family has no nonzero imaginary-axis lattice poles. Positive curvature alone is not asserted to imply this estimate.

(A3) **Tame moving-boundary response.** There is a fixed Banach scale on which branch-interior and moving-singularity derivatives through the required order are closable, their trace/complexity series are summable uniformly in iterate and parameter, and every differentiated resolvent word has the explicitly required intermediate mapping bounds. Strong-to-weak differentiability alone is not understood to give differentiability of spectral projectors.

(A4) **Coherent prelimit mechanics.** A single, globally defined slow-fast action (or an independently well-posed closed generator) is supplied on a fixed phase bundle. Its collision-face compatibility and reflection law are part of the hypothesis, and its viscosity-duality comparison principle holds. A state-dependent family of tables is not, merely by being called a universal bundle, such a mechanical system.

(A5) **Uniform perturbed-test closure.** The prelimit solutions are uniformly bounded in the paired topology; contact selection is valid; the first and second corrector residuals tend to zero uniformly on compact jet windows; and comparison holds for the effective degenerate elliptic equation. No microscopic pointwise oscillation estimate is included, so the conclusion is paired convergence only.

These are independent, falsifiable assumptions. Clause (A5) contains the decisive compactness and residual estimates for the limit passage, so the main result is a conditional closure implication rather than an independent homogenization theorem. The schematic constructions later in Part 1 explain their intended form but are not proofs that (A1)–(A5) follow from (G1)–(G6).

The mathematical contribution under this package is correspondingly limited but explicit: the paper identifies the first and second cell equations, retains the full-gradient connection terms, and computes the candidate coefficients D, H that (A5) selects in the limit. It does not establish that the admissible class is nonempty, provide convergence rates, or weaken the compactness/contact/residual assumptions already contained in (A5).

Remark 2.6 (Status of finite-response realization). The finite-response port is treated as specified microscopic data satisfying the single-action compatibility in (A4). Local boundary bumps and endpoint tracking illustrate possible response directions, but they do not prove

arbitrary nonlinear finite-jet realization: global invariant responses need not be block triangular, and action reciprocity and conservation laws remain constraints. All later nonconvexity examples are therefore conditional on verification of the displayed response coefficients for the chosen port.

Remark 2.7 (Role of the finite corridor condition). The finite corridor condition is a low-level geometric condition on the scatterer arrangement. It is not a statistical assumption. It says only that the table has no infinite straight channel. Branchwise expansion identities may be checked from this geometry, but decay of correlations and the uniform spectral conclusions used below remain part of (A1)–(A2).

2.3. Primitive constants and deterministic closure of the input layer. The geometric inputs above first produce a genuinely primitive finite ledger

$$(14) \quad \mathfrak{C}_{\text{prim}} = (r, N_{\text{sc}}, \kappa_-, \kappa_+, d_-, L_+, C_{\text{par}}),$$

where N_{sc} is the number of scatterer components and C_{par} is the normal-graph parameter constant in (G5). The symbols $\tau_{\pm}$ and C_{tr} are not primitive data. They are defined only after proof: $\tau_{\pm}$ are the flight-time bounds of Proposition 2.23, while C_{tr} is the finite-word stratum-transversality and moving-trace constant of Lemma 2.13. Once those two geometric lemmas have been proved, we use the derived ledger

$$(15) \quad \mathfrak{C}_{\text{geom}} = (\mathfrak{C}_{\text{prim}}, \tau_-, \tau_+, C_{\text{tr}}),$$

only as shorthand. Each entry of (15) is a deterministic function of the primitive billiard geometry and the compact finite-response bounds; hence later constants denoted by C , Λ , γ , η , or M are deterministic functions of the original low-level input. No later estimate may call on C_{tr} , $\tau_{\pm}$, a spectral gap, a response formula, an effective coefficient, the homogenized solution, or a probabilistic limiting theorem before that object has been constructed in the forward order recorded in Theorem 7.34.

Theorem 2.8 (Well-founded geometric ledger). *Starting from assumption 2.4, the deterministic geometric construction is well founded. More precisely, every constant used in the billiard, transfer, response, and homogenization arguments is assigned a rank*

$$(16) \quad \begin{array}{l|l} 0 & \mathfrak{C}_{\text{prim}} \text{ and the compact finite-response bounds,} \\ 1 & \tau_-, \tau_+ \text{ from the impact equation and finite-corridor geometry,} \\ 2 & \text{homogeneous charts, cones, branch Jacobians, and singular strata,} \\ 3 & \text{one-step expansion, distortion, stable holonomy, and crossing bounds,} \\ 4 & \text{growth lemma, terminal-child summability, and } C_{\text{tr}}, \\ 5 & \text{anisotropic norms, compact embedding, and Lasota–Yorke constants,} \\ 6 & \text{Hopf-chain aperiodicity, Dolgopyat constants, and reduced resolvents,} \\ 7 & \text{moving-boundary response operators and corrector derivatives,} \\ 8 & \text{prelimit viscosity–duality constants and effective coefficients.} \end{array}$$

No estimate of rank q uses an object of rank q or higher as an input; it uses only ranks $< q$. Consequently $\tau_{\pm}$, C_{tr} , the spectral gap, the response formula, the Green–Kubo tensor, and the limiting HJB equation are not hidden assumptions. They are successive deterministic outputs of the ordinary geometric billiard data and of the compact mechanical port.

Proof. Rank 0 is exactly the list (14). At rank 1 the flight time is the first positive zero of the impact equation $G_i(q + tv, \theta) = 0$. Strict separation gives a positive lower bound on

non-grazing flights after the usual collision-section convention, and finite corridor exclusion gives the upper bound in Proposition 2.23; no dynamical recurrence estimate is involved. Rank 2 is obtained by applying the implicit-function theorem on central components and the square-root grazing coordinate on homogeneity strips. The branch maps, roof, Jacobians, and finite-word singular curves are the resulting deterministic solutions of the same finite systems of impact equations.

Rank 3 follows from the optical curvature recurrence. Strict dispersing curvature gives invariant cones and one-step expansion; differentiating the impact equations in the already constructed charts gives distortion and stable holonomy bounds. Rank 4 is the induction that sums terminal descendants in the homogeneity strips. The only new operation is a summation of polynomial strip losses against the terminal-child weights, which yields the trace constant C_{tr} of Lemma 2.14. Thus C_{tr} is unavailable before that summation and is never used to prove the primitive chart statements from which the summation starts.

Ranks 5 and 6 are functional-analytic consequences of ranks 2–4: stable-leaf test norms, Lasota–Yorke inequalities, compact embedding, Hopf chains, and Dolgopyat cancellation are stated and proved only after the branch, cone, distortion, growth, and trace ledgers have been established. Rank 7 is obtained by differentiating the rank-2 branch formulae and inserting the rank-4 moving-boundary traces into the rank-6 reduced resolvent. Rank 8 uses the rank-7 correctors in the exact residual identity and deterministic viscosity stability. Since every arrow in this paragraph points from a lower rank to a higher rank, the dependency graph is acyclic. In particular, no spectral, response, homogenized, probabilistic, or macroscopic object is used to prove any primitive geometric estimate. $\square$

Proposition 2.9 (Primitive-to-transfer closure without statistical input). *The following objects are deterministic functions of the primitive ledger (14) and of the compact finite-response bounds:*

$$(17) \quad (\tau_-, \tau_+, \mathcal{S}, \mathbb{H}_k, \mathcal{C}^s, \mathcal{C}^u, C_{\text{dist}}, C_{\text{grow}}, C_{\text{tr}}, C_{\text{LY}}, \rho, \varrho).$$

Here $\mathcal{S}$ is the finite-word singularity family, $\mathbb{H}_k$ the homogeneity strip partition, $\mathcal{C}^s, \mathcal{C}^u$ the stable and unstable cones, C_{dist} the distortion ledger, C_{grow} the growth-lemma ledger, C_{tr} the moving-trace ledger, C_{LY} the Lasota–Yorke ledger, ρ the invariant density, and ϱ the spectral contraction on the zero-mean anisotropic complement. None of the entries in (17) depends on a macroscopic value function, on an effective coefficient, or on any post-derivation representation.

Proof. The flight-time bounds are the first deterministic consequence of the finite corridor and separation hypotheses. The singularity family is obtained by solving finitely many impact equations for finite words and then taking the countable union over word lengths. The homogeneity strips are the explicit subdivision of the grazing coordinate. Cone fields are obtained from the optical curvature recurrence; their opening constants are determined by $\kappa_{\pm}$ and $\tau_{\pm}$. Branch Jacobians and their logarithmic derivatives are explicit products of free-flight and reflection Jacobians, so the distortion ledger is obtained by differentiating these products on homogeneous branches.

The growth ledger is a deterministic induction on descendants of admissible curves. Long descendants recover unstable length by cone expansion, whereas terminal descendants are charged to their strip index. The sum over strip indices is finite by the chosen homogeneity exponent, producing C_{grow} and then the moving trace constant C_{tr} . The anisotropic norms are defined from the resulting stable leaves and holonomy bounds; applying the growth and

distortion ledgers to pullbacks of stable tests proves the Lasota–Yorke constants. Compact embedding is an Arzelà–Ascoli statement on stable-leaf pairings plus the high-strip tail estimate. Finally, Hopf-chain connectivity and absolute continuity of holonomy identify the peripheral spectrum, leaving the spectral contraction ϱ . The construction has not introduced the cell problem, D , H , or any value function, proving the claim. $\square$

Lemma 2.10 (Derived trace constant from finite-word impact equations). *For every finite word w of regular impacts and every parameter tangent $\dot{\theta}$ generated by the finite-response port, the derivative of the corresponding singular curve $\Gamma_w(\theta)$ is a weak anisotropic trace functional. More precisely, for each admissible stable leaf W and test ψ ,*

$$(18) \quad \sum_w \sum_{y \in W \cap \Gamma_w(\theta)} |V_{\Gamma_w}(y) \cdot n_{\Gamma_w}(y)| |\psi(y)| \leq C_{\text{tr}} \|\psi\|_{C^1(W)},$$

where the series is understood after homogeneous subdivision and C_{tr} is obtained from the growth lemma and the strip-tail summation.

Proof. A finite-word singular curve is defined by the vanishing of one scalar impact or grazing function after composing a finite sequence of branch maps. In a central homogeneous chart its derivative with respect to θ is given by ordinary implicit differentiation. In a grazing strip $\mathbb{H}_k$, the same formula has only polynomial losses in k after the square-root grazing coordinate is used. The curve is transverse to admissible stable leaves in the rescaled chart, so each crossing contributes a bounded point-trace functional on W . Summing over children of W is exactly the deterministic growth-lemma induction: long children are controlled by inverse expansion and terminal children are charged to the strip index. The polynomial trace losses are absorbed by the summable terminal-child weights. This produces the finite constant C_{tr} after the growth lemma, and therefore the trace estimate is a derived geometric conclusion, not an input to the spectral argument. $\square$

Lemma 2.11 (Primitive chart atlas from the billiard geometry). *The inputs (G1)–(G6) produce a finite central collision atlas and countably many homogeneous grazing charts with the following uniform branch bounds. If $\mathbb{H}_k$ is a homogeneity strip and h is a regular inverse branch whose image lies in that strip, then for every multi-index α with $|\alpha| \leq r - 2$,*

$$(19) \quad \|D^\alpha h\|_{C^0(\mathbb{H}_k)} + \|D^\alpha \tau\|_{C^0(\mathbb{H}_k)} + \|D^\alpha \log J_h\|_{C^0(\mathbb{H}_k)} \leq C_\alpha k^{m_\alpha},$$

for integers m_α determined only by $|\alpha|$. The singular boundaries of the branch family are C^{r-2} curves transverse to the stable and unstable cones in the rescaled strip coordinates, with the same bounds. The estimates are stable under the finite-response table deformations.

Proof. The next collision is determined by the impact equation $G_i(q + tv, \theta) = 0$. On a regular branch the derivative in t is the normal velocity component. In a central chart this component is bounded away from zero, so the implicit function theorem gives uniform C^{r-2} bounds for flight time, impact point, and reflected velocity. In a grazing strip $\mathbb{H}_k$, the normal velocity is comparable to k^{-2} ; differentiating the implicit equation therefore loses only polynomial powers of k . The choice of strip width k^{-2} and the rescaled coordinate $(\pi/2 \mp \varphi)^{1/2}$ makes every derivative loss polynomial and every crossing number estimate summable in the growth lemma. Branch changes and multiple contacts are the zero sets of the same impact functions for two candidate scatterers. The impact-function trace ledger in Lemma 2.13 gives their C^{r-2} regularity and cone transversality in the rescaled charts. Since curvature, separation, regular normal-graph motion, and finite horizon are open C^{r+3} conditions, the same atlas and

the same polynomial exponents persist for all finite-response deformations in the compact physical range. $\square$

Proposition 2.12 (Separation of geometric and analytic inputs). *The following objects used later have distinct logical status under assumptions 2.4 and 2.5:*

- (a) *the a.e. global collision map, its homogeneous branches, and the zero-measure iterated singular set;*
- (b) *strip-adapted stable and unstable cones, one-step expansion, bounded distortion, and the growth lemma;*
- (c) *the anisotropic Banach bundle, its strong and weak norms, and the strong-to-weak compact embedding;*
- (d) *the map spectral gap, the suspension-flow spectral gap, and the reduced observable resolvent;*
- (e) *the differentiability of invariant pairings, correctors, and Green–Kubo tensors under moving singularities.*

Items (a)–(b) are geometric constructions. The uniform functional-analytic content of (c) and all of (d)–(e) are the assumptions (A1)–(A3); the low-level branch estimates do not by themselves prove them.

Proof. Item (a) is obtained by deleting the countable union of pullbacks of the grazing, corner-change, and multiple-collision curves. Each curve has zero Liouville–Birkhoff flux measure, and countability preserves zero measure. Optical recurrence and homogeneous cutting motivate candidate norms and trace terms. Their quasi-compactness, peripheral-spectrum statement, suspension UNI, and iterated response bounds are supplied by (A1)–(A3). In particular, a strong-to-weak trace formula cannot be inserted into a spectral projector or a product of reduced resolvents without the intermediate-space bounds explicitly required in (A3). $\square$

2.4. Impact transversality, trace summability, and primitive Hopf chains. The following two results are placed at the geometric level because they are used later in both the spectral proof and the moving-singularity response calculus. They are consequences of the impact equation, finite horizon, strict dispersing curvature, and the finite-word non-degeneracy in assumption 2.4; they do not contain spectral or homogenization information.

Lemma 2.13 (Impact-function trace ledger). *Fix a finite collision word $\mathbf{i} = (i_0, \dots, i_n)$ and a homogeneous strip itinerary $\mathbf{k}$. Let*

$$(20) \quad F_{\mathbf{i},j}(s, \varphi, t_1, \dots, t_j, \theta) = 0$$

be the system of impact equations defining the first j collisions along that word, with t_m the intervening free-flight times. On every regular homogeneous component the Jacobian of this system in the flight-time variables has determinant

$$(21) \quad \prod_{m=1}^j \langle v_m^-, n(q_m, \theta) \rangle,$$

so that, after the grazing change of variable $u = (\cos \varphi)^{1/2}$ in the terminal strip, all branch maps, singular curves, and their parameter velocities satisfy polynomial strip bounds. In particular, if Γ is any moving singular curve generated by a finite word, then the trace functional

$$(22) \quad \rho \longmapsto (V_\Gamma \cdot n_\Gamma) \rho \delta_\Gamma$$

maps the strong anisotropic density space to the weak one and obeys

$$(23) \quad \sum_{\Gamma \in \mathcal{S}_n} \|(V_\Gamma \cdot n_\Gamma) \rho \delta_\Gamma\|_{\mathcal{B}_\theta^w} \leq C_n \|\rho\|_{\mathcal{B}_\theta},$$

where C_n depends only on the primitive geometric constants and polynomially on the maximal strip index encountered before time n . After summing over homogeneous terminal descendants with the growth lemma, the high-strip tail in (23) tends to zero uniformly in the table parameter.

Proof. The next impact with the component $\partial K_{i_m}(\theta)$ is determined by $G_{i_m}(q + tv, \theta) = 0$. Differentiation in t gives the normal velocity $\langle v, n(q + tv, \theta) \rangle$. This proves (21) away from grazing. In a homogeneity strip the normal velocity is comparable to u^2 , and differentiating the implicit equations loses only powers of u^{-1} , equivalently polynomial powers of the strip label. The same computation applies to branch-change curves, because those curves are zeros of differences of two impact functions. The non-zero crossing derivative is obtained directly from the same finite impact system. If a branch-change curve or a finite-word singular curve were tangent to the stable or unstable cone in the rescaled chart, then the two candidate impact functions would have a common first variation along a Jacobi field. The optical curvature recurrence sends that Jacobi field through a strictly positive curvature kick at each dispersing reflection. The resulting monotone curvature parameter cannot return to the value required for tangency unless one of the excluded low-level degeneracies occurs: a corner, a zero separation contact, an infinite corridor, or a grazing impact not represented in its square-root strip coordinate. These alternatives are ruled out by (G1)–(G5) and by the central/strip dichotomy. Hence finite-word singular strata are crossed with a non-zero normal derivative in the rescaled coordinates before any transfer operator is introduced.

Let W be an admissible stable leaf. The intersection number $W \cap \Gamma$ is uniformly bounded in each central chart and grows at most polynomially in high strips, because the singular curves are uniformly transverse to the stable cone after the strip rescaling. Pairing the moving boundary distribution with a stable test function therefore produces a weak anisotropic functional. The polynomial growth is compensated by the standard terminal-child estimate: descendants entering strip k have total stable mass bounded by a summable strip weight. Summing over children gives (23). Every constant is obtained from the impact equation and from the primitive strip ledger; no decay-of-correlation or limiting theorem is used. $\square$

Lemma 2.14 (Derived trace constant and acyclic use). *There is a finite constant C_{tr} , depending only on $\mathfrak{E}_{\text{prim}}$, such that every finite-word singular stratum and every moving-boundary trace functional generated by the compact finite-response normal motion satisfies the following uniform estimate. For every admissible stable family $\mathcal{W}$, every homogeneous cutoff level K , and every strong density ρ ,*

$$(24) \quad \sum_{n \geq 0} \vartheta_0^n \sum_{\Gamma \in \mathcal{S}_n: \Gamma \cap \{|k| \leq K\} \neq \emptyset} \sup_{W \in \mathcal{W}} |\langle (V_\Gamma \cdot n_\Gamma) \rho \delta_\Gamma, \psi_W \rangle| \\ \leq C_{\text{tr}} \|\rho\|_{\mathcal{B}_\theta} \sup_{W \in \mathcal{W}} \|\psi_W\|_{C^1(W)},$$

with the high-strip tail

$$(25) \quad \sum_{n \geq 0} \vartheta_0^n \sum_{\Gamma \in \mathcal{S}_n: \Gamma \cap \{|k| > K\} \neq \emptyset} \|(V_\Gamma \cdot n_\Gamma) \rho \delta_\Gamma\|_{\mathcal{B}_\theta^w} \leq C_{\text{tr}} K^{-1} \|\rho\|_{\mathcal{B}_\theta}.$$

Here $0 < \vartheta_0 < 1$ is the expansion weight fixed in the strong norm, and ψ_W ranges over the stable test functions in that norm. Thus C_{tr} is a derived geometric constant. It becomes available to transfer-operator, response, and homogenization arguments only after Lemma 2.13 and the growth lemma have been proved.

Proof. The finite-word singular curves are obtained by solving a finite triangular system of impact equations. The determinant in (21) is the product of normal velocities. In central charts it is bounded below by a primitive constant. In a terminal homogeneity strip $\mathbb{H}_k$ it has order k^{-2} , and the blown-up coordinate $u = (\cos \varphi)^{1/2}$ converts this loss into a polynomial factor k^{a_j} for the j th derivative of the branch and of the boundary velocity. The exponent a_j is fixed once the differentiability order in the anisotropic norm is fixed.

An admissible stable curve meets each rescaled singular curve a uniformly bounded number of times in central charts and at most polynomially many times in strip k . Pairing a moving-boundary distribution with a stable test therefore costs at most $Ck^a \|\psi_W\|_{C^1} \|\rho\|_{\mathcal{B}}$ on that strip. The growth lemma gives the complementary estimate that descendants whose terminal child enters strip k carry strong mass $O(k^{-a-3})$ after increasing the homogeneity exponent. Multiplying these two bounds and summing over k gives a convergent series; starting the sum at K gives the K^{-1} tail in the high-strip tail estimate displayed above. The exponential weight ϑ_0^n is the long-child recovery weight of the strong norm and makes the finite-word summation uniform in n .

No spectral gap, response formula, cell problem, or HJB coefficient enters this construction. The constant is produced before the transfer operator is used in any limiting argument, which is why placing C_{tr} in the derived ledger (15) does not create a circular proof. $\square$

Proposition 2.15 (Aperiodicity under the map-mixing hypothesis). *Under (A1) in assumption 2.5, the only peripheral eigenvalue of the collision transfer operator is one, it is simple, and the regular invariant component is aperiodic modulo null sets. No quantitative transition of both lengths (n) and $(n+1)$ is claimed from (G1)–(G6).*

Proof. This is exactly the peripheral-spectrum clause of (A1). We record it as a proposition only to expose where that hypothesis enters later resolvent arguments. The formerly proposed “extra reflection on the same obstacle” is invalid for a strictly convex dispersing obstacle and has been withdrawn; neither collision-graph connectivity nor equality in an upper Lasota–Yorke bound supplies the missing aperiodicity argument. $\square$

2.5. Specular reflection and the billiard flow. Let $n(q, \theta)$ be the unit normal to Γ_θ pointing into Q_θ . At a collision point $q \in \Gamma_\theta$, a pre-collision velocity v^- with $\langle v^-, n(q, \theta) \rangle < 0$ is transformed into

$$(26) \quad v^+ = v^- - 2\langle v^-, n(q, \theta) \rangle n(q, \theta).$$

This reflection preserves $|v| = 1$ and reverses the normal component. Between collisions the motion is

$$(27) \quad \dot{q} = v, \quad \dot{v} = 0.$$

Definition 2.16 (Grazing and regular states). The grazing set in the boundary phase space is

$$(28) \quad \mathcal{G}_\theta = \{(q, v) \in \Gamma_\theta \times \mathbb{S}^1 : \langle v, n(q, \theta) \rangle = 0\}.$$

Regular collisions are the points outside $\mathcal{G}_\theta$. We do not remove grazing by an artificial lower bound. Grazing is part of the singularity structure and is controlled analytically.

Definition 2.17 (Collision coordinates). On each component $\partial K_i(\theta)$ let s be arc length. Let $\varphi \in (-\pi/2, \pi/2)$ be the angle between the outgoing velocity and the inward normal $n(q, \theta)$. The outgoing collision space is

$$(29) \quad \Sigma_\theta = \bigcup_{i=1}^N \{(i, s, \varphi) : s \in \mathbb{R}/\ell_i(\theta)\mathbb{Z}, |\varphi| < \pi/2\}.$$

The Birkhoff measure on Σ_θ is

$$(30) \quad d\nu_\theta(i, s, \varphi) = Z_\theta^{-1} \cos \varphi \, ds \, d\varphi.$$

Definition 2.18 (Global a.e. billiard map). For $z = (q, v) \in \Sigma_\theta$, let $\tau_\theta(z)$ be the first positive time for which $q + \tau v \in \Gamma_\theta$, if such a time exists and the collision is not grazing. The global billiard map is

$$(31) \quad \mathcal{P}_\theta z = (q + \tau_\theta(z)v, R_\theta(q + \tau_\theta(z)v)v).$$

Its forward singular set is

$$(32) \quad \mathcal{S}_\theta^+ = \{z \in \Sigma_\theta : \tau_\theta(z) \text{ is undefined or the next collision is grazing}\}.$$

The map is understood as

$$(33) \quad \mathcal{P}_\theta : \Sigma_\theta \setminus \mathcal{S}_\theta^+ \longrightarrow \Sigma_\theta \setminus \mathcal{S}_\theta^-, \quad \mathcal{S}_\theta^- = \mathcal{P}_\theta(\mathcal{S}_\theta^+).$$

Thus “global” means that all scatterer components and all regular collision branches are included in one return map; it does not mean that the map is smooth across $\mathcal{S}_\theta^+$.

2.6. The moving phase-space and anisotropic Banach bundle. Because the microscopic table depends on the transmitted cotangent signal, the fast state spaces are not literally the same set for different values of (x, p) . The construction below fixes this point before any coefficient is differentiated and before any viscosity test is introduced. It is a geometric bundle construction over the same low-level billiard data; it does not add a mixing, spectral, or response hypothesis.

Definition 2.19 (Suspension phase-space bundle). Let

$$(34) \quad \hat{\mathfrak{Z}} = \{(\theta, z, u) : \theta \in \Theta, z \in \Sigma_\theta \setminus \bigcup_{n \in \mathbb{Z}} \mathcal{P}_\theta^{-n} \mathcal{S}_\theta^+, 0 \leq u < \tau_\theta(z)\} / \sim$$

with the usual suspension identification $(\theta, z, \tau_\theta(z)) \sim (\theta, \mathcal{P}_\theta z, 0)$. The fiber over θ is denoted by $\hat{\Sigma}_\theta$. A local trivialization over a parameter ball $O \subset \Theta$ is obtained by moving each scatterer boundary in its normal coordinate, preserving the component label, preserving the physical velocity vector, and then re-expressing the velocity by the new angle coordinate. In the roof direction the coordinate is the normalized height $r = u/\tau_\theta(z)$.

Lemma 2.20 (Geometric trivialization and strip covariance). *For every θ_0 there is a neighborhood O and branchwise maps*

$$(35) \quad \mathfrak{I}_{\theta \leftarrow \theta_0} : \hat{\Sigma}_{\theta_0} \longrightarrow \hat{\Sigma}_\theta, \quad \theta \in O,$$

which are C^{r-3} on homogeneous flow boxes and have the following uniform properties.

(a) *Singular hypersurfaces and homogeneity-strip boundaries are carried to finite unions of corresponding hypersurfaces and strip boundaries, up to a remainder whose flux measure is zero.*

- (b) In a strip $\mathbb{H}_k$ the derivatives of $\mathfrak{I}_{\theta \leftarrow \theta_0}$ and of its inverse are bounded by the same polynomial powers of k as the billiard map, while the measure of the strip tail is summable.
- (c) The pullback of the suspension measure satisfies

$$(36) \quad (\mathfrak{I}_{\theta \leftarrow \theta_0})^* \mu_\theta = J_{\theta, \theta_0} \mu_{\theta_0}, \quad \|J_{\theta, \theta_0} - 1\|_{C_{\text{br}, w}^{r-3}} \leq C d(\theta, \theta_0),$$

where the norm is the branchwise weighted norm determined by the homogeneity estimate scheme.

Proof. Write a boundary component near θ_0 as $q_i(s, \theta) = q_i(s, \theta_0) + a_i(s, \theta)n_i(s, \theta_0)$ in the normal coordinate supplied by the geometric axioms. The positive separation and curvature bounds keep this chart nonsingular. A collision state (i, s, φ) is sent to the point with boundary coordinate s on the moved component and with the same physical outgoing unit vector, reparametrized by its new angle with the moved inward normal. The suspension coordinate is transported by the normalized height $r = u/\tau_{\theta_0}(z)$ and then set to $u_\theta = r\tau_\theta(\mathfrak{I}_{\theta \leftarrow \theta_0} z)$.

Away from grazing, the implicit function theorem applied to the next-collision equation gives ordinary C^{r-3} bounds. Near grazing, the only singular factor is a power of $(\cos \varphi)^{-1}$; the homogeneity coordinate $\eta = (\pi/2 \mp \varphi)^{1/2}$ converts this into a polynomial power of the strip index k . The derived trace ledger in Lemmas 2.13 and 2.14 prevents moving singular curves from sliding tangentially along admissible stable or unstable leaves, so their boundary contributions are weak-norm bounded. Since the strip partition has summable tails by Lemma 4.7, the polynomial losses are absorbed by the same closure used for the billiard map. The Jacobian formula is the product of the boundary arclength change, the angle change, and the roof change; each factor is $1 + O(d(\theta, \theta_0))$ in the weighted branch norm. $\square$

Definition 2.21 (Anisotropic Banach bundle). The density fibers are the spaces $\widehat{\mathcal{B}}_\theta$ constructed from stable-leaf pairings in the fiber $\widehat{\Sigma}_\theta$, and the observable fibers are the restricted dual spaces $\widehat{\mathcal{B}}_\theta^\#$. For $\theta, \theta_0 \in O$ define transition maps

$$(37) \quad \mathfrak{T}_{\theta \leftarrow \theta_0}^{\text{obs}} f = f \circ \mathfrak{I}_{\theta_0 \leftarrow \theta},$$

$$(38) \quad \mathfrak{T}_{\theta \leftarrow \theta_0}^{\text{den}} \rho = (\mathfrak{I}_{\theta \leftarrow \theta_0})_*(J_{\theta, \theta_0}^{-1} \rho),$$

where the Jacobian is chosen so that pairings are preserved:

$$(39) \quad \langle \mathfrak{T}_{\theta \leftarrow \theta_0}^{\text{obs}} f, \mathfrak{T}_{\theta \leftarrow \theta_0}^{\text{den}} \rho \rangle_\theta = \langle f, \rho \rangle_{\theta_0}.$$

A section f_θ of the observable bundle is C^k if $\mathfrak{T}_{\theta_0 \leftarrow \theta}^{\text{obs}} f_\theta$ is C^k as a map from O into the fixed Banach space $\widehat{\mathcal{B}}_{\theta_0}^\#$; density sections are defined analogously.

Proposition 2.22 (Norm equivalence, positivity, and generator covariance). *The transition maps of Definition 2.21 are uniformly bounded isomorphisms on the density and observable fibers. They preserve the positive cone of admissible densities, up to a uniform change of the admissible constant M , and satisfy the covariance estimate*

$$(40) \quad \left\| \mathfrak{T}_{\theta_0 \leftarrow \theta}^{\text{obs}} \mathcal{A}_\theta \mathfrak{T}_{\theta \leftarrow \theta_0}^{\text{obs}} f - \mathcal{A}_{\theta_0} f \right\|_{\widehat{\mathcal{B}}_{\theta_0}^\#, w} \leq C d(\theta, \theta_0) \|f\|_{\text{Dom}(\mathcal{A}_{\theta_0})}.$$

Consequently every expression below involving $\widehat{\Sigma}_{\Theta(x, p)}$, $\text{Adm}_M(\Theta(x, p))$, $\mathcal{A}_{\Theta(x, p)}$, or a horizontal derivative is an expression in this Banach bundle, not an unidentified comparison of different phase spaces.

Proof. The stable, unstable, and weak norms are defined by pairings against admissible stable leaves and test functions. The trivialization maps stable leaves into a finite union of stable leaves with uniformly comparable curvature and length; when a leaf is cut by a moving singular curve the unmatched piece is estimated in the weak norm. The Jacobian estimate in Lemma 2.20 transfers the stable norm, the holonomy comparison transfers the unstable norm, and the summable high-strip tails transfer the weak norm. This gives the norm equivalence and the preservation of admissible positive densities in the distributional sense, because nonnegative branchwise C^1 observables pull back to nonnegative branchwise C^1 observables.

For (40), write the generator in a flow box as ∂_u plus the jump relation at the roof. The transition map changes the roof by a C^{r-3} branchwise amount and moves the jump hypersurface by a transversal displacement. Interior terms are therefore first-order Taylor remainders bounded by $d(\theta, \theta_0)\|f\|_{\text{Dom}(\mathcal{A})}$; boundary terms are weak anisotropic functionals supported on moving singular hypersurfaces and are bounded by the same strip-entry closure. This proves the covariance estimate and fixes the meaning of all later parameter and horizontal derivatives. $\square$

2.7. Finite horizon and zero-measure singularities.

Proposition 2.23 (Deterministic bounds on free-flight time). *Under assumption 2.4, there exist constants*

$$(41) \quad 0 < \tau_- \leq \tau_+ < \infty$$

such that every regular free flight between two consecutive non-grazing collisions satisfies

$$(42) \quad \tau_- \leq \tau_\theta(z) \leq \tau_+.$$

The constants are independent of z and θ .

Proof. The upper bound is a direct conversion of the finite corridor condition into time. Since $|v| = 1$, a free flight of duration t is a straight segment of length t . A segment of length L_+ must meet the interior of a scatterer, hence the first collision occurs before time L_+ . Thus $\tau_+ = L_+$.

For the lower bound, after reflection the particle leaves the obstacle into the exterior region. It cannot meet another scatterer before traveling at least the uniform separation distance d_- . It cannot immediately re-enter the same strictly convex scatterer because a line leaving a strictly convex boundary in an outward direction lies outside that scatterer for a one-sided interval, and the possible one-sided intervals have a uniform lower bound by compactness and the curvature bound. Combining the same-scatterer and different-scatterer alternatives gives $\tau_- > 0$. No lower bound on the angle $\cos \varphi$ is used; trajectories arbitrarily close to grazing remain allowed and belong to the regular domain until the limiting grazing state is reached. $\square$

Theorem 2.24 (Finite horizon). *The billiard flow generated by the free motion and specular reflection is defined for all real times for every initial state whose orbit avoids the grazing singularity. In every finite time interval the number of collisions is finite. The return map $\mathcal{P}_\theta$ is defined ν_θ -almost everywhere.*

Proof. The lower bound in Proposition 2.23 prevents accumulation of collision times in finite time. The upper bound prevents infinite free flights. The grazing condition is a scalar equation $\cos \varphi = 0$ on the smooth collision coordinates, hence has ν_θ -measure zero. The preimages of the grazing set under finitely many smooth branches are finite unions of smooth

curves, hence also have zero ν_θ -measure. The countable union over all iterates has zero measure. Therefore the flow and the return map are well-defined on a full-measure invariant set. $\square$

Proposition 2.25 (Birkhoff measure invariance). *For each θ , the global a.e. map $\mathcal{P}_\theta$ preserves ν_θ :*

$$(43) \quad \nu_\theta(\mathcal{P}_\theta^{-1}A) = \nu_\theta(A)$$

for every Borel set A for which the pullback is defined modulo the singular set.

Proof. The Liouville measure $dq dv$ on $Q_\theta \times \mathbb{S}^1$ is preserved by free flight because the motion is translation in q at fixed v . Specular reflection is an orthogonal transformation in the velocity variable and has Jacobian one. Taking the flux of $dq dv$ through the collision cross-section gives the density $\cos \varphi ds d\varphi$. Equality of incoming and outgoing fluxes proves invariance of ν_θ under the return map. The argument is local on each smooth branch, and the excluded singular curves have zero measure. $\square$

3. CONDITIONAL MICRO-ACTION AND THE PRELIMIT HAMILTON–JACOBI EQUATION

This section records the prelimit equation conditional on the coherent slow–fast action required in (A4). When the table parameter depends on a cotangent field that varies with the microscopic state, the family of ordinary billiard flows does not automatically define a single phase space, collision boundary, or specular Hamiltonian flow. The universal-bundle notation below is bookkeeping only; existence, collision-face compatibility, closedness of the generator, and comparison are supplied by (A4), not derived merely from the read-out map $\Theta(x, p)$.

Definition 3.1 (Low-level deterministic finite-response coupling). The coupling between the slow cotangent signal and the billiard cell is specified before any transfer operator, cell problem, or macroscopic equation is introduced. Its primitive data are the following finite-dimensional mechanical objects.

(M1) A compact internal energy surface $\mathcal{Y}$ carrying a C^r deterministic response flow

$$(44) \quad \mathcal{R}_s^{x,P} : \mathcal{Y} \longrightarrow \mathcal{Y}, \quad 0 \leq s \leq s_{\text{fr}},$$

depending C^r on the macroscopic position x and on the transmitted cotangent signal P . The compactness of $\mathcal{Y}$ is the finite-energy statement: the microscopic device that reads the cotangent signal has a bounded energy shell.

(M2) A calibrated deterministic endpoint map

$$(45) \quad \Pi_{\text{fr}} : \mathbb{R}^m \longrightarrow \mathbb{R}^m,$$

obtained by solving the internal Euler–Lagrange response on $\mathcal{Y}$ for the fixed response time s_{fr} . There is a compact physical window $K_{\text{phys}} \subset \mathbb{R}^m$ on which $\Pi_{\text{fr}}(p) = p$; outside the window the compact energy shell forces a smooth saturation. Thus Π_{fr} and all of its derivatives are bounded, but no local Taylor coefficient in K_{phys} is removed.

(M3) Smooth read-out maps on the fixed energy shell,

$$(46) \quad \vartheta : \mathcal{M} \times \mathcal{Y} \rightarrow \Theta, \quad \beta : \mathcal{M} \times \mathcal{Y} \times \hat{\Sigma}_{\vartheta(x,y)} \rightarrow \mathbb{R}^m, \quad \mathcal{V} : \mathcal{M} \times \mathcal{Y} \times \hat{\Sigma}_{\vartheta(x,y)} \rightarrow \mathbb{R},$$

which record, respectively, the deformed billiard geometry, the slow mechanical velocity, and the action density. They are branchwise C^r in the billiard variable and satisfy the same homogeneity-strip summability as the billiard maps.

With y_0 denoting the calibrated rest state and $y(x, P) = \mathcal{R}_{s_{\text{fr}}}^{x, P} y_0$, the coefficients used below are the read-outs

$$(47) \quad \begin{aligned} \Theta(x, p) &= \vartheta(x, y(x, \Pi_{\text{fr}}(p))), \\ B(x, p, z) &= \beta(x, y(x, \Pi_{\text{fr}}(p)), z), \\ V(x, p, z) &= \mathcal{V}(x, y(x, \Pi_{\text{fr}}(p)), z). \end{aligned}$$

They are therefore not macroscopic effective coefficients and are not chosen from the limiting HJB equation. They are restrictions of a deterministic finite-energy interaction on the microscopic phase space. Compactness of $\mathcal{Y}$, smooth dependence of $\mathcal{R}_s^{x, P}$, and the homogeneity strip bounds give

$$(48) \quad \sup_{x, p, \theta, z} \left| \partial_x^\alpha \partial_p^\beta \partial_z^\gamma (\Theta, B, V)(x, p, z) \right| < \infty, \quad |\alpha| + |\beta| + |\gamma| \leq r - 4,$$

where z -derivatives are taken branchwise in homogeneous flow boxes. Locally inside K_{phys} the transmitted signal is the genuine cotangent variable, so quadratic, quartic, and higher response terms in p are physical Taylor terms of the underlying deterministic interaction.

The slow drift used in the singular perturbation is the deterministic fluctuation around the instantaneous microcanonical transport,

$$(49) \quad b(x, p, z) = B(x, p, z) - \int B(x, p, \zeta) d\mu_{\Theta(x, p)}(\zeta).$$

Consequently

$$(50) \quad \int b(x, p, z) d\mu_{\Theta(x, p)}(z) = 0$$

identically. This centering is the deterministic change from the laboratory frame to the comoving slow frame after the ballistic transport velocity has been subtracted. It is not a statistical hypothesis and it is not inferred from the target HJB equation. The work term in the Hamilton–Jacobi equation is $b(x, p, z) \cdot p$, because the physical signal read by the transducer is the slow cotangent coordinate.

Proposition 3.2 (Finite-response coefficients are derived before homogenization). *The objects Θ , B , V , and b in Definition 3.1 are determined entirely by the low-level mechanical data $(\mathcal{Y}, \mathcal{R}, \vartheta, \beta, \mathcal{V}, \Pi_{\text{fr}})$ and by the invariant billiard measure at the instantaneous table. In particular, none of the following is used in their construction: the transfer spectral gap, the cell corrector, the Green–Kubo tensor, the effective Hamiltonian, or the homogenized solution u .*

Proof. For fixed (x, p) the internal response endpoint $y(x, \Pi_{\text{fr}}(p))$ is obtained by solving an ordinary deterministic flow on the compact energy surface $\mathcal{Y}$ for the finite time s_{fr} . The maps ϑ , β , and $\mathcal{V}$ are then evaluated at this endpoint and at the current billiard state z . This gives Θ , B , and V before any averaging in time is performed. The only invariant measure appearing in the definition of b is the Liouville–Birkhoff microcanonical measure of the already specified table $\Theta(x, p)$; its use subtracts the deterministic comoving velocity $\int B d\mu_{\Theta(x, p)}$ and does not invoke decay of correlations or a limiting law. The uniform bounds follow by differentiating the finite-time flow $\mathcal{R}_s^{x, P}$ and the read-out maps on compact sets and then using Π_{fr} for global saturation. Thus the finite-response structure is a low-level deterministic mechanical consequence and not a postulated property of the macroscopic limit. $\square$

Proposition 3.3 (Compatibility constraints for finite-response jets). *Any specified finite-response port used in the paper must satisfy the following necessary low-level constraints:*

- (i) *the scatterer motion is a normal C^r displacement small enough to preserve strict convexity, separation, finite horizon, and the derived grazing trace constants of Lemmas 2.13 and 2.14;*
- (ii) *the read-outs respect the specular reflection law and the parity of incoming and outgoing covectors on the collision section;*
- (iii) *the table displacement, slow force, and running cost are read from one finite-time Lagrangian response action, so their mixed derivatives satisfy the ordinary equality of mixed partials on regular branches;*
- (iv) *the zero-signal table and zero-action gauge are fixed before any cell problem is introduced, namely $\Theta(x, 0) = \Theta_0(x)$ and the normalized potential obeys (63);*
- (v) *every dependence on the cotangent signal passes through the saturated variable $\Pi_{\text{fr}}(p)$, so all derivatives required by the anisotropic Banach estimates are bounded globally;*
- (vi) *the amplitudes, supports, and endpoint coordinates are chosen on regular collision rectangles before the transfer operator, correctors, Green–Kubo tensor, effective Hamiltonian, or limiting solution are defined.*

These conditions are not asserted to be sufficient for realization. In particular, they do not establish local surjectivity of the endpoint map, independence of invariant-response directions, or compatibility of an arbitrary prescribed jet with reciprocity and conservation laws. They only ensure that a port already verified to exist is specified before D , H , any corrector, or the solution u is defined.

Proof. This follows directly from the definition in Definition 3.1. A compatible compact Lagrangian port moves the scatterers by normal boundary displacements, transports velocities by the ordinary specular reflection map, and evaluates force and action read-outs from the same finite-time action. Thus its Taylor coefficients automatically obey smallness, parity, gauge, and saturation requirements, and those requirements are stated entirely before any averaging or homogenization step. Nothing in this observation proves that prescribed coefficients can be realized; existence and compatibility remain application-specific hypotheses. $\square$

Theorem 3.4 (Finite-response port is not an effective-coefficient oracle). *Fix the primitive billiard geometry and a compact finite-response port satisfying Definition 3.1 and Proposition 3.3. Let $\mathcal{F}_{\text{port}}$ denote the finite list of data obtained before any transfer operator is introduced:*

$$(\mathcal{Y}, \mathcal{R}_s^{x,P}, \Pi_{\text{fr}}, \vartheta, \beta, \mathcal{V}, \text{scatterer charts, specular reflection, fixed energy}).$$

Then the effective objects

$$\Pi_\theta, \text{quad}(I - \mathcal{L}_\theta)^{-1} \Pi_\theta^\perp, \text{quad}(-\mathcal{A}_\theta|_{\ker \Pi_\theta})^{-1}, \text{quad}\chi_p, \text{quad}\chi_{p,X}^{(2)}, \text{quad}D(x, p), \text{quad}H(x, p)$$

are not coordinates of $\mathcal{F}_{\text{port}}$. They are obtained only by applying the already constructed singular-billiard transfer calculus to the coefficient functions read from $\mathcal{F}_{\text{port}}$. More concretely, the deterministic map from primitive data to the effective coefficients factors as

$$\mathcal{F}_{\text{port}} \longmapsto (\Theta, b, V) \longmapsto (\mathcal{P}_\theta, \mathcal{L}_\theta, \mathcal{A}_\theta, \Pi_\theta, R_\theta) \longmapsto (\chi_p, \chi_{p,X}^{(2)}) \longmapsto (D, H),$$

and no arrow in this factorization admits (D, H) or the homogenized solution as an input.

Proof. The first arrow is finite-dimensional. It consists of solving the internal Euler–Lagrange response for the finite time s_{fr} , reading the normal scatterer displacement, force, and action

density, and subtracting the Liouville–Birkhoff comoving velocity in (49). This arrow ends before a Banach space or cell equation has been defined.

The second arrow combines construction and hypotheses. From the table $\Theta(x, p)$ one solves impact equations and constructs homogeneous branches; (A1)–(A2) then supply the spectral gap and reduced resolvent for the resulting transfer operator. These operations are infinite-time analytic operations on the collision map; they cannot be read off from a finite endpoint coordinate of the port. They depend on the orbit structure of the resulting billiard table, not on an effective Hamiltonian chosen in advance.

The third and fourth arrows are likewise analytic. The first corrector is $-\mathcal{A}_\theta^{-1}\Pi_\theta^\perp(b \cdot p)$, the second corrector is the same reduced inverse applied to the centered full-gradient order-one observable, and D, H are invariant pairings of those objects. Thus changing the port changes the microscopic coefficients, and the transfer-resolvent calculation then determines the corresponding D, H . There is no operation in the reverse direction from D, H or from the macroscopic solution to the primitive port. This proves that finite-response flexibility is a microscopic realization statement rather than an oracle for the effective equation. $\square$

Proposition 3.5 (Admissibility template for a mechanical port). *Suppose a specified finite-response port is generated by a deterministic finite-time Lagrangian response system*

$$(51) \quad L_{\text{fr}}(x, P, y, \dot{y}) = \frac{1}{2}g_{ab}(y)\dot{y}^a\dot{y}^b - U_0(y) - U_1(x, \Pi_{\text{fr}}(P), y)$$

on a compact energy shell, with endpoint $y(x, P) = \mathcal{R}_{s_{\text{fr}}}^{x, P}y_0$ and the three read-outs $\vartheta, \beta, \mathcal{V}$ are obtained by differentiating the same microscopic action and by reading the induced scatterer displacement. In particular:

- (a) *the table parameter $\Theta(x, p)$ is the normal boundary displacement produced by the internal endpoint $y(x, \Pi_{\text{fr}}(p))$;*
- (b) *the velocity field $B(x, p, z)$ is the slow force conjugate to x in the same Lagrangian action, transported to the collision section by the Liouville flux map;*
- (c) *the running cost $V(x, p, z)$ is the branchwise action density of the same finite-time response and the billiard flight;*
- (d) *all global bounds in (48) follow from compactness of the internal energy shell and from the smooth saturation Π_{fr} ;*
- (e) *the particular response jet used in an application must be verified to satisfy the endpoint equations, action reciprocity, conservation laws, and billiard compatibility of (A4).*

No arbitrary finite-jet realization theorem is asserted. Non-convex and non-subadditive examples later in the paper are conditional examples until the specified port and its invariant-response coefficients are independently verified. This proposition is a compatibility checklist, not an existence or realization theorem. In particular, adjoining formal endpoint coordinates does not prove that the requested read-outs arise from one action.

Proof. For a port satisfying the premise, the three read-outs are derivatives or endpoint evaluations of the same action, so the listed reciprocity and gauge conditions are necessary. Compactness and saturation give bounded derivatives only after the endpoint flow and collision-face read-outs have been shown smooth on the chosen branches. Exact endpoint matching, global collision-face compatibility, conservation, and the required uniform bounds remain obligations of (A4); no extension theorem or formal tracking coordinate discharges them. $\square$

Lemma 3.6 (Conditional finite-time endpoint-jet construction). *Fix a compact cotangent window K_{phys} and a finite order $J \leq r - 6$. Let $\mathcal{J}(P)$ be a finite list of saturated polynomial jets. If the finite-dimensional endpoint map is locally onto this list while preserving the single-action constraints in (A4), then a compactly supported local construction realizes the prescribed J -jet on K_{phys} . Local surjectivity and compatibility are hypotheses, not consequences of disjoint bumps.*

Proof. Choose saturated monomials $M_\alpha(P) = \Pi_{\text{fr}}(P)^\alpha$, $|\alpha| \leq J$, which agree with ordinary monomials on K_{phys} . Work first in a compact coordinate ball of a torus factor of the internal phase space and take initial condition $(y, \dot{y}) = (0, 0)$. For a finite vector of amplitudes A define a potential

$$(52) \quad U_A(x, P, y) = - \sum_{\alpha, \ell} A_{\alpha\ell}(x) M_\alpha(P) y_\ell + \sum_{q=2}^{J+1} U_{A,q}(x, P, y) + U_{\text{conf}}(y),$$

where the confining term is flat on the chosen coordinate ball and positive outside it. The endpoint $E_A(x, P) = y(s_{\text{fr}}; x, P, A)$ has a Taylor expansion in A obtained by Picard iteration of the deterministic Euler–Lagrange ODE. The linear part is

$$(53) \quad E_A(x, P) = \frac{s_{\text{fr}}^2}{2} \sum_{\alpha, \ell} A_{\alpha\ell}(x) M_\alpha(P) e_\ell + O(|A|^2).$$

The higher correction potentials $U_{A,q}$ are chosen recursively. At degree q the lower-degree amplitudes determine a known polynomial error in the endpoint jet; adding a new internal coordinate with a linear force proportional to the negative of that error changes no lower-degree jet and cancels the degree- q error. Because only finitely many monomials and degrees are involved, the recursion terminates and the finite Jacobian from the newly introduced amplitudes to the degree- q endpoint coefficients is diagonal with entries $s_{\text{fr}}^2/2 \neq 0$.

After the required jet has been matched inside the coordinate ball, extend the local potential to the torus by a cutoff that is identically one on the ball and add the confining term so that the selected energy level is compact and regular. The endpoint orbit for all $P \in K_{\text{phys}}$ stays inside the ball when the response amplitudes are chosen within the smallness range required to preserve the billiard geometry. Outside the physical window the saturation Π_{fr} keeps the vector field and all derivatives bounded. The endpoint jet is thus exact on K_{phys} and globally smooth after saturation.

The argument is finite-dimensional deterministic controllability of an ODE endpoint map. It neither knows nor uses the transfer spectrum, the Green–Kubo tensor, the effective Hamiltonian, or a limiting solution. Those objects are computed only after the microscopic endpoint data have been fixed. $\square$

Proposition 3.7 (Single-action reciprocity of finite-response read-outs). *The finite-response port is generated by one deterministic microscopic action, not by three independent coefficient choices. On each homogeneous branch there is an action functional*

$$(54) \quad \mathcal{S}_{\text{fr}}(x, P, y_0; z) = \int_0^{s_{\text{fr}}} L_{\text{fr}}(x, P, y_s, \dot{y}_s) ds + \mathcal{S}_{\text{flight}}(\Theta(x, P), z)$$

such that, after the zero-energy gauge and the comoving Liouville subtraction,

$$(55) \quad \Theta(x, p) = \vartheta(x, y(x, \Pi_{\text{fr}}(p))),$$

$$(56) \quad B^a(x, p, z) = -\partial_{x_a} \mathcal{S}_{\text{fr}}(x, \Pi_{\text{fr}}(p), y_0; z) + \mathcal{A}_{\Theta(x, p)} R_{\text{fr}}^a(x, p, z),$$

$$(57) \quad V(x, p, z) = \partial_{s_{\text{fr}}} \mathcal{S}_{\text{fr}}(x, \Pi_{\text{fr}}(p), y_0; z) + \mathcal{A}_{\Theta(x, p)} R_{\text{fr}}^0(x, p, z),$$

$$(58) \quad D_p \Theta(x, p) = D_y \vartheta(x, y(x, \Pi_{\text{fr}}(p))) D_P y(x, \Pi_{\text{fr}}(p)) D \Pi_{\text{fr}}(p).$$

The branch functions R_{fr}^a and R_{fr}^0 are finite-response coboundary gauges with zero invariant averages. Consequently the allowed Taylor jets of Θ , B , and V obey the reciprocity identities obtained by differentiating (55); the constructive freedom of Proposition 3.5 is a finite-dimensional mechanical controllability statement and cannot insert an effective Hamiltonian coefficient after homogenization.

Proof. For fixed (x, P) the internal endpoint $y(x, P)$ is the endpoint of the Euler–Lagrange flow generated by L_{fr} . Hamilton’s principle gives the ordinary endpoint-variation identity for (54): variations in x produce the slow force, variations in response time produce the running action density, and variations in P are transmitted only through the endpoint map and the saturated signal $\Pi_{\text{fr}}(P)$. The first and fourth identities in (55) are therefore immediate from the definition of the boundary displacement read-out.

The billiard flight contribution is Liouville exact on every homogeneous branch: changing the branch action by a flight endpoint potential adds a generator coboundary. These endpoint potentials are the gauges R_{fr}^a and R_{fr}^0 . Their invariant averages vanish because $\int \mathcal{A}_{\Theta(x, p)} R d\mu_{\Theta(x, p)} = 0$ for generator-domain observables, with the identity justified branchwise and then closed in the weak anisotropic norm by Proposition 3.13. Hence the displayed formulas give the same centered drift and normalized potential used in the cell problem.

Differentiating the endpoint-variation formulas yields the mixed-derivative reciprocity relations that any admissible finite jet must satisfy. Whether a chosen endpoint family has enough independent directions while satisfying all these relations is a separate rank condition. The effective tensor and Hamiltonian are computed only after a compatible port has been fixed. $\square$

Proposition 3.8 (Conditional local-bump response directions). *Fix a finite jet order $J \leq r - 6$, a compact cotangent window K_{phys} , and a finite list of response coefficients satisfying the specular parity and the normalization at $p = 0$. Assume additionally that the full endpoint/response Jacobian, including global invariant-density response and single-action reciprocity, has the required rank. Then a compact finite-response port locally realizes these coefficients to order J . Disjoint bump supports alone do not imply this rank condition.*

Proof. Let F be the finite-dimensional map from admissible port parameters to the listed endpoint and invariant-response coefficients, restricted to the single-action compatibility manifold. The premise says that DF has the required rank at the reference port and that (A3)–(A4) provide the needed differentiability. The finite-dimensional submersion theorem therefore gives local realization of coefficients in the image of F . This argument does not prove the rank premise. In particular, disjoint supports do not eliminate normalization, recurrence, moving-singularity, reciprocity, or conservation couplings in DF . $\square$

Definition 3.9 (Acyclic jet graph for the microscopic action). Fix $\varepsilon > 0$ and a homogeneous flow box in the bundle of Definition 2.19. A deterministic slow–fast jet graph is a branchwise

C^1 submanifold

$$(59) \quad \Lambda_s^\varepsilon = \{(x, z, P_s(x, z), A_s(x, z)) : (x, z) \in \mathcal{M} \times \widehat{\Sigma}_{\Theta(x, P_s(x, z))}\}$$

where P_s is a cotangent signal and A_s is the accumulated action. The coefficient maps Θ , b , and V are evaluated at the independent jet coordinate P_s of this graph. The graph is propagated by the deterministic branch equations

$$(60) \quad \dot{z}_s = \varepsilon^{-2} F_{\Theta(x_s, P_s)}(z_s), \quad \dot{x}_s = \varepsilon^{-1} b(x_s, P_s, z_s), \quad \dot{A}_s = -V^\circ(x_s, P_s, z_s),$$

with specular reflection in the z variable and with P_s transported by branchwise differentiation of the same action graph:

$$(61) \quad P_s(x_s, z_s) = D_x A_s(x_s, z_s).$$

Equations (60)–(61) are a closed deterministic graph transform on finite time slabs. The cotangent signal is therefore not chosen from the macroscopic limit and is not an independent external control.

Lemma 3.10 (Deterministic graph-transform closure). *For every branchwise C^2 terminal graph with finite-response cotangent range, the map that sends a trial jet graph on a sufficiently short backward time slab to the graph obtained by integrating (60) and then differentiating the resulting action is a contraction in the C^1 graph norm after the usual exponential weight in time. The contraction constants depend only on the finite-response bounds, the billiard geometric constants, and ε on that fixed slab. Concatenating finitely many slabs constructs a unique branchwise jet graph on $[0, T]$.*

Proof. All coefficients in (60) are branchwise C^{r-4} and have bounded derivatives in the transmitted signal because of (48). In a fixed homogeneous flow box the difference of two trial graphs obeys a Gronwall inequality whose source is the Lipschitz difference of their cotangent signals. Differentiating the action identity with respect to x gives the same inequality for the cotangent part, because the horizontal derivatives of the billiard bundle are bounded by Proposition 2.22. Choosing the slab length so that the Gronwall factor times the coefficient Lipschitz constant is less than one gives a contraction. Specular reflections paste the branchwise estimates because the reflection map is C^{r-3} on each homogeneous component and preserves the Liouville action. If a branch reaches a singular hypersurface the trace is ignored in the graph norm, since the hypersurface has zero flux measure and its movement is controlled in the weak anisotropic norm. Compactness of the time interval gives the finite concatenation. $\square$

Definition 3.11 (Microscopic Hamilton principal function). For a terminal payoff $\phi \in C^1(\mathcal{M})$, the branchwise Hamilton principal function U^ε is the action component of the unique deterministic jet graph of Lemma 3.10 with terminal graph $A_T(x, z) = \phi(x)$ and $P_T(x, z) = D\phi(x)$. Equivalently, on every regular branch and for every sufficiently small $h > 0$,

$$(62) \quad U^\varepsilon(t, x, z) = U^\varepsilon(t + h, x_h, z_h) + \int_t^{t+h} V^\circ(x_s, P_s, z_s) ds + o(h),$$

where (x_s, z_s, P_s) is the closed jet graph trajectory starting from $(t, x, z, D_x U^\varepsilon(t, x, z))$. The zero-energy gauge is fixed directly from the microscopic running cost at zero transmitted signal:

$$(63) \quad C_0(x) = \int V(x, 0, z) d\mu_{\Theta(x, 0)}(z), \quad V^\circ(x, p, z) = V(x, p, z) - C_0(x).$$

This gauge is determined before the cell problem and before the macroscopic HJB equation are introduced.

Definition 3.12 (Finite-response branch generator). Let P be a branchwise finite-response cotangent field on a homogeneous flow box of the universal billiard bundle. The deterministic fast vector field with finite-response port P is denoted

$$(64) \quad \mathbf{F}_{x,P}^{\text{fr}}(z) := \mathbf{F}_{\Theta(x,P(z))}(z)$$

after transporting the state z to the corresponding moved billiard fiber by Lemma 2.20. On regular branches the associated observable generator is

$$(65) \quad \mathcal{A}_{x,P}^{\text{fr}} F(z) = \mathbf{F}_{x,P}^{\text{fr}}(z) \cdot D_z^{\text{br}} F(z),$$

with the specular reflection jump condition imposed at the moved boundary. If P is constant in z , this is exactly the fixed-table generator $\mathcal{A}_{\Theta(x,P)}$. If P is not constant, the expression is not treated as a fixed billiard flow with a fictitious z -dependent table; it is the branchwise deterministic vector field on the universal bundle. Its weak anisotropic closure is obtained by the same moving-boundary estimates as in Proposition 2.22.

Proposition 3.13 (Universal-bundle domain of the finite-response branch generator). *Let P be a branchwise C^1 finite-response cotangent field with bounded finite-response norm. The operator $\mathcal{A}_{x,P}^{\text{fr}}$ of Definition 3.12 has a closed weak anisotropic graph on the universal billiard bundle. Its domain consists of observable sections F such that*

- (a) F is branchwise C^1 in homogeneous flow boxes after trivialization;
- (b) the specular jump condition is satisfied on every moving collision face determined by $\Theta(x, P(z))$;
- (c) for every admissible density ρ the Green identity

$$(66) \quad \langle \mathcal{A}_{x,P}^{\text{fr}} F, \rho \rangle = -\langle F, \text{div}_{\mu_0}(X_{x,P}\rho) \rangle + \sum_{\Gamma \subset S(P)} \langle [F]_{\Gamma}, (X_{x,P} \cdot n_{\Gamma})\rho \rangle_{\Gamma}$$

is finite in the weak anisotropic norm.

Here $X_{x,P}$ is the pulled-back branch vector field on the fixed reference bundle, and the sum ranges over the moved singular hypersurfaces. The boundary term in (66) vanishes on true specular collision faces and is weak-norm bounded on grazing and branch-change faces. Moreover, if $P_{\eta} = p + \eta q$ then the first variation (255) is the derivative of this closed graph in the strong-to-weak topology.

Proof. Choose a reference table θ_0 and pull every nearby table to the reference fiber by the geometric trivialization of Lemma 2.20. For a non-constant field $P(z)$ this does not define a single fixed billiard table; it defines the ordinary first-order vector field

$$(67) \quad X_{x,P}(z) = D\mathfrak{J}_{\theta(z) \leftarrow \theta_0}^{-1} \mathbf{F}_{\theta(z)}(\mathfrak{J}_{\theta(z) \leftarrow \theta_0} z), \quad \theta(z) = \Theta(x, P(z)),$$

on each homogeneous branch of the universal bundle. Since P has bounded finite-response norm and Θ is globally bounded with bounded derivatives, $X_{x,P}$ has the same strip losses as the ordinary billiard vector field.

For branchwise smooth F and ρ , integration by parts with respect to the pulled-back Liouville volume gives (66). On physical collision faces the two traces of F are identified by specular reflection and the Liouville flux is preserved; therefore the corresponding boundary contributions cancel exactly. Grazing and branch-change hypersurfaces are not physical

boundaries of the Hamiltonian flow. Their contributions are trace functionals on stable leaves. The transversality of these hypersurfaces and the high-strip estimate of Lemma 4.7 imply

$$(68) \quad \sum_{\Gamma \subset S(P)} |\langle [F]_\Gamma, (X_{x,P} \cdot n_\Gamma) \rho \rangle_\Gamma| \leq C \|F\|_{\widehat{\mathcal{B}}^{\#,s}} \|\rho\|_{\widehat{\mathcal{B}}},$$

which gives weak graph boundedness. If $F_n \rightarrow F$ in the observable norm and $\mathcal{A}_{x,P}^{\text{fr}} F_n \rightarrow G$ in the weak norm, passing to the limit in (66) shows that F satisfies the same Green identity with generator G ; hence the graph is closed.

Finally let $P_\eta = p + \eta q$. The vector field, the roof identification, and the singular hypersurfaces vary C^1 in η in the strong-to-weak topology by exactly the moving-boundary computation of Theorem 6.6. Differentiating the Green identity at $\eta = 0$ gives (255). This proves that the full-gradient prelimit operator is a single deterministic branch operator on the universal bundle, not an ill-defined family of fixed billiards with a z -dependent parameter. $\square$

Theorem 3.14 (Exact deterministic prelimit Hamilton–Jacobi equation). *On every regular homogeneous flow branch, the principal function constructed by Lemma 3.10 satisfies*

$$(69) \quad -\partial_t U^\varepsilon - \varepsilon^{-2} \mathbf{F}_{x,p}^{\text{fr}}(z) \cdot D_z U^\varepsilon - \varepsilon^{-1} b(x, p, z) \cdot p - V^\circ(x, p, z) = 0, \quad p = D_x U^\varepsilon(t, x, z).$$

Equivalently, after completing branchwise observables in the anisotropic bundle and writing $\mathcal{A}_{x,p}^{\text{fr}}$ for the finite-response branch generator on the universal billiard bundle, (69) becomes

$$(70) \quad -\partial_t u^\varepsilon - \varepsilon^{-2} \mathcal{A}_{x,p}^{\text{fr}} u^\varepsilon - \varepsilon^{-1} b(x, p, z) \cdot p - V^\circ(x, p, z) = 0, \quad p = D_x u^\varepsilon(t, x, z).$$

The construction and the proof are deterministic graph-transform statements on the cotangent jet bundle. They do not use the macroscopic limit, the cell problem or any post-derivation representation.

Proof. The graph-transform lemma constructs U^ε and its cotangent signal first. Thus, when a derivative is written below, the identity $p = D_x U^\varepsilon$ is the closure relation of the already constructed graph, not an external prescription using an unknown limiting solution.

Fix a regular branch and start the graph trajectory from (t, x, z, p) with $p = D_x U^\varepsilon(t, x, z)$. Taylor expansion of the terminal side of (62) gives

$$(71) \quad U^\varepsilon(t + h, x_h, z_h) - U^\varepsilon(t, x, z) = h \partial_t U^\varepsilon + h \varepsilon^{-1} b(x, p, z) \cdot D_x U^\varepsilon$$

$$(72) \quad + h \varepsilon^{-2} \mathbf{F}_{x,p}^{\text{fr}}(z) \cdot D_z U^\varepsilon + o(h),$$

where the coefficient arguments are the jet coordinates at the starting point. Substituting into the action identity, dividing by h , and letting $h \downarrow 0$ gives (69). Specular reflection contributes no boundary action term: the tangential coordinate is continuous, the normal velocity changes sign, and the elastic reflection preserves the fixed energy shell. The singular hypersurfaces are excluded from pointwise evaluation and have zero flux measure; their traces are accounted for only through the anisotropic weak boundary terms of the bundle construction.

The passage to (70) is the definition of the observable generator on $\widehat{\mathcal{B}}_\theta^\#$, transported through the Banach-bundle charts of Proposition 2.22. Since the graph construction precedes the cell problem, there is no circular dependence between the prelimit equation and the homogenized HJB equation. $\square$

Theorem 3.15 (Closed deterministic construction of the prelimit jet graph). *Fix $\varepsilon > 0$, a terminal payoff $\phi \in C^4$, and a regular homogeneous branch rectangle in the universal*

billiard bundle. Before any macroscopic limit is considered, there is a unique branchwise exact Lagrangian jet graph

$$\Lambda_s^\varepsilon = \{(x_s, z_s, P_s, A_s) : P_s = D_{x_s} A_s\}$$

on its deterministic lifetime. It is generated by the finite-dimensional Hamilton system consisting of the slow transport, the fast billiard branch vector field, and the finite-response action read-out:

$$\begin{aligned} \dot{z}_s &= \varepsilon^{-2} \mathbf{F}_{x_s, P_s}^{\text{fr}}(z_s), \\ \dot{x}_s &= \varepsilon^{-1} b(x_s, P_s, z_s), \\ (73) \quad \dot{A}_s &= \varepsilon^{-1} b(x_s, P_s, z_s) \cdot P_s + V^\circ(x_s, P_s, z_s), \\ \dot{P}_s &= -D_x \left(\varepsilon^{-1} b(x_s, P_s, z_s) \cdot P_s + V^\circ(x_s, P_s, z_s) \right) - D_x \mathbf{F}_{x_s, P_s}^{\text{fr}}(z_s)^* \lambda_s. \end{aligned}$$

Here λ_s is the cotangent multiplier for the fast branch coordinate, transported through the specular reflection law. Across regular impacts the canonical one-form is preserved, and branch changes occur only through the zero-measure singular boundary whose derivative is represented later by weak trace functionals. Consequently the relation $P_s = D_x A_s$ is propagated by the branch dynamics itself; it is not an imposed fixed-point condition involving the limiting HJB solution.

Proof. On a fixed homogeneous branch all functions in (73) are branchwise C^{r-4} and globally bounded on compact jet windows by Equation (48) and the branch atlas. The finite-dimensional ODE therefore has a unique solution until it reaches the boundary of the chosen homogeneous rectangle. The terminal graph is exact because it is generated by $A_T(x, z) = \phi(x)$, so the canonical one-form on the graph is $dA_T = P_T dx$.

The Hamiltonian vector field in (73) is obtained from the same finite-response action as Theorem 3.14. Therefore the Lie derivative of the one-form $dA_s - P_s dx_s - \lambda_s dz_s$ along the branch flow is zero. Equivalently, differentiating the action identity with respect to the initial slow coordinate gives the adjoint equation for P_s . Thus exactness of the graph is transported from the terminal graph to every earlier regular time.

At a specular reflection, Lemma 3.17 shows that the jump in velocity is normal to the boundary, while allowed endpoint variations are tangent. Hence the canonical endpoint term is unchanged by the impact. The only points where the branch identity is not evaluated are grazing and multiple-collision singularities; they have zero flux measure and are accounted for, after completion, by the moving-boundary trace estimates. The construction of Λ_s^ε is therefore complete before the cell problem, the effective coefficients, or the homogenized solution are introduced. Substituting $P_s = D_x A_s$ in the branch action identity gives exactly (69). $\square$

Proposition 3.16 (Single universal-bundle meaning of a z -dependent contact covector). *Let $\Phi(t, x, z) = \varphi(t, x) + \varepsilon \Psi_1(t, x, z) + \varepsilon^2 \Psi_2(t, x, z)$ be a branchwise perturbed test function used in the viscosity-duality proof, and write*

$$(74) \quad P_\Phi(t, x, z) = D_x \Phi(t, x, z).$$

The prelimit operator evaluated at P_Φ is a single deterministic vector field on the pulled-back universal billiard bundle. More precisely, after normal-coordinate trivialization of the moving tables, the map

$$(75) \quad (x, z, P) \longmapsto \mathcal{A}_{x, P}^{\text{fr}} + \frac{1}{\varepsilon} b(x, P, z) \cdot P + V^\circ(x, P, z)$$

is branchwise C^{r-3} in (x, P, z) on homogeneous components, with strong-to-weak trace bounds across moving singular curves. Therefore a z -dependent contact covector does not mean that several billiard tables are run simultaneously; it means that the universal branch vector field is linearized along the graph $z \mapsto P_\Phi(t, x, z)$.

Proof. The mechanical port constructs scatterer deformations, finite-response drift, and potential as functions on the universal bundle (x, P, ζ) , where ζ is a reference collision coordinate. The normal-coordinate trivialization sends the table $\Theta(x, P)$ back to the reference section and expresses the reflection law, roof, and Jacobian as branchwise functions of (x, P, ζ) . The impact-function ledger Lemma 2.13 gives the stated regularity and the weak trace bounds. Substitution of the graph $P = P_\Phi(t, x, z)$ is then an ordinary composition in this deterministic bundle. The singular set is not a place where a pointwise equation is evaluated; it contributes only the already controlled trace functionals in the weak anisotropic topology. Consequently the full-gradient closure used later is not a change of dynamics after homogenization but the original branch vector field evaluated on a contact jet of the exact prelimit Hamilton–Jacobi equation. $\square$

Lemma 3.17 (Branchwise endpoint variation through specular reflection). *On a regular homogeneous branch, variation of the Hamilton principal action through a collision contributes only the standard endpoint cotangent terms; there is no unaccounted reflection boundary work. If q is the collision point and v^- , v^+ are the incoming and outgoing velocities, then for a boundary variation $\delta q \in T_q \partial Q_\theta$,*

$$(76) \quad \langle v^- - v^+, \delta q \rangle = 0,$$

and the normal component cancels because $v^+ = v^- - 2\langle v^-, n \rangle n$. The only boundary terms left by differentiating a moving family of branches are the moving-singularity trace terms of Lemma 2.13.

Proof. The tangential component of velocity is preserved by elastic reflection, while the normal component changes sign. Since a regular endpoint variation of the collision point is tangent to the boundary, its scalar product with the jump $v^- - v^+$, which is purely normal, vanishes. If the branch boundary itself moves, the variation is not an endpoint term of a regular trajectory but the derivative of a characteristic function selecting a homogeneous branch. This is exactly the trace distribution estimated in Lemma 2.13. $\square$

Remark 3.18 (Acyclic role of the coupling). The maps Θ , B , and V are low-level constitutive data. The spectral gap and response bounds are independent clauses of (A1)–(A3). Conditional on them, the cell resolvent, Green–Kubo tensor, and HJB coefficients are defined by the stated transfer and two-scale formulae.

4. THE GLOBAL BILLIARD MAP AS A HYPERBOLIC MAP WITH SINGULARITIES

4.1. Homogeneity strips and smooth branches. The derivative of $\mathcal{P}_\theta$ becomes singular near $|\varphi| = \pi/2$. Instead of excluding such states, we use the standard homogeneity decomposition. For $k \geq k_0$ define

$$(77) \quad \mathbb{H}_k^\pm = \left\{ (s, \varphi) : \frac{1}{(k+1)^2} < \frac{\pi}{2} \mp \varphi < \frac{1}{k^2} \right\},$$

and let $\mathbb{H}_0$ be the complement of the union of these boundary strips. A homogeneous component is a connected component of the complement of the singular curves and the homogeneity strip boundaries.

Proposition 4.1 (Piecewise smoothness of the global map). *For every θ , the map $\mathcal{P}_\theta$ is C^r on each homogeneous component. On such a component it is a diffeomorphism onto its image. The singular set is a countable union of C^r curves whose slopes are uniformly transverse to the stable and unstable cones defined below.*

Proof. Fix a regular branch. The next impact point is determined by the equation

$$(78) \quad F(s, \varphi, t, \theta) = G(q(s, \theta) + tv(s, \varphi, \theta), \theta) = 0.$$

At a non-grazing collision,

$$(79) \quad \partial_t F = \langle v, n(q + tv, \theta) \rangle \neq 0.$$

The implicit function theorem gives a C^r flight time $t = \tau_\theta(s, \varphi)$ on that branch. The impact point and reflected velocity are compositions of C^r functions, hence $\mathcal{P}_\theta$ is C^r . The inverse branch is obtained by time reversal, so it is also C^r . Singularities occur only when $\partial_t F = 0$ or when the identity of the next scatterer changes. The impact-function trace ledger Lemma 2.13 gives C^r singular curves after the square-root grazing blow-up. Homogeneity strip boundaries are smooth by construction. Uniform transversality follows from curvature and the angle-coordinate definition, and is therefore a derived geometric estimate rather than an input. $\square$

4.2. Cone fields. The billiard metric used for expansion is a length element on unstable and stable curves, not a norm on the full tangent plane. For a non-vertical tangent vector $u = (ds, d\varphi)$ tangent to a homogeneous unstable curve we write

$$(80) \quad |u|_p = \cos \varphi |ds|,$$

and for estimates involving arbitrary tangent vectors we use the genuine auxiliary norm

$$(81) \quad |u|_* = \cos \varphi |ds| + |d\varphi|.$$

For $ds \neq 0$ set

$$(82) \quad \mathcal{B}(u) = \frac{d\varphi}{ds} + \kappa(s, \theta).$$

This is the curvature of the infinitesimal post-collision wave front represented by u . The upper size of $\mathcal{B}$ is not uniformly bounded near grazing; indeed a dispersing reflection contains the term $2\kappa/\cos \varphi$. The correct invariant object is therefore a lower cone together with a strip estimate scheme.

Definition 4.2 (Strip-adapted stable and unstable cones). Choose $a_- > 0$ depending only on the finite horizon and curvature constants. If $z \in \mathbb{H}_k^\pm$, write $k(z) = k$, and put $k(z) = 1$ on the central region $\mathbb{H}_0$. The unstable cone is

$$(83) \quad C_\theta^u(z) = \{u : ds \neq 0, \mathcal{B}(u) \geq a_-\}.$$

Its homogeneous-strip truncation is

$$(84) \quad C_\theta^u(z; k) = \{u \in C_\theta^u(z) : \mathcal{B}(u) \leq a_+ k(z)^2\},$$

where a_+ is chosen larger than the universal constants in (86) below. Stable cones are defined by time reversal:

$$(85) \quad C_\theta^s(z) = \{u : ds \neq 0, \mathcal{B}^-(u) \leq -a_-\},$$

where $\mathcal{B}^-$ is the corresponding pre-collision curvature. The full cone is the union over strips of the truncated cones. Thus grazing is kept in the phase space and appears only through the polynomial strip factor k^2 .

Lemma 4.3 (Deterministic optical recurrence). *Let $\mathcal{B}_n^+$ be the post-reflection curvature of a wave front at the n th collision and let $\mathcal{B}_{n+1}^-$ be the pre-reflection curvature at the next collision. On every regular homogeneous branch,*

$$(86) \quad \mathcal{B}_{n+1}^- = \frac{\mathcal{B}_n^+}{1 + \tau_n \mathcal{B}_n^+}, \quad \mathcal{B}_{n+1}^+ = \mathcal{B}_{n+1}^- + \frac{2\kappa_{n+1}}{\cos \varphi_{n+1}}.$$

Moreover,

$$(87) \quad \frac{|D\mathcal{P}_\theta(z)u|_p}{|u|_p} = 1 + \tau_n \mathcal{B}_n^+.$$

Proof. The first formula is the elementary geometry of parallel rays during a free flight: if the infinitesimal radius of curvature is $R_n^+ = 1/\mathcal{B}_n^+$, then after a flight of length τ_n the radius is $R_n^+ + \tau_n$, hence the curvature is $(R_n^+ + \tau_n)^{-1}$. The reflection formula follows by differentiating the specular reflection map in the normal section; a convex scatterer adds twice the boundary curvature divided by the normal velocity component. The Jacobian of projection of the transported front onto the next collision section is exactly $1 + \tau_n \mathcal{B}_n^+$ in the p -length (80). These are branchwise identities before any averaging or limiting operation. $\square$

Theorem 4.4 (Invariant cones and deterministic expansion). *There exists $\Lambda > 1$ such that for every θ , every regular point z , and every $u \in C_\theta^u(z)$ for which $D\mathcal{P}_\theta(z)u$ is defined,*

$$(88) \quad D\mathcal{P}_\theta(z)u \in C_\theta^u(\mathcal{P}_\theta z), \quad |D\mathcal{P}_\theta(z)u|_p \geq \Lambda |u|_p.$$

In addition, if $\mathcal{P}_\theta z \in \mathbb{H}_k^\pm$, then

$$(89) \quad \mathcal{B}(D\mathcal{P}_\theta(z)u) \leq a_+ k^2$$

for the homogeneous truncation after standard subdivision. The analogous statement holds for $D\mathcal{P}_\theta^{-1}$ on stable cones.

Proof. Let $u \in C_\theta^u(z)$ and write $\mathcal{B}_n^+ = \mathcal{B}(u) \geq a_-$. By finite horizon, $\tau_n \in [\tau_-, \tau_+]$. The free-flight part of (86) gives

$$(90) \quad 0 < \frac{a_-}{1 + \tau_+ a_-} \leq \mathcal{B}_{n+1}^- \leq \frac{1}{\tau_-}.$$

After reflection,

$$(91) \quad \mathcal{B}_{n+1}^+ \geq \frac{a_-}{1 + \tau_+ a_-} + 2\kappa_-.$$

Choosing a_- smaller than the right hand side gives strict invariance of the lower cone. If the next point lies in the central region, $\cos \varphi$ is bounded below and the same recurrence gives a fixed upper bound. If it lies in $\mathbb{H}_k^\pm$, then $\cos \varphi \asymp k^{-2}$ and hence the reflection term is at most

Ck^2 ; choosing a_+ larger than $1/\tau_- + C$ proves (89). Thus the sharp strip-dependent bound is obtained without imposing a positive-angle cutoff. Finally (87) gives

$$(92) \quad |D\mathcal{P}_\theta u|_p \geq (1 + \tau_- a_-)|u|_p,$$

so $\Lambda = 1 + \tau_- a_- > 1$. Time reversal gives the stable cone statement. $\square$

4.3. Distortion and complexity.

Definition 4.5 (Admissible unstable curves). An unstable curve W is admissible if it is a C^2 curve contained in one homogeneous component, tangent to C^u , with length at most a fixed small δ_0 , and with curvature bounded by a constant fixed from the geometric data. Stable curves are defined analogously.

Lemma 4.6 (Bounded distortion). *There are constants $C_D > 0$ and $\alpha \in (0, 1)$ such that, for every homogeneous unstable curve W and all $x, y \in W$ for which $\mathcal{P}_\theta^n W$ remains homogeneous,*

$$(93) \quad |\log J_W \mathcal{P}_\theta^n(x) - \log J_W \mathcal{P}_\theta^n(y)| \leq C_D d_W(x, y)^\alpha.$$

Proof. On a homogeneous component, the derivative of $\mathcal{P}_\theta$ is a rational expression in τ , curvature, and $\cos \varphi$. The only unbounded factor is a power of $(\cos \varphi)^{-1}$ near grazing. The homogeneity strips convert this into a controlled power of the strip index k . Distances inside $\mathbb{H}_k$ scale as k^{-3} in the angular coordinate, while the logarithmic derivative grows at most polynomially in k . Choosing the strip exponent 2 makes the product summable over images of an unstable curve. The curvature and free-flight bounds control all non-grazing terms. Summing the one-step logarithmic distortion along the orbit gives the stated Hölder bound. $\square$

Lemma 4.7 (Strip-entry and terminal-child closure). *There exist constants $C > 0$ and $\delta_* > 0$, depending only on the geometric data, such that for every admissible unstable curve W with $|W|_p \leq \delta_*$ the homogeneous children of $\mathcal{P}_\theta W$ in the strips $\mathbb{H}_k^\pm$ satisfy the following two estimates.*

$$(94) \quad N_k^{\text{long}}(W) \leq C(1 + k|W|_p),$$

$$(95) \quad m_W\{x : \mathcal{P}_\theta x \text{ belongs to a terminal child in a strip } k > K\} \leq C|W|_p K^{-1}.$$

Here a child is called terminal if, after the standard subdivision used in the growth lemma, its image length is below the recovery scale; terminal children are estimated in the weak boundary layer rather than counted as long descendants.

Proof. Use the strip coordinate $\eta = (\pi/2 \mp \varphi)^{1/2}$ near the two grazing sides. The boundary of $\mathbb{H}_k^\pm$ is given by $\eta \asymp k^{-1}$. Along an unstable curve, (89) and the lower cone bound imply bounded variation of η in the rescaled coordinate $k\eta$; a crossing of two successive strip boundaries therefore requires either one genuine transverse intersection or a sliding interval whose number is proportional to $k|W|_p$. This gives (94). For the terminal estimate, observe that a component in a strip k has unstable Jacobian at least ck^2 . If the image component is shorter than the recovery scale, its preimage length is bounded by Ck^{-2} times that recovery length. Summing the monotone strip widths from K upward gives $C|W|_p K^{-1}$ after the same bounded-distortion normalization. All constants are uniform because curvature, horizon, and singularity transversality are uniform by the derived trace ledger of Lemmas 2.13 and 2.14. $\square$

Lemma 4.8 (One-step expansion). *For the admissible length δ_0 chosen sufficiently small and for a fixed adapted iterate n_* , there exists $\vartheta \in (0, 1)$ such that for every admissible unstable curve W , if $\{W_j\}$ are the long homogeneous components of $\mathcal{P}_\theta^{n_*}W$ after standard subdivision, then*

$$(96) \quad \sum_j \left(\inf_{x \in W_j} J_W \mathcal{P}_\theta^{n_*}(x) \right)^{-1} \leq \vartheta.$$

The terminal children omitted from the sum satisfy the boundary-layer estimate (95) at the last subdivision time.

Proof. First choose a central strip cutoff k_0 . On the complement of the high strips there are only finitely many branch cuts for one iterate, and uniform expansion from Theorem 4.4 makes the inverse-Jacobian contribution of an adapted iterate n_* smaller than $1/4$.

For high strips, Lemma 4.7 gives for the long children

$$(97) \quad \sum_{k \geq k_0} N_k^{\text{long}}(W)(ck^2)^{-1} \leq C \sum_{k \geq k_0} k^{-2} + C|W|_p \sum_{k_0 \leq k \leq K(W)} k^{-1}.$$

The upper limit is not an assumption. A child with $k > K(W) := C_0|W|_p^{-1}$ is terminal by (95) and is sent to the weak boundary layer. Therefore the long-child contribution is bounded by

$$(98) \quad Ck_0^{-1} + C|W|_p \log(C_0|W|_p^{-1}).$$

Choose k_0 so large that the first term is $< 1/4$, and then choose the standard admissible length δ_0 so small that the second term is $< 1/4$. The terminal children contribute only to the boundary term in the growth lemma, not to the long-descendant inverse-expansion sum. Bounded distortion permits replacement of pointwise Jacobians by their componentwise infima. Combining central and high-strip estimates gives (96) with some $\vartheta < 1$. $\square$

Theorem 4.9 (Growth lemma). *Let $r_n(x)$ be the distance, measured along $\mathcal{P}_\theta^n W$, from $\mathcal{P}_\theta^n x$ to the nearest endpoint of its homogeneous component. There are constants $C > 0$ and $\rho \in (0, 1)$ such that for every admissible unstable curve W and every $\varepsilon > 0$,*

$$(99) \quad m_W\{x \in W : r_n(x) < \varepsilon\} \leq C\varepsilon + C\rho^n m_W(W).$$

Proof. Construct the standard family tree. At each adapted generation n_* , long children are propagated and terminal children are placed in the boundary layer. For long children the total inverse expansion is at most ϑ by Lemma 4.8. Thus the mass of descendants that have not yet recovered to standard size is bounded by $C\vartheta^q$ after q adapted generations. At the final generation, points within ε of a fresh endpoint have preimage length bounded by $C\varepsilon$ because distortion on each homogeneous child is bounded. Terminal high-strip children satisfy (95); summing the resulting geometric series with the long-child contraction changes only the constants. Taking $\rho \in (\vartheta^{1/n_*}, 1)$ gives the stated estimate for all n after absorbing the finitely many remainder iterates into C . $\square$

Theorem 4.10 (Hyperbolic map with singularities). *For every $\theta \in \Theta$, the global a.e. billiard map*

$$(100) \quad (\Sigma_\theta, \mathcal{P}_\theta, \nu_\theta)$$

is a uniformly hyperbolic map with singularities. All constants in the cone, distortion, and growth estimates are uniform in θ .

Proof. The map is piecewise C^r by Proposition 4.1. Cone invariance and expansion are proved in Theorem 4.4. Distortion is Lemma 4.6. Complexity control is Lemma 4.8 and Theorem 4.9. Each estimate uses only the uniform constants in assumption 2.4; hence the constants are uniform in θ . $\square$

5. TRANSFER OPERATORS AND DETERMINISTIC SPECTRAL REGULARIZATION

The conservative billiard flow has two dual analytic faces. On densities one uses the transfer operator; on value functions and correctors one uses the Koopman-dual observable operator. The two are paired by the invariant billiard measure. This distinction is essential: the prelimit HJB equation and the cell correctors live on the observable side, while the spectral compactness is proved most efficiently on the transfer side. The $L^2(\nu_\theta)$ Koopman operator is unitary and is never used as a source of strong decay.

5.1. Anisotropic norms.

Definition 5.1 (Admissible stable and unstable curves). Fix constants $L_0 > 0$, $K_0 > 0$, and exponents $0 < \beta < \alpha < 1$. A stable admissible curve is a connected C^2 curve contained in one homogeneous component, tangent to the stable cone, of length at most L_0 , and with curvature at most K_0 in the rescaled homogeneous coordinates. Unstable admissible curves are defined analogously with the unstable cone. If a curve meets a singular curve or a homogeneity-strip boundary, it is subdivided and the children are measured with the terminal-child weights of (94). Stable holonomy is used only between curves in the same homogeneous chart; curves in different charts are never compared in the unstable seminorm.

Let $\mathcal{W}_\theta^s$ and $\mathcal{W}_\theta^u$ denote the stable and unstable admissible curves obtained from the strip-adapted cones. Curves are always contained in one homogeneous component and have uniformly bounded curvature in the rescaled strip coordinates. For a branchwise smooth density h define

$$(101) \quad |h|_{w,\theta} = \sup_{W \in \mathcal{W}_\theta^s} \sup_{\|\psi\|_{C^1(W)} \leq 1} \left| \int_W h \psi \, dm_W \right|.$$

The stable norm is

$$(102) \quad \|h\|_{s,\theta} = \sup_{W \in \mathcal{W}_\theta^s} \sup_{\|\psi\|_{C^\alpha(W)} \leq 1} \left| \int_W h \psi \, dm_W \right|,$$

and the unstable norm compares these stable-curve pairings by stable holonomy. If $W_1, W_2 \in \mathcal{W}_\theta^s$ lie in the same homogeneous chart, $\mathfrak{h} : W_1 \rightarrow W_2$ is the stable holonomy, and $d(W_1, W_2)$ is the C^1 distance of graph representations in the rescaled chart, set

$$(103) \quad \|h\|_{u,\theta} = \sup_{W_1, W_2} \sup_{\psi_i} \frac{1}{d(W_1, W_2)^\beta} \left| \int_{W_1} h \psi_1 \, dm_{W_1} - \int_{W_2} h \psi_2 \, J\mathfrak{h}^{-1} \, dm_{W_2} \right|,$$

where $0 < \beta < \alpha < 1$ and

$$(104) \quad \|\psi_i\|_{C^\alpha(W_i)} \leq 1, \quad \|\psi_1 - \psi_2 \circ \mathfrak{h}\|_{C^0(W_1)} \leq d(W_1, W_2)^\beta.$$

Curves in different homogeneous charts are not compared in the unstable seminorm; their difference is paid for by the stable and weak norms after standard subdivision. With fixed small weights $b_0, c_0 > 0$ put

$$(105) \quad \|h\|_{\mathcal{B}_\theta} = \|h\|_{s,\theta} + b_0 \|h\|_{u,\theta} + c_0 |h|_{w,\theta}.$$

The transfer space $\mathcal{B}_\theta$ is the completion of branchwise C^{r-3} densities of finite norm, modulo equality of all stable-curve pairings. The flow space $\widehat{\mathcal{B}}_\theta$ is obtained by applying the same construction in suspension flow boxes and integrating in the roof coordinate, with roof singularities measured by the same homogeneity estimate scheme.

The observable space is the restricted dual generated by physical observables, not an unspecified algebraic dual. Let $C_{\text{br}}^{r-3}(\Sigma_\theta)$ be the branchwise C^{r-3} observables with the strip weights appearing in the finite-response condition. Define

$$(106) \quad \|f\|_{\mathcal{B}_\theta^\#} = \sup_{\|h\|_{\mathcal{B}_\theta} \leq 1} |\langle f, h \rangle_\theta|,$$

and let $\mathcal{B}_\theta^\#$ be the completion of $C_{\text{br}}^{r-3}(\Sigma_\theta)$ under this norm, modulo constants when the zero-mean quotient is used. The flow observable space $\widehat{\mathcal{B}}_\theta^\#$ is defined analogously. Pairings are written $\langle f, h \rangle_\theta = \int f h d\nu_\theta$ or $\int f h d\mu_\theta$ when both entries are smooth and extended continuously thereafter.

Proposition 5.2 (Intrinsic anisotropic space and separating dual). *The spaces $\mathcal{B}_\theta$, $\widehat{\mathcal{B}}_\theta$, and their restricted observable duals are intrinsic to the singular billiard atlas. They have the following properties, uniformly on the compact finite-response family:*

- (a) *the norm is independent, up to uniform equivalence, of the finite choice of central charts, homogeneity cutoffs, and reference lengths;*
- (b) *smooth branch densities are dense in $\mathcal{B}_\theta$, and physical branch observables are dense in $\mathcal{B}_\theta^\#$;*
- (c) *positive unit admissible densities separate observable classes modulo the canonical zero-measure iterated singular set;*
- (d) *multiplication by a finite-response observable, stable holonomy transport, and normal-coordinate transport between nearby tables are bounded operators in the strong-to-weak topology. Consequently every transfer, response, cell, and viscosity pairing used later is made in a fixed Banach fiber after a specified deterministic transport; no unspecified algebraic dual or pointwise value on a grazing set is used.*

Proof. Changing central charts only replaces a stable leaf by finitely many stable subleaves with uniformly bounded Jacobians. Changing the homogeneity cutoff alters finitely many central strips and a tail whose contribution is bounded by $\sum_{k \geq K} k^{-2}$. Thus the strong and weak norms before and after such a change dominate each other with constants depending only on the primitive geometric ledger.

Density follows by smoothing along stable leaves inside one homogeneous chart, then summing over a finite central chart family and finally letting the high-strip cutoff tend to infinity. The weak tail bound gives convergence in $\mathcal{B}_\theta$. The same smoothing argument, applied to observables and read through the defining dual norm, gives density in the restricted observable dual.

For separation, take a non-zero observable class and a regular point at which a branch representative has non-zero leafwise density value. A stable-unstable product rectangle around that point, smoothed along stable leaves and truncated inside the same homogeneous component, gives a positive admissible density whose pairing detects the class. A countable differentiation basis covers all regular points, and the omitted iterated singular set has zero flux measure.

Finally, multiplication and holonomy transport are bounded because branchwise C^{r-3} norms and stable-holonomy Jacobians are uniformly controlled by the distortion estimates. Normal-coordinate transport of nearby tables produces interior branch derivatives and moving-boundary traces; these are the strong-to-weak terms bounded by Lemma 6.5. This proves all four assertions. $\square$

Definition 5.3 (Transfer and observable operators). For h smooth on homogeneous components, the transfer operator is the push-forward

$$(107) \quad \int f \mathcal{L}_\theta h \, d\nu_\theta = \int f \circ \mathcal{P}_\theta h \, d\nu_\theta$$

for all smooth observables f . Since ν_θ is invariant, $\mathcal{L}_\theta \mathbf{1} = \mathbf{1}$. The observable operator is

$$(108) \quad \mathcal{U}_\theta f = f \circ \mathcal{P}_\theta, \quad \langle \mathcal{U}_\theta f, h \rangle_\theta = \langle f, \mathcal{L}_\theta h \rangle_\theta.$$

For the suspension flow we write $\mathcal{L}_{\theta,t}$ and $\mathcal{U}_{\theta,t}$. The generator used in the HJB equation is the observable generator

$$(109) \quad \mathcal{A}_\theta f = \lim_{t \downarrow 0} \frac{\mathcal{U}_{\theta,t} f - f}{t} \quad \text{on } \text{Dom}(\mathcal{A}_\theta) \subset \widehat{\mathcal{B}}_\theta^\#.$$

The transfer generator is its Banach adjoint under the above pairing. This convention fixes the sign of every corrector below.

Lemma 5.4 (Compact embedding of strong into weak anisotropic topology). *The unit ball of $\mathcal{B}_\theta$ is relatively compact in the weak norm $|\cdot|_{w,\theta}$, uniformly for θ in compact parameter families satisfying assumption 2.4. The same statement holds for $\widehat{\mathcal{B}}_\theta$.*

Proof. Fix finitely many central homogeneous charts. On each chart, the stable-curve pairings of a strong-norm bounded family are equicontinuous in the curve parameter and in the C^1 test function by the unstable norm and the holonomy Jacobian estimate. Arzelà–Ascoli gives compactness on the central part. In the high strips, the strip-adapted cone and the growth lemma give a tail bound of the form $C \sum_{k \geq K} k^{-2} + C \delta_0 \sum_{k \leq C/\delta_0} k^{-1}$ after standard subdivision; choosing first K large and then the recovery scale δ_0 small makes the tail uniformly small in the weak norm. The suspension version follows by the finite-horizon roof bounds and the same argument in flow boxes. $\square$

Lemma 5.5 (Admissible products and horizontal derivatives). *Let $a(x, p, z)$ be a finite-response branchwise C^{r-3} observable and let $\chi(x, p, \cdot) \in \widehat{\mathcal{B}}_{\Theta(x,p)}^{\#,0}$ be a corrector obtained from the deterministic reduced resolvent. Multiplication by a , pairing against admissible densities, and the horizontal derivative $D_x^{\text{hor}} \chi$ are continuous in the observable anisotropic norm. More precisely, for every compact jet range and every admissible density $\rho \in \text{Adm}_M(\Theta(x, p))$,*

$$(110) \quad |\langle a\chi, \rho \rangle_{\Theta(x,p)}| + |\langle D_x^{\text{hor}} \chi, \rho \rangle_{\Theta(x,p)}| \leq C_M \|a\|_{C_{\text{br}}^{r-3}} \|\chi\|_{\widehat{\mathcal{B}}_{\Theta(x,p)}^\#}.$$

The derivative D_x^{hor} is taken after identifying nearby billiard tables by normal-coordinate charts and holding the cotangent covector p fixed. It therefore differentiates the billiard geometry, the invariant projection, and $\Theta(x, p)$, but not a composite field $p = \nabla \varphi(x)$.

Proof. On a homogeneous flow box, multiplication by a sends each stable test function to a stable test function with norm bounded by the branchwise C^{r-3} norm of a . In the unstable seminorm the only additional factor is the stable-holonomy Jacobian, already controlled by bounded distortion. Summing over homogeneous strips is legitimate because each derivative

of a satisfies the same polynomial strip estimate scheme as the billiard map, while the inverse expansion and terminal-child estimates give a summable tail. Horizontal differentiation is computed in the normal-coordinate identification of nearby tables. Derivatives of branch charts produce boundary terms on moving singularity curves; the derived trace bounds in Lemmas 2.13 and 2.14 make these terms weak-norm bounded, and the strip-entry closure makes their high-strip tails summable. Derivatives of the invariant projection and reduced resolvent are controlled by Theorem 6.4. Since p is fixed in the horizontal chart, the chain-rule term $D_p \chi D_x p$ is absent by definition, not by cancellation. Completion gives the asserted estimate. $\square$

Remark 5.6 (How distributional correctors enter viscosity tests). A corrector is an element of the restricted observable dual space. The perturbed test function in the singular variable is therefore the scalar functional $\rho \mapsto \langle \chi, \rho \rangle$ on admissible microscopic densities. No pointwise representative on the grazing set is used. When a smooth branchwise representative exists, it gives the same pairings; when only the anisotropic class exists, the viscosity–duality definition below is the canonical interpretation.

5.2. Lasota–Yorke inequalities.

Theorem 5.7 (Lasota–Yorke verification data). *For the anisotropic spaces specified in (A1), the following estimates are part of the uniform verification data: there exist $C > 0$ and $\sigma \in (0, 1)$, uniform in θ , such that for all $n \geq 0$ and all $h \in \mathcal{B}_\theta$,*

$$(111) \quad |\mathcal{L}_\theta^n h|_{w,\theta} \leq C|h|_{w,\theta},$$

$$(112) \quad \|\mathcal{L}_\theta^n h\|_{s,\theta} \leq C\sigma^n \|h\|_{s,\theta} + C|h|_{w,\theta},$$

$$(113) \quad \|\mathcal{L}_\theta^n h\|_{u,\theta} \leq C\sigma^n \|h\|_{u,\theta} + C\|h\|_{s,\theta}.$$

The dual estimates hold for $\mathcal{U}_\theta^n$ on $\mathcal{B}_\theta^\#$ after quotienting constants.

Proof. The estimates are recorded as the quantitative Banach-space form underlying (A1). The branch decomposition below explains the terms that a concrete proof must bound, but its schematic strip summation is not a derivation of the required uniform constants from (G1)–(G6) alone.

We prove the estimate for one adapted iterate and then iterate. Pair $\mathcal{L}^{n*} h$ with a C^α test ψ on a stable curve W . Branchwise duality gives

$$(114) \quad \int_W \mathcal{L}^{n*} h \psi \, dm_W = \sum_j \int_{W_j} h (\psi \circ \mathcal{P}^{n*}) J_{W_j} \mathcal{P}^{n*} \, dm_{W_j},$$

where W_j are the homogeneous preimage components. Stable contraction makes $\psi \circ \mathcal{P}^{n*}$ a test with Hölder seminorm multiplied by σ_0^{n*} after normalization, and bounded distortion controls the Jacobian factor. Long children therefore contribute the strong term $C\sigma\|h\|_{s,\theta}$.

The singular cuts are controlled by the one-step closure. Central children are finite in number. In high strips, $J^u \mathcal{P} \geq ck^2$ and the long-child count is (94). The contribution of long high-strip children is

$$(115) \quad C \sum_{k \geq k_0} k^{-2} + C|W|_p \sum_{k_0 \leq k \leq C/|W|_p} k^{-1},$$

which is made smaller than the prescribed contraction margin by the choice of k_0 and of the standard length δ_0 . Terminal children are precisely the boundary-layer pieces of Lemma 4.7;

their pairings are estimated by $|h|_{w,\theta}$. This proves the stable Lasota–Yorke inequality for the adapted iterate. The finitely many remainder iterates are absorbed into C .

For the unstable norm one compares two stable curves through the stable holonomy. The two pullback families can be matched except for components cut by singularities. Matched components gain the same contraction as above, while the unmatched components are controlled by the stable norm using the absolute continuity and Hölder Jacobian of stable holonomy. The weak estimate follows from invariance of the reference measure and bounded branch distortion. Dual estimates are obtained by Banach duality between $\mathcal{L}_\theta$ and $\mathcal{U}_\theta$. Every constant is expressed in terms of the curvature, horizon, separation, and transversality constants in the geometric and analytic verification data. Uniformity in θ is required as part of (A1). $\square$

Proposition 5.8 (Geometric quasi-compactness obligation). *The displayed Lasota–Yorke and compact-embedding estimates reduce quasi-compactness to standard operator theory and bound the essential spectral radius below one. They do not identify the peripheral spectrum. In particular, leafwise seminorm inequalities and connectivity of the collision graph do not exclude periodic components or nontrivial roots of unity.*

Proof. The essential-radius conclusion is the usual Hennion consequence of the strong-to-weak inequality and compact embedding. The missing peripheral elimination is exactly why simplicity and aperiodicity are retained in (A1). $\square$

Theorem 5.9 (Conditional map spectral gap). *Under (A1), the operator $\mathcal{L}_\theta : \mathcal{B}_\theta \rightarrow \mathcal{B}_\theta$ has simple eigenvalue one and, on the zero-mean subspace, constants $C > 0$ and $\rho \in (0, 1)$, uniform in θ , such that*

$$(116) \quad \|\mathcal{L}_\theta^n h\|_{\mathcal{B}_\theta} \leq C\rho^n \|h\|_{\mathcal{B}_\theta}.$$

The same contraction holds on the zero-mean observable quotient.

Proof. This is the map-mixing clause of (A1). The geometric estimates above make the hypothesis concrete and potentially verifiable, but no implication from (G1)–(G6) to aperiodicity is asserted. $\square$

Proposition 5.10 (Map spectral consequence of the mixing hypothesis). *Under (A1), uniformly on the compact parameter window, the collision transfer operator has the decomposition*

$$(117) \quad \mathcal{L}_\theta^n = \Pi_\theta + N_\theta^n, \quad \|N_\theta^n\|_{\widehat{\mathcal{B}}_\theta \rightarrow \widehat{\mathcal{B}}_\theta} \leq C\varrho^n, \quad 0 < \varrho < 1,$$

and hence

$$(118) \quad (I - \mathcal{L}_\theta)^{-1} \Pi_\theta^\perp = \sum_{n \geq 0} N_\theta^n$$

is a bounded reduced resolvent. The preceding cone, distortion, and Lasota–Yorke estimates describe geometric verification obligations, but do not establish the simplicity or aperiodicity clause of (A1).

Proof. This is the spectral decomposition and geometric-series consequence of (A1). Connectivity in (G6) is not used to eliminate peripheral eigenvalues. $\square$

Theorem 5.11 (Observable reduced resolvent and deterministic correlation calculus). *The zero-mean observable quotient $\widehat{\mathcal{B}}_\theta^{\#,0} = \widehat{\mathcal{B}}_\theta^\# / \mathbb{R}$ carries a bounded inverse of the suspension*

generator,

$$(119) \quad \mathcal{R}_\theta = (-\mathcal{A}_\theta|_{\widehat{\mathcal{B}}_\theta^{\#,0}})^{-1}, \quad \|\mathcal{R}_\theta f\|_{\widehat{\mathcal{B}}_\theta^{\#,0}} \leq C\|f\|_{\widehat{\mathcal{B}}_\theta^{\#,0}},$$

uniformly in the compact finite-response family. For smooth centered observables f, g the bilinear form

$$(120) \quad \mathfrak{C}_\theta(f, g) = \int_0^\infty \langle f \mathcal{U}_{\theta,t} g, \rho_\theta \rangle_\theta dt = \langle f, \mathcal{R}_\theta g \rangle_\theta$$

is obtained by deterministic contour inversion. If $f = g$, then the symmetric part satisfies

$$(121) \quad 2\mathfrak{C}_\theta^{\text{sym}}(f, f) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{\widehat{\Sigma}_\theta} \left(\int_0^T f(\Phi_\theta^t z) dt \right)^2 d\mu_\theta(z) \geq 0,$$

where the limit is taken in the deterministic anisotropic-duality sense. Thus the Green–Kubo tensor used below is positive semidefinite as a consequence of Liouville invariance and transfer spectral regularization.

Proof. The map spectral gap gives the reduced inverse of $I - \mathcal{L}_\theta$ on the zero-mean transfer complement. The suspension generator is obtained from the weighted transfer family $\mathcal{L}_{\theta,s} = \mathcal{L}_\theta(e^{-s\tau_\theta} \cdot)$. The roof is bounded above and below and has the same homogeneous branch estimates as the map. The Dolgopyat estimate gives a uniform bound for $(I - \mathcal{L}_{\theta,i\omega})^{-1}$ on high vertical lines after the invariant pole at $s = 0$ has been removed. Near $s = 0$, analytic perturbation of the simple eigenvalue separates the invariant direction from the zero-mean quotient. Inverting the Laplace transform on the shifted contour gives (119). This is an operator identity in the Banach bundle and uses only the deterministic transfer family.

For smooth centered f, g , the finite-time identity

$$(122) \quad \left\langle f, \int_0^T \mathcal{U}_{\theta,t} g dt \right\rangle = \int_0^T \langle f \mathcal{U}_{\theta,t} g, \rho_\theta \rangle dt$$

is merely duality between observables and densities. Exponential contraction on the zero-mean quotient makes the integral converge as $T \rightarrow \infty$ and identifies the limit with $\langle f, \mathcal{R}_\theta g \rangle$. To obtain the square identity, expand the square and use Liouville invariance:

$$(123) \quad \begin{aligned} \frac{1}{T} \int \left(\int_0^T f(\Phi^t z) dt \right)^2 d\mu &= \frac{1}{T} \int_0^T \int_0^T \langle f \mathcal{U}_{|t-s|} f, \rho \rangle dt ds \\ &= 2 \int_0^T \left(1 - \frac{r}{T} \right) \langle f \mathcal{U}_r f, \rho \rangle dr. \end{aligned}$$

The exponential transfer bound permits passage to the limit and gives (121). Positivity is the positivity of an ordinary square under the invariant Liouville measure; no external stochastic limit theorem is invoked. $\square$

Corollary 5.12 (Cell problem solvability from the observable resolvent). *Every centered finite-response observable f in the generator response domain has the unique zero-mean corrector*

$$(124) \quad \chi_f = \mathcal{R}_{\Theta(x,p)} f, \quad \mathcal{A}_{\Theta(x,p)} \chi_f = -f, \quad \int \chi_f d\mu_{\Theta(x,p)} = 0.$$

The solution depends C^{r-5} on (x, p) on compact jet windows after Banach bundle transport, and this dependence is obtained by differentiating the branchwise transfer family and the contour-resolvent formula.

Proof. The inverse in (119) solves the cell equation on the zero-mean quotient. Uniqueness follows because the kernel of $\mathcal{A}_\theta$ on the generator domain consists only of constants, a consequence of the same Hopf-chain argument that removed the peripheral spectrum. Parameter regularity follows by the strong-to-weak derivative formula for the weighted transfer family and by differentiating the contour integrals defining the spectral projection and reduced inverse. The centering condition fixes the representative in the quotient. $\square$

5.3. The suspension flow. The continuous-time billiard flow is the suspension over $\mathcal{P}_\theta$ with roof function τ_θ . Let

$$(125) \quad \widehat{\Sigma}_\theta = \{(z, u) : z \in \Sigma_\theta, 0 \leq u < \tau_\theta(z)\} / (z, \tau_\theta(z)) \sim (\mathcal{P}_\theta z, 0).$$

The invariant measure is

$$(126) \quad d\mu_\theta = \frac{d\nu_\theta(z) du}{\int \tau_\theta d\nu_\theta}.$$

Let Φ_θ^t be the suspension flow.

Lemma 5.13 (Uniform non-integrability hypothesis in branch form). *Under (A2) there exist constants $n_{\text{UNI}} \in \mathbb{N}$ and $c_{\text{UNI}} > 0$, uniform on the admitted parameter family, with the following property. Every standard unstable family contains a homogeneous subrectangle and two inverse branches h_1, h_2 of $\mathcal{P}_\theta^{n_{\text{UNI}}}$, joined by stable holonomy, such that the temporal displacement*

$$(127) \quad \Delta(y) = \sum_{j=0}^{n_{\text{UNI}}-1} \tau_\theta(\mathcal{P}_\theta^j h_1 y) - \sum_{j=0}^{n_{\text{UNI}}-1} \tau_\theta(\mathcal{P}_\theta^j h_2 y)$$

satisfies

$$(128) \quad |D_u \Delta(y)| \geq c_{\text{UNI}}$$

in the rescaled unstable coordinate on that subrectangle.

Proof. This is the branchwise formulation of (A2). Curvature identities can help verify it in a concrete family, but a pointwise nonzero temporal displacement does not yield a uniform lower bound over the countably many grazing branches, and compactness of finitely many central charts does not close that gap. The earlier curvature-only proof is therefore not used. $\square$

Lemma 5.14 (Temporal displacement derivative in blown-up coordinates). *Let h_1, h_2 be two homogeneous inverse branches of $\mathcal{P}_\theta^n$ whose images are matched by stable holonomy over a standard unstable base. In the rescaled unstable coordinate u , the temporal displacement $\Delta_n = \tau_n \circ h_1 - \tau_n \circ h_2$ satisfies*

$$(129) \quad D_u \Delta_n = \sum_{j=0}^{n-1} \left(\frac{2\kappa(q_j)}{\cos \varphi_j} \Xi_j + \tau_j \mathcal{E}_j + \mathcal{H}_j \right).$$

Here Ξ_j is the signed transverse separation of the two transported wavefronts at the j th reflection, $\mathcal{E}_j$ is a bounded expression in pre-reflection wavefront curvatures, and $\mathcal{H}_j$ is the derivative of the stable-holonomy time shift. In a strip $\mathbb{H}_k$, all three quantities are bounded in the blown-up chart, while the displayed curvature term has a monotone angular derivative bounded below by a positive multiple of κ_- on a fixed substrip.

Proof. For one flight, differentiating the impact equation $G(q + \tau v, \theta) = 0$ gives

$$(130) \quad D\tau(w) = -\frac{\langle Dq(w) + \tau Dv(w), n(q') \rangle}{\langle v, n(q') \rangle}.$$

Transport of a wavefront followed by elastic reflection is governed by the optical recurrence (86). The reflection part adds $2\kappa/\cos\varphi$ to the post-collision curvature. When two inverse branches are compared through stable holonomy, the common stable timing term is absorbed into $\mathcal{H}_j$, and the curvature contribution is multiplied by the signed transverse separation Ξ_j . Free-flight curvature transport gives the bounded error $\tau_j \mathcal{E}_j$. Since $\cos\varphi \asymp k^{-2}$ and $(\pi/2 \mp \varphi)^{1/2} \asymp k^{-1}$ on $\mathbb{H}_k$, the unbounded coefficient becomes a controlled monotone term in the rescaled coordinate. The bounded distortion and stable-holonomy estimates control the remainder. $\square$

Lemma 5.15 (Curvature excludes temporal coboundaries). *The roof τ_θ cannot have vanishing Dolgopyat temporal displacement on all stable-matched inverse-branch pairs. Equivalently, the identities $D_u \Delta_n \equiv 0$ for every standard rectangle and every matched branch pair are incompatible with strict dispersing curvature.*

Proof. If all such derivatives vanished, the stable and unstable holonomy time changes would be path-independent on Hopf rectangles and would define a C^1 temporal potential ψ with roof displacement equal to a stable-plus-unstable coboundary. Pull this identity through a single dispersing reflection and differentiate in the blown-up angular coordinate. The coboundary side has bounded rescaled variation because the stable and unstable holonomies have bounded Jacobian. The reflection side contains the monotone term $2\kappa/\cos\varphi$ from Lemma 5.14; its rescaled derivative has a fixed sign and a positive lower bound on a substrip because $\kappa \geq \kappa_-$. This contradiction is quantitative after compactness of the central charts and normalization of high strips. $\square$

Lemma 5.16 (Dolgopyat contraction required by (A2)). *Under (A2), let $\mathcal{L}_{\theta,s}h = \mathcal{L}_\theta(e^{-s\tau_\theta}h)$. There are $N \in \mathbb{N}$, $\eta > 0$, $C > 0$, and $b_0 > 0$, uniform in θ , such that for $s = a + ib$, $a \geq 0$, $|b| \geq b_0$, and every zero-mean density h ,*

$$(131) \quad \|\mathcal{L}_{\theta,s}^N h\|_{\mathcal{B}_\theta} \leq (1 - \eta)\|h\|_{\mathcal{B}_\theta} + C|b|^{-1}|h|_{w,\theta}.$$

Proof. Partition each unstable curve into intervals on which the phase difference $b\Delta$ of Lemma 5.13 changes by an amount between two fixed constants. On paired inverse branches the two complex weights $e^{-ib\tau_N \circ h_1}$ and $e^{-ib\tau_N \circ h_2}$ cannot have the same argument throughout such an interval; a C^1 partition of unity therefore gives a fixed cancellation in the strong stable norm. Intervals for which a pair is cut by a singular curve are terminal children and are estimated by the weak norm using Lemma 4.7. Near-resonant intervals have length $O(|b|^{-1})$ in the rescaled unstable coordinate and hence contribute the term $C|b|^{-1}|h|_w$. Combining this cancellation with the Lasota–Yorke recovery estimate for long children gives the displayed contraction. $\square$

Theorem 5.17 (Conditional suspension spectral gap). *Assume (A1)–(A2). The zero-mean suspension transfer semigroup $\mathcal{L}_{\theta,t}$ decays exponentially on the specified anisotropic spaces, uniformly on the compact parameter window. Equivalently, its Laplace-family resolvent has the pole-free strip and bounds stipulated in (A2), and the zero-mean observable generator has a bounded reduced inverse.*

Proof. The map gap is (A1) and the quantitative non-lattice/UNI and Dolgopyat estimates are (A2). Standard suspension resolvent inversion then gives the stated decay. The branch computations preceding this theorem are verification templates only: dispersing curvature and finite horizon do not by themselves establish UNI. $\square$

Proposition 5.18 (Standard-pair form of the (A2) obligation). *Under (A2), let $\mathfrak{S}$ be any standard family whose total standard-family norm is one. There exist integers $n_\star, N_\star$, constants $\eta_\star, c_\star, C_\star > 0$, and a finite partition of the long recovered children of $\mathcal{P}_\theta^{-N_\star}\mathfrak{S}$ into paired and terminal classes, with uniform constants supplied by (A2), such that for $s = a + ib$, $a \geq 0$, $|b| \geq b_0$,*

$$(132) \quad \sum_{W' \in \mathfrak{S}_{\text{paired}}} \left| \int_{W'} e^{-s\tau_{N_\star}} g_{W'} d\ell_{W'} \right| \leq (1 - \eta_\star) \sum_{W' \in \mathfrak{S}_{\text{paired}}} \int_{W'} |g_{W'}| d\ell_{W'} + C_\star |b|^{-1} \|g\|_{\mathcal{B}},$$

and

$$(133) \quad \sum_{W' \in \mathfrak{S}_{\text{terminal}}} \int_{W'} |g_{W'}| d\ell_{W'} \leq C_\star |g|_w + \frac{\eta_\star}{4} \|g\|_{\mathcal{B}}.$$

These estimates state the standard-pair content that must be verified to invoke the high-frequency resolvent bound in Theorem 5.17.

Proof. Choose n_{UNI} and c_{UNI} from Lemma 5.13. First iterate any standard family until the growth lemma recovers a fixed fraction of its mass in long homogeneous components. On each recovered component the UNI lemma supplies two inverse descendants joined by stable holonomy and a temporal displacement Δ with $|D_u \Delta| \geq c_{\text{UNI}}$. Subdivide the base interval into phase intervals of length between $(8|b|c_{\text{UNI}})^{-1}$ and $(4|b|c_{\text{UNI}})^{-1}$, except for a boundary remainder of total length $O(|b|^{-1})$. On every non-remainder interval the two phases $e^{-ib\tau_n \circ h_1}$ and $e^{-ib\tau_n \circ h_2}$ rotate by a definite relative amount. A two-branch trigonometric inequality gives

$$(134) \quad |Ae^{i\alpha} + Be^{i\beta}| \leq (1 - \eta_0)(A + B) + C|A - B|$$

for nonnegative amplitudes when $|\alpha - \beta|$ stays in the prescribed phase window. Bounded distortion and stable holonomy absolute continuity control $|A - B|$ by the unstable variation seminorm. This gives the first term on the right side of (132).

Children cut by grazing curves, multiple-collision curves, or strip boundaries are not paired. The trace ledger Lemma 2.13 and the growth lemma give the terminal estimate (133). The near-resonant and boundary phase intervals have total length $O(|b|^{-1})$, which produces the weak-norm error. Finally choose the recovery time and the strip cutoff so that the terminal strong contribution is below $\eta_\star/4$, and concatenate the pairing step with the deterministic Lasota–Yorke inequality. This yields the operator contraction of Lemma 5.16 with constants uniform over the compact finite-response family. $\square$

Theorem 5.19 (Conditional Dolgopyat strip resolvent). *Under (A1)–(A2), let $\mathcal{L}_{\theta,s} = \mathcal{L}_\theta(e^{-s\tau_\theta} \cdot)$ be the weighted collision operator. There are constants $b_0, C, \gamma > 0$, depending on the constants in (A1)–(A2), such that for every allowed table parameter θ , every $s = a + ib$ with $|b| \geq b_0$ and $|a| \leq \gamma$, and every zero-mass density h ,*

$$(135) \quad \|(I - \mathcal{L}_{\theta,s})^{-1} h\|_{\mathcal{B}_\theta} \leq C \|h\|_{\mathcal{B}_\theta}.$$

This is the high-frequency resolvent consequence stipulated by (A2).

Proof. Apply Proposition 5.18 to the standard-family decomposition of a density in the anisotropic unit ball. The paired descendants obey the cancellation estimate (132); the terminal descendants obey (133). Choose the recovery time, strip cutoff, and adapted iterate so that the terminal strong contribution is less than one quarter of the cancellation margin. The weak term is then returned to the strong norm by one deterministic Lasota–Yorke recovery step, while the $|b|^{-1}$ near-resonant remainder is absorbed by choosing b_0 sufficiently large. This gives, for one adapted iterate N ,

$$(136) \quad \|\mathcal{L}_{\theta, a+ib}^N h\|_{\mathcal{B}_\theta} \leq (1 - \eta) \|h\|_{\mathcal{B}_\theta}$$

on the zero-mass quotient, uniformly for $|a| \leq \gamma$ after decreasing γ so that the multiplier $e^{-a\tau}$ changes the constants by less than $\eta/4$. The finitely many intermediate iterates are bounded by the weighted Lasota–Yorke estimate because the roof and all of its branchwise derivatives have the same homogeneity-strip bounds as the collision map.

The Neumann series for the adapted iterate gives a bounded inverse for $I - \mathcal{L}_{\theta, s}^N$. The algebraic identity

$$(137) \quad (I - \mathcal{L}_{\theta, s})^{-1} = (I + \mathcal{L}_{\theta, s} + \cdots + \mathcal{L}_{\theta, s}^{N-1})(I - \mathcal{L}_{\theta, s}^N)^{-1}$$

then gives (135), conditional on the uniform pairing, terminal-tail, and recovery estimates supplied by (A1)–(A2). The displayed induction does not derive those uniform estimates from curvature alone. $\square$

Lemma 5.20 (Conditional low-frequency pole removal). *Under (A1)–(A2), the only pole of the Laplace transform of the suspension transfer semigroup in a neighborhood of the imaginary axis is the simple invariant pole at $s = 0$. On the zero-mean quotient this pole is absent, and the inverse Laplace contour may be shifted to $\Re s = -\gamma$.*

Proof. For $|s|$ small the weighted map operator $\mathcal{L}_{\theta, s} = \mathcal{L}_\theta(e^{-s\tau_\theta} \cdot)$ is an analytic strong-to-weak perturbation of $\mathcal{L}_\theta$. The map spectral theorem gives a simple eigenvalue $\lambda(s)$ near one. Its first derivative is

$$(138) \quad \lambda'(0) = - \int \tau_\theta d\nu_\theta < 0,$$

because the roof is bounded below. Thus $1 - \lambda(s)$ has a simple zero at $s = 0$. The primitive Hopf-chain aperiodicity of (A1)–(A2) prevent any other unit-modulus map eigenvalue or roof lattice pole. Projecting away constants removes the residue at $s = 0$. The high-frequency strip is supplied by Proposition 5.18; compactness covers the bounded remaining frequencies. Hence the shifted inverse Laplace contour is legitimate on the zero-mean observable bundle. $\square$

Theorem 5.21 (Conditional geometry-to-resolvent dependency chain). *Under assumption 2.4 and (A1)–(A2), the reduced observable resolvent used in every cell equation has the following dependency chain; the labeled assumption arrows are inputs, not derived implications:*

$$(139) \quad \begin{aligned} & \text{finite horizon and dispersing curvature} \implies \text{strip-adapted cones, distortion, and growth} \\ & \implies \text{branchwise verification ledger} \\ & \xrightarrow{(A1)} \text{simple invariant density and map spectral gap} \\ & \xrightarrow{(A2)} \text{suspension resolvent strip} \\ & \implies (-\mathcal{A}_\theta|_{\ker \Pi_\theta})^{-1} : \hat{\mathcal{B}}_\theta^{\#,0} \rightarrow \hat{\mathcal{B}}_\theta^{\#,0}. \end{aligned}$$

In particular, ordinary curvature and finite horizon do not establish the two assumption arrows.

Proof. Geometry defines the branch map and the optical/strip quantities. The map gap is exactly (A1) and the suspension strip is exactly (A2). Integration of the exponential zero-mean semigroup bound then defines the stated reduced inverse. No converse implication is asserted. $\square$

Theorem 5.22 (Conditional spectral and response package). *Assume (A1)–(A3) for the finite-horizon dispersing billiard family and the specified compatible finite-response port. After quotienting constants, there exist constants $C > 0$, $\vartheta \in (0, 1)$, and $\gamma > 0$, depending only on the primitive constants, such that for every allowed table parameter θ :*

$$(140) \quad \|\mathcal{L}_\theta^n h\|_{\mathcal{B}_\theta} \leq C\vartheta^n \|h\|_{\mathcal{B}_\theta} \quad (\langle 1, h \rangle = 0),$$

$$(141) \quad \|\mathcal{U}_{\theta,t} f\|_{\widehat{\mathcal{B}}_\theta^{\#,0}} \leq C e^{-\gamma t} \|f\|_{\widehat{\mathcal{B}}_\theta^{\#,0}},$$

$$(142) \quad \left\| \int_0^\infty \mathcal{U}_{\theta,t} f dt \right\|_{\widehat{\mathcal{B}}_\theta^{\#}} \leq C \|f\|_{\widehat{\mathcal{B}}_\theta^{\#,0}}.$$

Moreover, for every admitted finite-response parameter curve θ_s the operator paths

$$(143) \quad s \mapsto \Pi_{\theta_s}, \quad s \mapsto (-\mathcal{A}_{\theta_s}|_{\ker \Pi_{\theta_s}})^{-1}$$

are C^{r-4} as strong-to-weak operator-valued maps, with derivatives given by finite sums of branch-interior derivatives and moving-singularity trace functionals, with the intermediate mapping bounds stipulated by (A3). These are consequences of the analytic package, not of ordinary geometry alone.

Proof. The map estimate is (A1), the suspension estimate and time-integrated resolvent bound are (A2), and the differentiability statements are (A3). The preceding branch formulae are a verification ledger for these hypotheses and do not independently prove them. $\square$

6. CELL PROBLEMS ON ANISOTROPIC OBSERVABLE SPACES

The cell problem is solved for observables, because correctors are inserted into test functions. Under (A1)–(A2), the observable resolvent is the dual reduced resolvent of the transfer semigroup. This avoids the L^2 Koopman obstruction and also avoids using a density generator on a value function.

Definition 6.1 (Projection and zero-mean part). For $f \in \widehat{\mathcal{B}}_\theta^\#$, define

$$(144) \quad \Pi_\theta f = \int f d\mu_\theta, \quad \Pi_\theta^\perp f = f - \Pi_\theta f.$$

Theorem 6.2 (Deterministic solution of the cell problem). *For every $f \in \widehat{\mathcal{B}}_\theta^{\#,0}$, the improper integral*

$$(145) \quad \chi_\theta[f] = \int_0^\infty \mathcal{U}_{\theta,t} f dt$$

converges in $\widehat{\mathcal{B}}_\theta^\#$ and is the unique zero-mean solution of

$$(146) \quad \mathcal{A}_\theta \chi = -f, \quad \int \chi d\mu_\theta = 0.$$

Moreover,

$$(147) \quad \|\chi_\theta[f]\|_{\widehat{\mathcal{B}}_\theta^\#} \leq C\|f\|_{\widehat{\mathcal{B}}_\theta^\#}.$$

For every admissible density ρ , the scalar $\langle \chi_\theta[f], \rho \rangle$ is defined by anisotropic duality and is the quantity used in perturbed viscosity tests.

Proof. The dual flow gap from Theorem 5.17 gives

$$(148) \quad \|\mathcal{U}_{\theta,t}f\|_{\widehat{\mathcal{B}}_\theta^\#} \leq Ce^{-\gamma t}\|f\|_{\widehat{\mathcal{B}}_\theta^\#}$$

for zero-mean f , so the integral converges in the observable anisotropic space. For $T < \infty$ the generator identity on the domain gives

$$(149) \quad \mathcal{A}_\theta \left(\int_0^T \mathcal{U}_{\theta,t}f dt \right) = \mathcal{U}_{\theta,T}f - f.$$

The convergence of the integral and of $\mathcal{U}_{\theta,T}f$ to zero on the zero-mean quotient allow $T \rightarrow \infty$, yielding $\mathcal{A}_\theta\chi = -f$ in $\widehat{\mathcal{B}}_\theta^\#$. If $\mathcal{A}_\theta\chi = 0$ and $\int \chi d\mu_\theta = 0$, then $\mathcal{U}_{\theta,t}\chi = \chi$ for all t ; the same dual gap forces $\chi = 0$. The norm estimate is the integrated exponential estimate. Since χ is an element of the restricted observable dual space, every occurrence of χ in the HJB proof is a continuous scalar pairing against a density, not a pointwise evaluation at a singular microscopic state. $\square$

Remark 6.3 (No L^2 inversion). The formula above is not an L^2 Bochner integral for the conservative Koopman group. It is the observable side of the anisotropic transfer resolvent. Hyperbolicity is visible only after the stable and unstable directions are measured anisotropically.

6.1. Parameter dependence. The effective Hamiltonian depends on θ through the table, invariant measure, and reduced resolvent. This dependence is derived from the same branchwise geometry.

Theorem 6.4 (Smooth deterministic response). *Let f_θ be a C^r family of finite-response observables that is C^r on homogeneous components and has uniformly controlled singular behavior compatible with the homogeneity strips. Then the maps*

$$(150) \quad \theta \mapsto \int f_\theta d\mu_\theta, \quad \theta \mapsto \chi_\theta[\Pi_\theta^\perp f_\theta]$$

are C^{r-3} as maps into the corresponding weak observable anisotropic spaces. Their derivatives are obtained by differentiating the spectral projection and the reduced observable resolvent

$$(151) \quad (-\mathcal{A}_\theta|_{\ker \Pi_\theta})^{-1}.$$

Proof. Identify nearby collision sections by the normal-coordinate charts of the moving scatterer boundaries. On every homogeneous branch, the map, roof, Jacobian, and observable are C^r in θ . Differentiating the transfer operator produces two kinds of terms. The interior terms are ordinary branchwise derivatives of inverse branches and Jacobians. The boundary terms come from motion of singular curves and strip boundaries. For a stable test curve W , a boundary term is an integral over $W \cap S_{\alpha,\theta}$ with coefficient proportional to the normal velocity of the moving singular curve. The derived finite-word trace ledger Lemmas 2.13 and 2.14 makes this coefficient uniformly bounded in the rescaled chart; the strip-entry closure makes the sum of all high-strip boundary terms finite. Thus, for $k \leq r - 3$,

$$(152) \quad \partial_\theta^k \mathcal{L}_\theta : \mathcal{B}_\theta \longrightarrow \mathcal{B}_{\theta,w}$$

is a bounded operator, uniformly in θ .

The Lasota–Yorke constants, the compact embedding, and the peripheral spectral separation are uniform in θ . Choose a contour Γ_1 enclosing the simple eigenvalue one and a contour Γ_0 enclosing the zero-mean spectral complement, both independent of θ . The spectral projection and reduced resolvent are

$$(153) \quad \Pi_\theta = \frac{1}{2\pi i} \int_{\Gamma_1} (z - \mathcal{L}_\theta)^{-1} dz, \quad R_\theta = \frac{1}{2\pi i} \int_{\Gamma_0} \frac{1}{1-z} (z - \mathcal{L}_\theta)^{-1} dz.$$

The derivative bound (152) and the resolvent identity differentiate these contour integrals in the strong-to-weak operator topology. Iterating the identity gives C^{r-3} regularity. Passing to the suspension generator is the same argument applied to the weighted Laplace family $\mathcal{L}_{\theta,s}$ and then inverting the Laplace transform; the Dolgopyat bounds are uniform in θ . Dualizing gives the response of observables. On the moving zero-mean bundle the first derivative is

$$(154) \quad \nabla_\theta(-\mathcal{A}_\theta)^{-1} = (-\mathcal{A}_\theta)^{-1}(\nabla_\theta \mathcal{A}_\theta)(-\mathcal{A}_\theta)^{-1} - (-\mathcal{A}_\theta)^{-1}(\nabla_\theta \Pi_\theta),$$

where the last term is the covariant correction keeping the input centered. Higher derivatives are finite sums of the same branchwise differentiated operators and are controlled by the strip estimate scheme. This proves the stated smooth deterministic response without assuming a response theorem as an external hypothesis. $\square$

Lemma 6.5 (Explicit moving-boundary operator). *Let $D_h(\theta)$ be a homogeneous branch domain and let $\Gamma_s \subset \partial D_h(\theta_s)$ be a moving singular curve with normal velocity V_Γ in the blown-up strip chart. For a stable test leaf W , a test function ψ , and a strong density ρ , the derivative of the branch characteristic function is the weak functional*

$$(155) \quad \langle \mathcal{B}_\Gamma \rho, \psi \rangle_W = \sum_{y \in W \cap \Gamma_0} (V_\Gamma \cdot n_\Gamma)(y) \rho_W(y) [\psi J_h]_\Gamma(y),$$

where $[\psi J_h]_\Gamma$ is the jump of the pulled-back test-Jacobian factor across the two adjacent homogeneous branches. In a strip $\mathbb{H}_k$, the coefficient in (155) grows at most like k^{a_1} , while the total stable mass of descendants entering that strip is summable. Consequently

$$(156) \quad \sum_h \sum_{\Gamma \subset \partial D_h} \|\mathcal{B}_\Gamma \rho\|_{\mathcal{B}^w} \leq C \|\rho\|_{\mathcal{B}}.$$

Proof. Write the branch contribution in fixed coordinates as $1_{D_h(\theta_s)} J_{h_s}^{-1} \rho \circ h_s$. Differentiating the indicator of a moving domain gives $-V_\Gamma \cdot n_\Gamma \delta_{\Gamma_0}$. Pairing with a stable test leaf reduces this distribution to the finite sum over intersections $W \cap \Gamma_0$, which is (155). The finite-word impact equation gives polynomial bounds for V_Γ in the blown-up coordinate; transversality to stable leaves bounds the intersection number in each homogeneous chart. The high-strip descendants have summable stable mass by the terminal-child estimate of the growth lemma. Summing central charts and then the high-strip tail proves (156). $\square$

Theorem 6.6 (Moving-singularity response formula in the Banach bundle). *Assume (A1)–(A3). Let θ_s be a C^{r-4} curve of billiard tables generated by the mechanical port, and let f_s be a C^1 observable section of the observable Banach bundle. Then the invariant pairing*

$$(157) \quad I(s) = \langle f_s, \rho_{\theta_s} \rangle_{\theta_s}$$

Every inverse and product in this formula is interpreted on the intermediate spaces and with the derivative-loss bounds stipulated in (A3). The formula does not follow from a bare

strong-to-weak derivative. is differentiable, and at $s = 0$

$$(158) \quad I'(0) = \langle \nabla_s f_s|_0, \rho_{\theta_0} \rangle + \langle (I - \mathcal{L}_{\theta_0})^{-1} \Pi_{\theta_0}^\perp (\nabla_s \mathcal{L}_{\theta_s}|_0 \rho_{\theta_0}), f_0 \rangle.$$

The derivative of the transfer operator is a strong-to-weak operator and has the branch formula

$$(159) \quad \nabla_s \mathcal{L}_{\theta_s}|_0 = \sum_{h \in \mathbb{H}} \mathcal{L}_{\theta_0, h} (\mathcal{D}_h^{\text{int}} + \mathcal{D}_h^{\text{Jac}}) + \sum_{\Gamma \subset \mathcal{S}} \mathcal{B}_\Gamma.$$

Here $\mathcal{D}_h^{\text{int}}$ differentiates the inverse branch in the interior, $\mathcal{D}_h^{\text{Jac}}$ differentiates the Jacobian and roof factors, and $\mathcal{B}_\Gamma$ is the boundary functional coming from the motion of a singular curve. The boundary series is absolutely bounded in the weak anisotropic norm:

$$(160) \quad \sum_{\Gamma \subset \mathcal{S}} \|\mathcal{B}_\Gamma \rho\|_{\widehat{\mathcal{B}}_{\theta_0}^w} \leq C \|\rho\|_{\widehat{\mathcal{B}}_{\theta_0}}.$$

Thus invariant averages, correctors, and Green–Kubo tensors are differentiated by deterministic moving-boundary calculus rather than by an external response hypothesis.

Proof. Trivialize the moving fibers by Definition 2.21 and write all operators on the fixed fiber over θ_0 . On a homogeneous inverse branch h_s the transfer operator has the form

$$(161) \quad \mathcal{L}_{\theta_s, h_s} \rho = J_{h_s}^{-1} \rho \circ h_s$$

defining a density on the image component. Differentiating in s gives two interior terms: the derivative of h_s applied to the branchwise derivative of ρ , and the derivative of $\log J_{h_s}$. These are exactly $\mathcal{D}_h^{\text{int}}$ and $\mathcal{D}_h^{\text{Jac}}$ in (159). The differentiability of the flight time, reflection law, and Jacobian on the branch follows from the implicit impact equation used in Proposition 4.1.

The domain of a branch also moves. If a boundary curve Γ_s moves with normal velocity V_Γ , differentiating the characteristic function of the branch gives the distribution

$$(162) \quad (V_\Gamma \cdot n_\Gamma) \rho \delta_{\Gamma_0},$$

paired against the jump of the test observable across the branch boundary. In ordinary C^1 norms this is singular; in the billiard anisotropic norm it is a weak stable-leaf functional. Transversality of singular curves to stable and unstable cones bounds the crossing number of an admissible leaf, and Lemma 4.7 gives a summable high-strip tail. This proves (160) and shows that $\nabla_s \mathcal{L}_{\theta_s}|_0$ maps strong densities to weak densities.

The spectral projector is

$$(163) \quad \Pi_{\theta_s} = \frac{1}{2\pi i} \int_{|\zeta-1|=r} (\zeta - \mathcal{L}_{\theta_s})^{-1} d\zeta,$$

where r is smaller than the uniform gap from Proposition 5.10. The strong-to-weak differentiability of $\mathcal{L}_{\theta_s}$ and the uniform resolvent bound justify differentiating this contour integral. Since $\rho_{\theta_s} = \Pi_{\theta_s} 1$ after mass normalization, differentiating $\mathcal{L}_{\theta_s} \rho_{\theta_s} = \rho_{\theta_s}$ and imposing zero mass on $\nabla_s \rho_{\theta_s}$ gives

$$(164) \quad (I - \mathcal{L}_{\theta_0}) \nabla_s \rho_{\theta_s}|_0 = \Pi_{\theta_0}^\perp (\nabla_s \mathcal{L}_{\theta_s}|_0 \rho_{\theta_0}).$$

The reduced resolvent from Proposition 5.10 solves this identity. Pairing the result with f_0 and adding the trivialized derivative of f_s gives (158). The same formula applied to the resolvent representation of the corrector differentiates all cell-problem coefficients. No stochastic response principle is used. $\square$

Proposition 6.7 (Uniform multi-response bounds for moving singularities). *For every integer $1 \leq \ell \leq r - 5$, the ℓ th parameter derivative of the collision-map transfer operator, the weighted suspension transfer family, the spectral projector, and the reduced observable resolvent exists in the strong-to-weak anisotropic topology. More precisely, in every Banach-bundle chart and uniformly over the compact parameter manifold,*

$$(165) \quad \|\nabla_\theta^\ell \mathcal{L}_\theta h\|_{\mathcal{B}_\theta^w} \leq C_\ell \|h\|_{\mathcal{B}_\theta},$$

$$(166) \quad \|\nabla_\theta^\ell \Pi_\theta h\|_{\mathcal{B}_\theta^w} \leq C_\ell \|h\|_{\mathcal{B}_\theta},$$

$$(167) \quad \|\nabla_\theta^\ell (-\mathcal{A}_\theta|_{\ker \Pi_\theta})^{-1} f\|_{\widehat{\mathcal{B}}_\theta^{\#,w}} \leq C_\ell \|f\|_{\widehat{\mathcal{B}}_\theta^\#}.$$

The derivatives are finite sums of branch-interior terms, Jacobian/roof terms, and moving-boundary distributions on singular hypersurfaces. Each boundary series is absolutely summable in the weak anisotropic norm. The constants C_ℓ are determined only by the geometric constants in assumption 2.4, the finite-response bounds, and the chosen homogeneity exponents.

Proof. On a homogeneous strip $\mathbb{H}_k$, the implicit impact equation gives ℓ th derivatives of the flight time, inverse branch, reflection map, and Jacobian bounded by $C_\ell k^{a_\ell}$ for an exponent a_ℓ depending only on ℓ and the strip convention. Differentiating a branch contribution therefore gives terms of the form

$$(168) \quad \mathcal{L}_{\theta,h}(A_{h,\ell} D_{\text{br}}^j h), \quad j \leq \ell,$$

with $A_{h,\ell}$ satisfying the same polynomial strip bound. The stable norm pays one inverse expansion factor, the unstable seminorm pays the holonomy Jacobian difference, and the weak norm pays only the stable-leaf trace. The one-step expansion and terminal-child estimate imply

$$(169) \quad \sum_{k \geq k_0} k^{a_\ell} \text{Mass}(\text{children entering } \mathbb{H}_k) \leq C_\ell k_0^{-1},$$

for the chosen homogeneity exponent after increasing the initial strip cutoff. Thus all interior and Jacobian/roof derivative terms are strong-to-weak bounded.

When a branch boundary moves, differentiating its characteristic function produces a trace functional on the corresponding singular hypersurface. The derived trace ledger Lemmas 2.13 and 2.14 bounds the number of intersections with an admissible stable leaf in the rescaled chart, and the same high-strip summability gives the displayed tail estimate. Repeated differentiation produces products of normal velocities, curvature derivatives, and lower-order boundary traces; each is controlled by the same polynomial strip bound and is therefore absolutely summable in the weak norm. This proves (165); the weighted suspension family is identical with the additional multiplier $e^{-s\tau_\theta}$, whose derivatives obey the same roof estimates.

The spectral projector and the reduced map resolvent are contour integrals of $(z - \mathcal{L}_\theta)^{-1}$ over contours fixed by the uniform spectral gap. The differentiated resolvent identity expresses each derivative as a finite sum of products of uniformly bounded resolvents and the operator derivatives just estimated. This proves (166). For the suspension observable generator, apply the same argument to the Laplace family and use the uniform Dolgopyat strip to shift the inverse Laplace contour. Dualizing yields (167). No response estimate is imported as an external assumption; it is the differentiated form of the branchwise singular-billiard transfer calculus. $\square$

Definition 6.8 (Fixed strong-weak response domain). Choose a reference table θ_0 and a compact parameter ball U . The response domain $\mathfrak{D}_U \subset \mathcal{B}_{\theta_0}$ is the completion of finite linear combinations of homogeneous branch densities whose pullbacks to all $\theta \in U$ have uniformly bounded strong anisotropic norm and whose first $r - 5$ moving-boundary traces have uniformly summable strip tails. The weak response codomain $\mathfrak{D}_U^w$ is defined in the same trivialization by stable-leaf testing with one fewer derivative. The inclusions

$$(170) \quad \mathfrak{D}_U \hookrightarrow \mathcal{B}_{\theta_0}, \quad \mathcal{B}_{\theta_0}^w \hookrightarrow \mathfrak{D}_U^w$$

are continuous and all transfer, weighted transfer, projector, and reduced resolvent derivatives below are maps $\mathfrak{D}_U \rightarrow \mathfrak{D}_U^w$.

Proposition 6.9 (Faà di Bruno response expansion with singular traces). *For $1 \leq \ell \leq r - 5$, the derivative $\partial_\theta^\ell \mathcal{L}_\theta$ in the fixed response domain is a finite sum of terms of the form*

$$(171) \quad \sum_{h \in \mathbb{H}} \mathcal{L}_{\theta,h} \left(A_{h,\mathbf{a}} D_{\text{br}}^{j_1} h \cdots D_{\text{br}}^{j_m} h \right) + \sum_{\Gamma \subset S} B_{\Gamma,\mathbf{a}} \partial_\tau^q \delta_\Gamma,$$

where $\sum j_m + q \leq \ell$. The coefficients $A_{h,\mathbf{a}}$ and $B_{\Gamma,\mathbf{a}}$ obey polynomial strip bounds, and the complete series in (171) is absolutely bounded from $\mathfrak{D}_U$ to $\mathfrak{D}_U^w$. Consequently every differentiated contour integral defining Π_θ , the map reduced resolvent, and the suspension reduced generator resolvent is a genuine bounded strong-weak operator, not a formal derivative.

Proof. Differentiate a branch contribution written in fixed coordinates as $J_h(\theta)^{-1} \rho(h_\theta(\cdot)) 1_{D_h(\theta)}$. The ordinary Faà di Bruno formula gives the interior part: derivatives may hit the inverse branch, the Jacobian, the roof weight, or the observable. The impact-equation bounds of Lemma 2.13 give polynomial strip estimates for every factor. Derivatives of the characteristic function $1_{D_h(\theta)}$ give distributions supported on moving branch boundaries; second and higher derivatives give tangential derivatives of such distributions and products of normal velocities. Because the singular curves cross stable tests transversally in the rescaled coordinates, each trace is a weak anisotropic functional. The polynomial strip losses are summable by the growth lemma after the homogeneity exponent is fixed large enough.

The differentiated resolvent identity is then a finite sum of products

$$(172) \quad (z - \mathcal{L}_\theta)^{-1} (\partial_\theta^{\ell_1} \mathcal{L}_\theta) (z - \mathcal{L}_\theta)^{-1} \cdots (\partial_\theta^{\ell_m} \mathcal{L}_\theta) (z - \mathcal{L}_\theta)^{-1},$$

with $\ell_1 + \cdots + \ell_m = \ell$. Uniform spectral separation fixes the contours and bounds the resolvents from weak back to strong after inserting the Lasota–Yorke recovery step. The suspension case is the same computation for $\mathcal{L}_{\theta,s}$, with the Dolgopyat strip providing the uniform vertical-frequency bound before inverse Laplace inversion. This proves the proposition. $\square$

Theorem 6.10 (Finite regularity-loss budget for the response calculus). *Fix a compact finite-response parameter ball U and the response domain $\mathfrak{D}_U$ of Definition 6.8. For $j \geq 0$, let $\mathfrak{D}_U^{(j)}$ denote the same domain with j additional controlled branch derivatives and j additional moving-boundary traces in the blown-up homogeneity-strip charts; let $\mathfrak{D}_U^{w,(j)}$ be the corresponding weak stable-leaf testing space. For every $0 \leq \ell \leq r - 8$, the response operators used later satisfy the quantitative loss rules*

$$(173) \quad \partial_\theta^\ell \mathcal{L}_\theta : \mathfrak{D}_U^{(\ell+2)} \longrightarrow \mathfrak{D}_U^{w,(0)},$$

$$(174) \quad \partial_\theta^\ell \Pi_\theta, \partial_\theta^\ell (I - \mathcal{L}_\theta)^{-1} \Pi_\theta^\perp : \mathfrak{D}_U^{(\ell+3)} \longrightarrow \mathfrak{D}_U^{w,(0)},$$

$$(175) \quad \partial_\theta^\ell (-\mathcal{A}_\theta|_{\ker \Pi_\theta})^{-1} : \widehat{\mathfrak{D}}_U^{\#,(\ell+4),0} \longrightarrow \widehat{\mathfrak{D}}_U^{\#,w,(0)}.$$

Consequently the first corrector, the second corrector, the Green–Kubo tensor, the effective Hamiltonian, and the complete covariant work response used in the perturbed-test proof have all (x, p) derivatives required by Lemma 7.25, Proposition 7.32, and Theorem 7.33 inside the weak observable bundle. The maximal loss along any path entering the main theorem is eight primitive derivatives: two for the branch/trace linearization, one for centering, one for the suspension resolvent, and four for the C^4 contact-jet Taylor expansion. Since the primitive scatterer family is C^{r+3} with $r \geq 12$, the response and cell calculations never exhaust the ordinary geometric regularity budget.

Proof. The transfer-operator derivative is the branch-and-trace expansion of Proposition 6.9. Each parameter derivative can hit an inverse branch, a Jacobian, a roof, or a moving branch indicator. The interior terms require one branch derivative of the density, while a moving-indicator derivative is paired with a stable test leaf and therefore lands in the weak space. The two extra derivatives in (173) are the fixed overhead for the blown-up grazing coordinate and the stable-holonomy comparison; the terminal-child summability absorbs the resulting polynomial strip factors.

For the projector and map resolvent, differentiate the contour formula. Every differentiated word alternates uniformly bounded resolvents with one $\partial_\theta^q \mathcal{L}_\theta$. The contour is fixed by the spectral gap, so no additional singular set is created; the single extra derivative in (174) is the centering derivative $\partial_\theta \Pi_\theta$, which keeps the input in the zero-mean bundle. For the suspension generator, the Laplace family adds the roof multiplier and the inverse-Laplace contour. The roof derivative is controlled by the same finite horizon and strip bounds, and the Dolgopyat strip gives the uniform vertical resolvent bound; this accounts for the fourth overhead derivative in (175).

The coefficients in the main theorem are finite algebraic combinations of finite-response read-outs, invariant pairings, reduced-resolvent correctors, and correlation integrals. Products are continuous in the response domain by the anisotropic multiplication bounds, and correlation integrals are dominated by the flow spectral gap. The C^4 viscosity test contributes at most four macroscopic derivatives, each entering through the already bounded horizontal chain rule. Hence the full-gradient residual uses no derivative outside the budget displayed above. This is a bookkeeping theorem about deterministic branch calculus and trace summability, not an additional response assumption. $\square$

Theorem 6.11 (Closed response chain rule for moving billiard coefficients). *Let $(x, p) \mapsto \theta = \Theta(x, p)$ be the table parameter generated by the finite-response port, and let $F(x, p, \cdot)$ be any coefficient, corrector, or finite product of such objects used in the effective HJB derivation. On each compact jet window the covariant derivative has the exact decomposition*

$$(176) \quad D_{x,p}^{\text{hor}} F = D_{x,p}^{\text{br}} F + (D_\theta^{\text{int}} F) D_{x,p} \Theta + (D_\theta^{\text{bd}} F) D_{x,p} \Theta + D_{x,p}^{\text{proj}} F,$$

where the four terms are, respectively, the branchwise derivative of the observable formula, the interior derivative of inverse branches/roofs/Jacobians, the moving-boundary trace derivative on singular hypersurfaces, and the centering/projection derivative required to remain in the zero-mean bundle. All four terms are bounded from strong to weak anisotropic spaces. The right hand side is constructed before the homogenized solution is introduced and is the only chain rule used in the full-gradient residual calculation.

Proof. Transport all objects to a fixed Banach-bundle chart. A coefficient or corrector is obtained by finitely many algebraic operations, invariant projections, and reduced resolvents

applied to branchwise finite-response observables. Differentiating a branchwise observable gives $D_{x,p}^{\text{br}}F$. When the billiard table changes, the inverse branch, Jacobian, roof, and reflection law change; differentiating their explicit impact-equation formulae gives $D_\theta^{\text{int}}F$. The domains of the branches also move; differentiating their characteristic functions gives the trace operator $D_\theta^{\text{bd}}F$. Its boundedness is exactly the moving-boundary estimate (160) and its higher-order version (171). Finally, centered objects must remain centered as the invariant density moves. Differentiating $\Pi_\theta^\perp F$ by the contour formula for Π_θ gives $D_{x,p}^{\text{proj}}F$.

Each term is a previously constructed deterministic operator: branch differentiation uses the finite-response C^r bounds, interior table differentiation uses the impact-equation derivatives, boundary differentiation uses the trace ledger, and projection differentiation uses the spectral gap. The composition with $D_{x,p}\Theta$ is legitimate because Θ is the endpoint map of the compact finite-response port and has globally bounded derivatives after saturation. Therefore (176) is a closed deterministic chain rule and cannot import information from the limiting HJB solution. $\square$

Proposition 6.12 (No loss of centering under coefficient differentiation). *If $F(x, p, \cdot)$ is centered for every (x, p) , then the covariant derivative $D_{x,p}^{\text{hor}}F$ decomposes as*

$$(177) \quad D_{x,p}^{\text{hor}}F = \Pi_{\Theta(x,p)}^\perp D_{x,p}^{\text{hor}}F + c_{x,p}, \quad c_{x,p} = \int D_{x,p}^{\text{hor}}F d\mu_{\Theta(x,p)},$$

and the zero-mean component belongs to the range of $\mathcal{A}_{\Theta(x,p)}$ on the observable generator domain. The scalar $c_{x,p}$ is retained as part of the effective coefficient whenever it is not zero.

Proof. Differentiate the identity $\int F(x, p, z) d\mu_{\Theta(x,p)}(z) = 0$ in the Banach bundle. The derivative of the measure is the reduced-transfer response of Theorem 6.6; the derivative of F is the closed chain rule (176). Their sum is zero, which identifies the constant component $c_{x,p}$. The primitive spectral gap gives the inverse of the generator on the zero-mean quotient, so $\Pi^\perp D_{x,p}^{\text{hor}}F$ is a deterministic coboundary. The constant component is not thrown away; in the full-gradient expansion it is precisely the part that contributes to H or to the Green–Kubo Hessian contraction. $\square$

Corollary 6.13 (Response regularity of all coefficients used in the HJB). *The maps $(x, p) \mapsto D(x, p)$, $(x, p) \mapsto H(x, p)$, the first and second correctors, and the complete covariant work response are C^{r-5} as sections of the transported weak observable bundle on compact jet windows. Their derivatives are obtained by applying Proposition 6.9 to the explicit branch formulae and then applying the reduced observable resolvent. No derivative of an effective coefficient is used before the corresponding branchwise response expression has been constructed.*

Proof. Each coefficient is a finite algebraic combination of invariant pairings, centered observables, correctors, and correlation integrals. Invariant pairings are differentiated by the projector formula in Theorem 6.6. Correctors are differentiated by the reduced generator resolvent, whose derivatives are bounded by Proposition 6.9. Correlation integrals are absolutely convergent because of the flow spectral gap, and their parameter derivatives are obtained by differentiating the integrand and the semigroup and then using the same exponential bound to dominate the time integral. The centering constants are differentiated at the end to keep the inputs in the zero-mean bundle. The construction is ordered at the level of branchwise operators, so it cannot be circular in the effective HJB coefficients. $\square$

Corollary 6.14 (Differentiated cell problems are deterministic coboundaries). *For every centered observable section $g_{x,p}$ generated by the drift, potential, or finite-response connection,*

the differentiated corrector $\nabla_{x,p}\chi[g_{x,p}]$ is the unique zero-mean solution of

$$(178) \quad \mathcal{A}_{\Theta(x,p)} \nabla_{x,p}\chi[g_{x,p}] = -\nabla_{x,p}g_{x,p} - (\nabla_{x,p}\mathcal{A}_{\Theta(x,p)})\chi[g_{x,p}] + c_{x,p},$$

where $c_{x,p}$ is the invariant average required to make the right-hand side zero mean. The term $(\nabla_{x,p}\mathcal{A}_{\Theta(x,p)})\chi[g_{x,p}]$ is the moving-singularity derivative of Theorem 6.6 and is bounded in the weak observable norm by strip summability.

Proof. Differentiate the identity $\mathcal{A}_{\Theta(x,p)}\chi[g_{x,p}] = -g_{x,p} + \langle g_{x,p} \rangle$ in the Banach bundle. The derivative of the generator is the suspension version of (159); the zero-mean constant is fixed by pairing with the invariant density. The reduced generator resolvent then solves the differentiated equation uniquely on the zero-mean subspace. $\square$

Proposition 6.15 (Covariant centering of differentiated generator terms). *Let $\theta(s) = \Theta(x, p + sq)$ be a finite-response cotangent variation and let F_s be a C^1 observable section in the moving generator domain with $F_0 = F$. Then the covariant derivative of the invariant generator identity is*

$$(179) \quad \int (\nabla_s \mathcal{A}_{\theta(s)}) F d\mu_{\theta_0} + \int \mathcal{A}_{\theta_0} (\nabla_s F_s|_0) d\mu_{\theta_0} + \langle \mathcal{A}_{\theta_0} F, \nabla_s \rho_{\theta(s)}|_0 \rangle = 0.$$

If F is a zero-mean corrector solving $\mathcal{A}_{\theta_0} F = -g$ and if the observable g_s is centered for every s , then

$$(180) \quad (\nabla_s \mathcal{A}_{\theta(s)}) F + \nabla_s g_s|_0 - \int ((\nabla_s \mathcal{A}_{\theta(s)}) F + \nabla_s g_s|_0) d\mu_{\theta_0}$$

is a zero-mean observable in the weak anisotropic space. Consequently it is in the range of the reduced deterministic generator resolvent. In particular, the zero-mean projections of the $\partial_p \Theta$, $\partial_p b$, moving-boundary, and moving-projection terms appearing in the full-gradient perturbed-test expansion are deterministic coboundaries before the homogenization limit is taken; their invariant projections are not discarded but are recorded in $\Lambda_{x,p}$ and therefore in the coefficients H and D .

Proof. For every s , Liouville invariance gives $\int \mathcal{A}_{\theta(s)} F_s d\mu_{\theta(s)} = 0$. Pull the three factors to the fixed Banach fiber at $s = 0$ and differentiate. The derivative of the operator is the strong-to-weak moving-singularity derivative of Theorem 6.6; the derivative of the measure is obtained by the reduced transfer resolvent; and the derivative of the section is ordinary covariant differentiation. This gives (179). If $\mathcal{A}_{\theta(s)} F_s = -g_s + \langle g_s \rangle_{\theta(s)}$ and $\langle g_s \rangle_{\theta(s)} = 0$ for all s , differentiating the centering identity cancels the last term in (179) against the derivative of g_s . The centered expression (180) is therefore orthogonal to the invariant density. The primitive spectral resolvent theorem then solves a cell equation for it. No cancellation is asserted by notation alone: it is the differentiated invariance identity in the moving billiard bundle. $\square$

7. DETERMINISTIC SCALE ANALYSIS AND HOMOGENIZATION

The scale analysis is purely analytic. It uses the observable generator $\mathcal{A}_\theta$, the deterministic transfer-resolvent estimates, deterministic correlation identities, and viscosity stability. Only the deterministic microscopic model and deterministic analytic estimates appear in this part.

7.1. Effective coefficients. Let $b(x, p, z)$ be the microscopic drift observable, smooth in (x, p) and piecewise C^r in z , and let

$$(181) \quad \theta = \Theta(x, p)$$

with Θ of class C^{r-3} . The deterministic comoving-frame centering is

$$(182) \quad \int b(x, p, z) d\mu_{\Theta(x, p)}(z) = 0.$$

For each coordinate vector e_i , define the observable corrector

$$(183) \quad \chi_i(x, p, \cdot) = \chi_{\Theta(x, p)}[b_i(x, p, \cdot)], \quad \mathcal{A}_{\Theta(x, p)}\chi_i = -b_i(x, p, \cdot), \quad \int \chi_i d\mu_\theta = 0,$$

and set

$$(184) \quad \chi_p(x, p, z) = \sum_{i=1}^m p_i \chi_i(x, p, z).$$

The derivative ∇_x^{fr} is the horizontal derivative in the moving billiard bundle with p held fixed:

$$(185) \quad \partial_{x_j}^{\text{fr}} \chi_p(x, p, z) = \left. \frac{d}{ds} \right|_{s=0} \chi_p(x + se_j, p, z), \quad Q_x \chi_p = (\partial_{x_1}^{\text{fr}} \chi_p, \dots, \partial_{x_m}^{\text{fr}} \chi_p).$$

This convention is not shorthand for differentiating the composite $x \mapsto \chi_p(x, \nabla\varphi(x), z)$. In the viscosity proof the coefficient parameter is frozen at the actual contact jet, while the complete slow gradient of the perturbed test is still inserted in the prelimit equation.

Definition 7.1 (Complete covariant work response). Fix (x, p) , put $\theta = \Theta(x, p)$, and let $q = (q_1, \dots, q_m)$ be a branchwise finite-response observable vector. The raw coefficient-response part is

$$(186) \quad \mathcal{R}_{x, p}(q) := (\nabla_\theta \mathcal{A}_\theta)[D_p \Theta(x, p)q] \chi_p + (\nabla_p b(x, p, \cdot)[q]) \cdot p.$$

The complete first-variation work observable is

$$(187) \quad \mathfrak{W}_{x, p}(q) := b(x, p, \cdot) \cdot q + \mathcal{R}_{x, p}(q),$$

and its invariant projection is the deterministic linear functional

$$(188) \quad \Lambda_{x, p}(q) := \int \mathfrak{W}_{x, p}(q) d\mu_\theta.$$

The centered complete response is

$$(189) \quad \mathfrak{W}_{x, p}^\circ(q) := \mathfrak{W}_{x, p}(q) - \Lambda_{x, p}(q), \quad \int \mathfrak{W}_{x, p}^\circ(q) d\mu_\theta = 0.$$

Thus no term produced by a z -dependent full gradient is discarded: its invariant component is recorded by $\Lambda_{x, p}$ and its zero-mean component is sent to the deterministic second cell equation.

Lemma 7.2 (Hessian-sector reciprocity of the finite-response connection). For $q_{ij} = \chi_i e_j$ define

$$(190) \quad \Gamma_{ij}(x, p) = \int \mathcal{R}_{x, p}(q_{ij}) d\mu_{\Theta(x, p)}.$$

The single-action finite-response port implies

$$(191) \quad \Gamma_{ij}(x, p) + \Gamma_{ji}(x, p) = 0.$$

Consequently, for every symmetric matrix X ,

$$(192) \quad \sum_{i,j} X_{ij} \Lambda_{x,p}(\chi_i e_j) = \frac{1}{2} \sum_{i,j} X_{ij} \int (b_j \chi_i + b_i \chi_j) d\mu_{\Theta(x,p)}.$$

Proof. The proof is the covariant mixed-variation identity of the microscopic action. For constant cotangent directions e_i, e_j , differentiate the first cell identity

$$(193) \quad \mathcal{A}_{\Theta(x,p)} \chi_p + b(x, p, \cdot) \cdot p = 0$$

in the moving Banach bundle. The derivative in the e_j direction is

$$(194) \quad (\nabla_{\theta} \mathcal{A}_{\theta})[D_p \Theta e_j] \chi_p + \mathcal{A}_{\theta}(\nabla_{p_j}^{\text{cov}} \chi_p) + (\nabla_{p_j} b) \cdot p + b_j = 0.$$

Multiply (194) by χ_i , pair with μ_{θ} , and repeat with i and j interchanged. The terms containing $\mathcal{A}_{\theta}(\nabla_{p_j}^{\text{cov}} \chi_p)$ are integrated by the deterministic Green identity; the physical collision faces cancel by specular reflection and the grazing and branch-change traces are summable by the moving-singularity bounds. The sum of the two resulting identities is the symmetric second variation, with respect to p_i and p_j , of the same finite-response action functional in (54). Since the port is single-action, mixed partials commute on every regular branch, and the remaining Liouville-exact endpoint gauges have zero invariant average by Proposition 3.7. Hence the symmetric part of Γ vanishes, giving (191).

Now

$$(195) \quad \Lambda_{x,p}(\chi_i e_j) = \int b_j \chi_i d\mu_{\theta} + \Gamma_{ij}.$$

Contracting with a symmetric matrix and using (191) removes the skew connection part and gives (192). The argument is an identity of deterministic branch actions, Green formulas, and weak trace summations; no probabilistic approximation is used. $\square$

Definition 7.3 (Covariant Green–Kubo tensor from deterministic correlations). The effective diffusivity is the symmetric matrix

$$(196) \quad D_{ij}(x, p) = \frac{1}{2} \left(\Lambda_{x,p}(\chi_i e_j) + \Lambda_{x,p}(\chi_j e_i) \right), \quad \theta = \Theta(x, p).$$

By Lemma 7.2, this is equivalently the deterministic Green–Kubo tensor of the centered microscopic velocity field,

$$(197) \quad D_{ij}(x, p) = \frac{1}{2} \int_0^{\infty} \int \left(b_i(x, p, z) b_j(x, p, \Phi_{\theta}^t z) + b_j(x, p, z) b_i(x, p, \Phi_{\theta}^t z) \right) d\mu_{\theta}(z) dt,$$

or, using $\chi_j = \int_0^{\infty} \mathcal{U}_{\theta,t} b_j dt$,

$$(198) \quad D_{ij}(x, p) = \frac{1}{2} \int (b_i \chi_j + b_j \chi_i) d\mu_{\theta}.$$

The first formula (196) is the one that appears in the full-gradient expansion; the correlation formulas are its reciprocity-reduced deterministic form.

Proposition 7.4 (Positivity). *For every (x, p) , the matrix $D(x, p)$ is symmetric and positive semidefinite.*

Proof. Symmetry is built into (196). For $a \in \mathbb{R}^m$, set $f_a = a \cdot b(x, p, \cdot)$ and $\theta = \Theta(x, p)$. By Lemma 7.2, the covariant Hessian-sector definition gives the same quadratic form as the Green–Kubo correlation:

$$(199) \quad a^T D(x, p) a = \int_0^{\infty} \int f_a(z) f_a(\Phi_{\theta}^t z) d\mu_{\theta}(z) dt.$$

For $T > 0$, invariance of μ_θ gives the exact finite-time identity

$$(200) \quad \frac{1}{2T} \int \left(\int_0^T f_a(\Phi_\theta^t z) dt \right)^2 d\mu_\theta(z) = \int_0^T \left(1 - \frac{s}{T} \right) \int f_a(z) f_a(\Phi_\theta^s z) d\mu_\theta(z) ds.$$

The right-hand side converges to the Green–Kubo integral by the deterministic exponential correlation bound from Theorem 5.17. The left-hand side is a square divided by $2T$, hence non-negative. This is a deterministic identity for time integrals along the billiard flow, proved by deterministic correlation bounds and independent of probabilistic representation input. $\square$

Definition 7.5 (Effective potential, normalization, and Hamiltonian). Let $V(x, p, z)$ be the microscopic potential. The raw deterministic potential is

$$(201) \quad H_{\text{raw}}(x, p) = \int V(x, p, z) d\mu_{\Theta(x, p)}(z) + \Lambda_{x, p}(Q_x \chi_p),$$

where the second term is the full covariant horizontal work response from Definition 7.1. Written out, it contains both the physical work $b \cdot Q_x \chi_p$ and the moving-table/centered-drift connection terms generated by $\partial_p \Theta$ and $\partial_p b$. The expectation gauge is the deterministic normalization

$$(202) \quad H(x, p) = H_{\text{raw}}(x, p) - H_{\text{raw}}(x, 0), \quad H(x, 0) = 0.$$

Equivalently, the microscopic running potential in the prelimit equation is replaced by

$$(203) \quad V^\circ(x, p, z) = V(x, p, z) - H_{\text{raw}}(x, 0).$$

This subtraction is a change of zero energy at each macroscopic point; it does not alter the deterministic corrector mechanism. The θ -Hamiltonian is

$$(204) \quad H_\theta(x, p, X) = \text{Tr}(D(x, p)X) + H(x, p).$$

Lemma 7.6 (Contact-jet-frozen corrector calculus). *Let $\varphi \in C^3$ and let $(t_\varepsilon, x_\varepsilon)$ be a contact point in the viscosity–duality argument for the ε -problem. Set*

$$(205) \quad p_\varepsilon = D_x \varphi(t_\varepsilon, x_\varepsilon), \quad X_\varepsilon = D_x^2 \varphi(t_\varepsilon, x_\varepsilon), \quad \theta_\varepsilon = \Theta(x_\varepsilon, p_\varepsilon).$$

The admissible first perturbed test is frozen at this actual contact jet in the coefficient arguments but is multiplied by the current macroscopic gradient of φ :

$$(206) \quad \Xi_\varepsilon[\varphi](t, x, z) = \sum_{i=1}^m \partial_i \varphi(t, x) \chi_i(x, p_\varepsilon, z), \quad \varphi^\varepsilon(t, x, z) = \varphi(t, x) + \varepsilon \Xi_\varepsilon[\varphi](t, x, z) + \varepsilon^2 \chi_\varepsilon^{(2)}(x, z).$$

At the contact point,

$$(207) \quad D_x \Xi_\varepsilon[\varphi](t_\varepsilon, x_\varepsilon, z) = D_x^{\text{fr}} \chi_{p_\varepsilon}(x_\varepsilon, p_\varepsilon, z) + \sum_{i,j=1}^m X_{\varepsilon, ij} \chi_i(x_\varepsilon, p_\varepsilon, z) e_j.$$

The order ε^{-1} part cancels exactly at the contact point,

$$(208) \quad -b(x_\varepsilon, p_\varepsilon, z) \cdot p_\varepsilon - \mathcal{A}_{\theta_\varepsilon} \left(\sum_i p_{\varepsilon, i} \chi_i(x_\varepsilon, p_\varepsilon, \cdot) \right) (z) = 0,$$

and the complete covariant Hessian coefficient produced at order one is

$$(209) \quad \sum_{i,j} X_{\varepsilon, ij} \Lambda_{x_\varepsilon, p_\varepsilon} \left(\chi_i(x_\varepsilon, p_\varepsilon, \cdot) e_j \right) = \text{Tr}(D(x_\varepsilon, p_\varepsilon) X_\varepsilon).$$

Equivalently, by the Hessian-sector reciprocity lemma, this is the usual symmetrized Green–Kubo contraction. No uncontrolled contribution of the form $\varepsilon^{-1}D_p\chi(p_\varepsilon - p_0)$ or $D_p\chi D_x^2\varphi$ appears; every coefficient-response contribution is included in Λ before the second cell problem is solved.

Proof. The contact point for the ε -problem is chosen first, and the cell problem is then solved with the coefficient parameters $(x, p) = (x_\varepsilon, p_\varepsilon)$ frozen in the singular part. This gives (208) before any passage to the limit and removes the possible ε^{-1} parameter error. The proof never tests the prelimit equation at $(x_\varepsilon, p_\varepsilon)$ with a corrector frozen at the limiting jet (x_0, p_0) .

The multiplier $\partial_i\varphi(t, x)$ in (206) is not frozen. Differentiating the first corrector while holding the coefficient argument p_ε fixed gives

$$(210) \quad D_x\Xi_\varepsilon[\varphi] = \sum_i \partial_i\varphi(t, x) D_x^{\text{fr}}\chi_i(x, p_\varepsilon, z) + \sum_i \chi_i(x, p_\varepsilon, z) D_x\partial_i\varphi(t, x).$$

Evaluation at $(t_\varepsilon, x_\varepsilon)$ gives (207). The second sum in that identity produces the Hessian vector field

$$(211) \quad q_X(z) = \sum_{i,j} X_{\varepsilon,ij}\chi_i(x_\varepsilon, p_\varepsilon, z)e_j.$$

When the prelimit coefficients are evaluated at the complete gradient, the order-one work generated by this vector field is not merely $b \cdot q_X$; it is the complete covariant work $\mathfrak{W}_{x_\varepsilon, p_\varepsilon}(q_X)$. Its invariant projection is

$$(212) \quad \sum_{i,j} X_{\varepsilon,ij}\Lambda_{x_\varepsilon, p_\varepsilon}(\chi_i e_j),$$

which equals $\text{Tr}(DX_\varepsilon)$ by Lemma 7.2 and Definition 7.3. The skew part of the finite-response connection is thus retained in the calculation but disappears when contracted with the symmetric Hessian. A composite corrector $\chi_i(x, D_x\varphi(t, x), z)$ would create the chain-rule term $D_p\chi_i D_x^2\varphi$; the contact-jet-frozen corrector uses $\chi_i(x, p_\varepsilon, z)$ instead. Parameter derivatives of χ_i enter only later through ordinary continuity as $(x_\varepsilon, p_\varepsilon, X_\varepsilon) \rightarrow (x_0, p_0, X_0)$, with no singular factor. This proves the lemma. $\square$

Lemma 7.7 (Correlation Hilbert bundle and Lipschitz diffusion factor). *Fix a compact cotangent window K and a coordinate ball $O \subset \mathcal{M}$. For $p \in K$ and $x \in O$, let $\mathfrak{c}_{x,p}$ be the deterministic correlation form on zero-mean finite-response observables,*

$$(213) \quad \mathfrak{c}_{x,p}(f, g) = \frac{1}{2} \int_0^\infty \int \left(f(z)g(\Phi_{\Theta(x,p)}^t z) + g(z)f(\Phi_{\Theta(x,p)}^t z) \right) d\mu_{\Theta(x,p)}(z) dt.$$

After quotienting the null space and completing, these forms define a correlation Hilbert bundle $\mathcal{H}_{x,p}$. Under the Banach-bundle transport of Definition 2.21 there is a common Hilbert model $\mathcal{H}_{O,K}$ and finite-rank maps

$$(214) \quad \Sigma(x, p)^* e_i = [b_i(x, p, \cdot)]_{\mathcal{H}_{x,p}}, \quad i = 1, \dots, m,$$

for which

$$(215) \quad D(x, p) = \Sigma(x, p)\Sigma(x, p)^*.$$

Moreover,

$$(216) \quad \|\Sigma(x, p) - \Sigma(y, p)\|_{HS(\mathcal{H}_{O,K}, \mathbb{R}^m)} \leq C_K |x - y|,$$

where the constant is independent of $p \in K$. The estimate is valid even at points where $D(x, p)$ is degenerate; no matrix square root is taken.

Proof. The form $\mathbf{c}_{x,p}$ is finite by the deterministic exponential flow gap. Positivity follows from the finite-time identity

$$(217) \quad \mathbf{c}_{x,p}(f, f) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int \left(\int_0^T f(\Phi_{\Theta(x,p)}^t z) dt \right)^2 d\mu_{\Theta(x,p)}(z),$$

where convergence is justified by the same correlation bound; this is an identity of deterministic time integrals. The quotient by the null space and completion therefore gives a Hilbert fiber.

To compare nearby fibers, transport observables and densities to a reference fiber over $x_0 \in O$. The multi-response bounds Proposition 6.7 imply the bilinear estimate

$$(218) \quad |\mathbf{c}_{x,p}(f, g) - \mathbf{c}_{y,p}(f, g)| \leq C_K |x - y| \|f\|_{\widehat{\mathcal{B}}^\#} \|g\|_{\widehat{\mathcal{B}}^\#}$$

for finite-response observables in the local chart. Since $b_i(x, p, \cdot)$ is finite-response and C^1 in x with uniform bounds,

$$(219) \quad \|b_i(x, p, \cdot) - b_i(y, p, \cdot)\|_{\widehat{\mathcal{B}}^\#} \leq C_K |x - y|.$$

Hence, in the transported quotient Hilbert norm,

$$(220) \quad \|[b_i(x, p)] - [b_i(y, p)]\|_{\mathcal{H}_{O,K}}^2 \leq C_K \|b_i(x, p) - b_i(y, p)\|_{\widehat{\mathcal{B}}^\#}^2$$

$$(221) \quad \leq C_K |x - y|^2.$$

The derivative of the form itself contributes only the harmless term in (218), which is absorbed by choosing the parallel transport in the correlation bundle through the Gram–Schmidt quotient connection generated by $\nabla_x \mathbf{c}_{x,p}$. Summing over the finite coordinate index $i = 1, \dots, m$ gives the Hilbert–Schmidt estimate (216). Since the construction uses the classes $[b_i]$ in the correlation Hilbert space, degeneracy merely enlarges the null space and does not create a non-Lipschitz square-root singularity. $\square$

Proposition 7.8 (Concrete common Hilbert transport for the diffusion factor). *Let $O \times K$ be a compact macroscopic coordinate window. The correlation Hilbert fibers $\mathcal{H}_{x,p}$ of Lemma 7.7 admit deterministic parallel transports*

$$(222) \quad J_{y \rightarrow x}^p : \mathcal{H}_{y,p} \longrightarrow \mathcal{H}_{x,p}$$

with the following properties:

- (a) $J_{x \rightarrow x}^p = I$ and $J_{z \rightarrow x}^p J_{y \rightarrow z}^p = J_{y \rightarrow x}^p$ along coordinate line segments;
- (b) $J_{y \rightarrow x}^p$ is an isometry for the correlation inner products;
- (c) for the drift-factor maps $\Sigma(x, p)^* e_i = [b_i(x, p, \cdot)]_{\mathcal{H}_{x,p}}$ one has

$$(223) \quad \|\Sigma(x, p) - J_{y \rightarrow x}^p \Sigma(y, p)\|_{HS(\mathbb{R}^m, \mathcal{H}_{x,p})} \leq C_K |x - y|.$$

Consequently

$$(224) \quad D(x, p) = \Sigma(x, p) \Sigma(x, p)^*, \quad D(y, p) = \Sigma(y, p) \Sigma(y, p)^*,$$

may be compared in the single Hilbert fiber $\mathcal{H}_{x,p}$ by replacing $\Sigma(y, p)$ with $J_{y \rightarrow x}^p \Sigma(y, p)$. No expression of the form $(I - G_{x,p})^{1/2}$ and no matrix square root at a degenerate covariance is used.

Proof. Start with the dense finite-response observable module and the correlation form $\mathbf{c}_{x,p}$. The response theorem gives

$$(225) \quad |\partial_{x_a} \mathbf{c}_{x,p}(f, g)| \leq C_K \|f\|_{\mathcal{H}_{x,p}} \|g\|_{\mathcal{H}_{x,p}}$$

for representatives modulo the null space. Define a metric connection on the quotient module by declaring the covariant derivative $\nabla_a f$ to be the ordinary Banach-bundle derivative of the representative minus the unique finite-rank correction, orthogonal to the null space, for which

$$(226) \quad \partial_{x_a} \mathbf{c}_{x,p}(f, g) = \mathbf{c}_{x,p}(\nabla_a f, g) + \mathbf{c}_{x,p}(f, \nabla_a g).$$

The correction is obtained by Riesz representation in the quotient Hilbert fiber; if the null space changes, the equation is read on the quotient and then closed by density. The strong-to-weak response estimates make the connection bounded on the finite-response module, and completion gives a metric connection on the Hilbert bundle.

Solving the linear ODE for this connection along the segment from y to x gives the transport $J_{y \rightarrow x}^p$. Equation (226) implies conservation of the correlation inner product along the transported pair, hence $J_{y \rightarrow x}^p$ is an isometry. Differentiating $[b_i(x, p, \cdot)]_{\mathcal{H}_{x,p}}$ covariantly gives a bounded vector because b_i is a finite-response observable and the moving singularity terms are strong-to-weak bounded. Integrating this covariant derivative along the segment proves (223).

The covariance identity follows directly from the Green–Kubo definition in each fiber:

$$(227) \quad e_i \cdot D(x, p) e_j = \langle \Sigma(x, p)^* e_i, \Sigma(x, p)^* e_j \rangle_{\mathcal{H}_{x,p}}.$$

Because the transport is an isometry, moving the y -factor into $\mathcal{H}_{x,p}$ preserves $D(y, p)$. Thus the Crandall–Ishii comparison uses a genuine common Hilbert space for the two points, while avoiding any non-Lipschitz square-root selection at degenerate diffusion matrices. $\square$

Proposition 7.9 (Coefficient regularity and comparison structure). *The geometric assumptions and the finite-response mechanical coupling imply the following deterministic structure.*

- (i) D is symmetric positive semidefinite, bounded, and C^{r-4} in (x, p) . Its factorization in the correlation Hilbert bundle is Lipschitz in x , uniformly in the transmitted cotangent signal p .
- (ii) H is C^{r-4} and satisfies the finite-response bounds

$$(228) \quad |H(x, p)| \leq C(1 + |p|), \quad |H(x, p) - H(y, p)| \leq C(1 + |p|)|x - y|,$$

together with the corresponding local derivative bounds used in the viscosity argument.

- (iii) The operator

$$(229) \quad F(x, p, X) = -\text{Tr}(D(x, p)X) - H(x, p)$$

is degenerate elliptic and satisfies the Crandall–Ishii structural condition on the compact torus: there is a modulus ω such that, whenever

$$(230) \quad -3\alpha \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} \leq \begin{pmatrix} X & 0 \\ 0 & -Y \end{pmatrix} \leq 3\alpha \begin{pmatrix} I & -I \\ -I & I \end{pmatrix},$$

one has

$$(231) \quad F(y, \alpha(x - y), Y) - F(x, \alpha(x - y), X) \leq \omega(\alpha|x - y|^2 + |x - y|).$$

Consequently the terminal-value problem (291) has comparison and uniqueness in the bounded uniformly continuous class.

Proof. The proof is divided into four deterministic steps.

Step 1: regularity of the averaged objects. The observable b , the potential V , the projection $f \mapsto \Pi_\theta^\perp f$, and the reduced observable resolvent are sections of the Banach bundle of Definition 2.21. The response formula in Theorem 6.4 is obtained by differentiating the branchwise transfer operator and its contour resolvent in the strong-to-weak anisotropic

topology. Composing this response with the smooth finite-response map $(x, p) \mapsto \Theta(x, p)$ gives C^{r-4} dependence of $\chi_i(x, p, \cdot)$, of $\mu_{\Theta(x, p)}$, of the complete work response $\mathfrak{W}_{x, p}$, and hence of D and H . The finite-response saturation gives bounds uniform in p for the microscopic coefficients, for the moving-generator and centered-drift derivatives entering $\mathfrak{W}_{x, p}$, and for their parameter derivatives. The affine growth in H comes from the factor $\chi_p = \sum_i p_i \chi_i$ inside the covariant horizontal work response $\Lambda_{x, p}(Q_x \chi_p)$ in (201), yielding (228).

Step 2: positivity and the deterministic correlation factor. Positivity is Proposition 7.4. More is needed for comparison: a Lipschitz factor, not merely a Lipschitz matrix. This factor is supplied by the correlation Hilbert bundle constructed in Lemma 7.7. In that bundle

$$(232) \quad \Sigma(x, p)^* e_i = [b_i(x, p, \cdot)]_{\mathcal{H}_{x, p}}, \quad D(x, p) = \Sigma(x, p) \Sigma(x, p)^*,$$

and the deterministic response bounds for the branch transfer operator give

$$(233) \quad \|\Sigma(x, p) - \Sigma(y, p)\|_{HS(\mathcal{H}_{O, K}, \mathbb{R}^m)} \leq C|x - y|,$$

with C independent of p on compact cotangent windows. This is the required square-root structure; it is not obtained by taking a possibly non-Lipschitz matrix square root at a degenerate point.

Step 3: the diffusion part of the Crandall–Ishii inequality. Let $p = \alpha(x - y)$ and let X, Y satisfy Ishii’s matrix inequality. Apply that matrix inequality to the vectors $(\Sigma(x, p)^* \xi, \Sigma(y, p)^* \xi)$ and sum over an orthonormal finite-rank projection of the correlation Hilbert space. The finite-rank statement gives

$$(234) \quad -\text{Tr}(D(x, p)X - D(y, p)Y) \leq 3\alpha \|\Sigma(x, p) - \Sigma(y, p)\|_{HS}^2$$

for the projected factor. The positive correlation forms increase monotonically with the projection, and the exponential transfer bound gives a projection-tail estimate independent of x, y, p . Passing to the limit and using (233) yields

$$(235) \quad -\text{Tr}(D(x, p)X - D(y, p)Y) \leq C\alpha|x - y|^2.$$

Step 4: the Hamiltonian part and comparison. By (228), again with $p = \alpha(x - y)$,

$$(236) \quad |H(x, p) - H(y, p)| \leq C|x - y| + C\alpha|x - y|^2.$$

Combining this with (235) gives (231) with $\omega(r) = Cr$ after increasing C on the compact torus. Degenerate ellipticity follows from $D \geq 0$. The deterministic comparison proof in Appendix A.1 is then the standard doubling-of-variables argument for a proper degenerate parabolic equation with the structural modulus just derived. Only deterministic transfer, correlation, and comparison estimates enter this step. $\square$

7.2. The analytic two-scale equation. For $\varepsilon > 0$, the prelimit deterministic equation is posed in the anisotropic observable space. Its branchwise classical expression is the full-gradient completion of (70):

$$(237) \quad -\partial_t u^\varepsilon - \varepsilon^{-2} \mathcal{A}_{x, P}^{\text{fr}} u^\varepsilon - \varepsilon^{-1} b(x, P, z) \cdot P - V^\circ(x, P, z) = 0,$$

$$(238) \quad P = D_x u^\varepsilon(t, x, z), \quad u^\varepsilon(T, x, z) = \phi(x).$$

The cotangent variable P in this display is the complete slow gradient of the branchwise test touching the value function. Thus a microscopic corrector gauge contributes to P through its horizontal derivative. The anisotropic-duality definition below is the completion of this

branchwise equation and keeps this full-gradient dependence; it does not replace P by a split macroscopic jet.

Definition 7.10 (Admissible microscopic densities). For $M > 0$ and $\theta \in \Theta$, let

$$(239) \quad \text{Adm}_M(\theta) = \{\rho \in \widehat{\mathcal{B}}_\theta : \rho \geq 0, \langle 1, \rho \rangle_\theta = 1, \|\rho\|_{\widehat{\mathcal{B}}_\theta} \leq M\}.$$

The positivity is understood by testing against nonnegative C^1 observables on homogeneous flow boxes. Standard-family densities produced by the billiard flow belong to $\text{Adm}_M(\theta)$ for a uniform M after the deterministic growth lemma.

Proposition 7.11 (The paired anisotropic order is separating). *Let F be a bounded branchwise continuous observable section with summable homogeneity-strip tails. If*

$$(240) \quad \langle F, \rho \rangle_\theta \leq 0 \quad \text{for every } \rho \in \text{Adm}_M(\theta) \text{ and every finite } M,$$

then $F \leq 0$ at every regular point at which the stable-unstable density basis differentiates F . In particular, the paired order used in Definition 7.16 agrees with the ordinary a.e. order on regular microscopic states after the canonical singular set is discarded.

Proof. Assume that $F(z_0) > \eta > 0$ at a regular density point. Choose a homogeneous stable-unstable rectangle $R_\delta(z_0)$ avoiding the finite singular skeleton up to the corresponding order and so small that the oscillation of F on the rectangle is less than $\eta/2$. Smooth the normalized Liouville restriction to $R_\delta(z_0)$ along stable leaves and cut off in the unstable direction. The resulting density is nonnegative, has unit mass, and has finite anisotropic norm $M(\delta)$; the strip tail removed by the cutoff has arbitrarily small mass by the growth lemma. Pairing with this density gives $\langle F, \rho_\delta \rangle \geq \eta/3$ for sufficiently small smoothing scale, contradicting the assumed paired inequality with $M = M(\delta)$. The exceptional set is the union of the iterated singular set and the null set where the product differentiation basis fails. The argument is a deterministic density-basis argument; it does not use random sampling or a limit theorem. $\square$

Lemma 7.12 (Compactness and deterministic transport of admissible densities). *Let $\Theta_* \Subset \Theta$ be a compact set of billiard parameters generated by the finite-response port, and let $M > 0$. After trivializing the anisotropic Banach bundle on a parameter chart, the family*

$$(241) \quad \bigcup_{\theta \in \Theta_*} \{\theta\} \times \text{Adm}_M(\theta)$$

is sequentially compact in the weak anisotropic topology. More precisely, if $\theta_n \rightarrow \theta$ and $\rho_n \in \text{Adm}_M(\theta_n)$, then a subsequence of the transported densities converges to a density $\rho \in \text{Adm}_{M'}(\theta)$, where M' depends only on the bundle chart and on M . For every branchwise C^{r-3} observable section F_θ continuous in the moving-billiard Banach bundle,

$$(242) \quad \langle F_{\theta_n}, \rho_n \rangle_{\theta_n} \longrightarrow \langle F_\theta, \rho \rangle_\theta$$

along the convergent subsequence. The same assertion holds uniformly for the windowed family $\text{Adm}_{M,K}(x)$ when (x, p) ranges in a compact macroscopic cotangent cylinder.

Proof. Choose a finite system of homogeneous flow boxes and Banach-bundle trivializations covering Θ_* . In each box the strong anisotropic unit ball is compactly embedded into the weak anisotropic dual topology because the stable test functions gain a positive Hölder exponent while the unstable variation is bounded by the growth-lemma norm. The uniform bound $\|\rho_n\|_{\widehat{\mathcal{B}}_{\theta_n}} \leq M$ therefore gives a weakly convergent subsequence after transporting

all densities to the fixed chart. The transition maps between charts are the deterministic moving-boundary maps of Theorem 6.6; their operator norms are uniformly bounded by Proposition 6.7, so the limit has bounded norm M' .

Mass and positivity pass to the limit because both are tested against nonnegative branchwise C^1 functions and the pull-back of the constant observable is the constant observable. Singular boundaries do not contribute extra mass: the homogeneity-strip estimate in the growth lemma gives a summable high-strip tail, while the moving singular hypersurfaces have zero Liouville flux and their boundary distributions are bounded in the weak anisotropic norm by (160). Thus the limit belongs to $\text{Adm}_{M'}(\theta)$.

Finally, for a continuous observable section F_θ , decompose the pairing into a finite low-strip part and a high-strip tail. On the low-strip part the transport maps and F_θ converge in the branchwise C^{r-3} topology, so the weak convergence of the transported densities gives convergence of the pairings. The tail is uniformly smaller than any prescribed number by strip summability. This proves (242). The finite-response saturation makes the set of parameters generated by $(x, p) \in K$ compact, giving the stated uniform version. $\square$

Definition 7.13 (Admissible density bundle over a compact jet window). Let $K \Subset T^*\mathcal{M}$ be a compact cotangent window and put

$$(243) \quad \Theta_K(x) = \{\Theta(x, p) : p \in K_x\}, \quad K_x = K \cap T_x^*\mathcal{M}.$$

The admissible density bundle over K is

$$(244) \quad \text{Adm}_{M,K}(x) = \{(p, \rho) : p \in K_x, \rho \in \text{Adm}_M(\Theta(x, p))\},$$

with all pairings transported to a fixed local Banach-bundle chart by Definition 2.21. For an observable section F define

$$(245) \quad \|F(x, \cdot)\|_{\text{Adm}_{M,K}} = \sup_{(p, \rho) \in \text{Adm}_{M,K}(x)} |\langle F_{\Theta(x, p)}, \rho \rangle_{\Theta(x, p)}|.$$

This notation is used only to state uniform convergence and comparison on a compact range of contact jets. At an actual viscosity contact the parameter p is not free: it is the contact gradient $D_x\varphi$.

Lemma 7.14 (Compactness and continuity of admissible density windows). *For every finite M and compact cotangent window K , the transported family $\text{Adm}_{M,K}(x)$ is compact in the weak anisotropic topology after placement in any reference fiber of the Banach bundle. If $(x_n, p_n, \rho_n) \rightarrow (x, p, \rho)$ with $(p_n, \rho_n) \in \text{Adm}_{M,K}(x_n)$ and if F is a Banach-bundle continuous observable section, then*

$$(246) \quad \langle F_{\Theta(x_n, p_n)}, \rho_n \rangle_{\Theta(x_n, p_n)} \longrightarrow \langle F_{\Theta(x, p)}, \rho \rangle_{\Theta(x, p)}.$$

The same continuity holds for the weak moving-boundary trace functionals that appear in differentiated singularity terms.

Proof. Transport all densities to a fixed reference fiber. The strong anisotropic norm bounds stable-leaf pairings and unstable variations, while positivity and unit mass give a uniform weak bound. The compact embedding of Lemma 5.4 makes the transported bounded set relatively compact in the weak topology; positivity and mass are closed under testing against nonnegative branchwise observables, so the admissible set is compact. The transport maps depend C^{r-4} on the finite-response parameters by Proposition 2.22 and Theorem 6.6. Thus

the difference between the two sides of (246) splits into weak convergence of ρ_n and Banach-bundle continuity of F . Boundary traces are continuous because their normal velocities are bounded in rescaled singular charts and their high-strip tails are summable. $\square$

Definition 7.15 (Normalized dual parabolic jets). Let U^ε be an observable-valued function and let $\rho \in \text{Adm}_M(\Theta(x_0, p))$. A quadruple (a, p, X, Ψ) belongs to the upper normalized dual jet $J_{\#}^{2,+}U^\varepsilon(t_0, x_0; \rho)$ if Ψ is a local observable section of the Banach bundle, $\Psi_0 := \Psi(t_0, x_0, \cdot)$ belongs to the generator domain after the relevant fiber trivialization, and

$$(247) \quad \begin{aligned} \langle U^\varepsilon(t, x, \cdot) - U^\varepsilon(t_0, x_0, \cdot), \rho \rangle &\leq a(t - t_0) + p \cdot (x - x_0) + \frac{1}{2}(x - x_0)^T X (x - x_0) \\ &\quad + \langle \Psi(t, x, \cdot) - \Psi(t_0, x_0, \cdot), \rho \rangle + o(|t - t_0| + |x - x_0|^2). \end{aligned}$$

The lower jet $J_{\#}^{2,-}$ is defined by reversing the inequality. The horizontal derivatives $\partial_t^{\text{hor}}\Psi_0$ and $D_x^{\text{hor}}\Psi_0$ are taken in the moving Banach bundle. The complete cotangent field associated with the test is

$$(248) \quad P_\Psi(z) = p + D_x^{\text{hor}}\Psi_0(z).$$

At a normalized contact the value of the microscopic gauge is subtracted, but its horizontal derivative is not subtracted; the derivative is part of the physical slow gradient in (237).

Definition 7.16 (Viscosity–duality solution of the prelimit equation). A bounded observable-valued function U^ε is a subsolution of (237) if, for every M , every normalized upper dual test $(a, p, X, \Psi) \in J_{\#}^{2,+}U^\varepsilon(t_0, x_0; \rho)$, and every admissible density transported to the fiber determined by P_Ψ through the bundle charts,

$$(249) \quad \begin{aligned} &-a - \langle \partial_t^{\text{hor}}\Psi_0, \rho \rangle - \varepsilon^{-2} \langle \mathcal{A}_{x_0, P_\Psi}^{\text{fr}} \Psi_0, \rho \rangle \\ &- \varepsilon^{-1} \langle b(x_0, P_\Psi(\cdot), \cdot) \cdot P_\Psi(\cdot), \rho \rangle - \langle V^\circ(x_0, P_\Psi(\cdot), \cdot), \rho \rangle \leq 0. \end{aligned}$$

It is a supersolution if the reverse inequality holds for every lower normalized dual jet. A viscosity–duality solution is both a subsolution and a supersolution and satisfies the terminal condition after pairing against every admissible density. On homogeneous branches $\mathcal{A}_{x_0, P_\Psi}^{\text{fr}}$ is the first-order deterministic branch generator on the universal billiard bundle with its finite-response port evaluated at the pointwise complete cotangent field $P_\Psi(z)$. When P_Ψ is constant this is the ordinary fixed-table generator $\mathcal{A}_{\Theta(x_0, P_\Psi)}$; for non-constant finite-response P_Ψ it is defined by the same branch vector field, reflection law, and moving-boundary weak terms used in Proposition 2.22. Its Taylor expansion at a frozen contact jet is (255). The definition therefore matches the exact branchwise action equation of Theorem 3.14 without pretending that a z -dependent parameter is a single fixed billiard table.

Definition 7.17 (Paired anisotropic order and semicontinuity). Let U and V be observable-valued sections on a macroscopic cylinder. We write $U \preceq_{\#} V$ if, for every compact cotangent window K , every finite admissibility constant M , every (t, x) , every $p \in K$, and every positive unit density $\rho \in \text{Adm}_{M, K}(x)$ transported to the fiber $\Theta(x, p)$,

$$(250) \quad \langle U_{\Theta(x, p)}(t, x, \cdot) - V_{\Theta(x, p)}(t, x, \cdot), \rho \rangle \leq 0.$$

Upper semicontinuity of U in the paired topology means that every transported pairing

$$(251) \quad (t, x, p, \rho) \longmapsto \langle U_{\Theta(x, p)}(t, x, \cdot), \rho \rangle$$

has the usual upper semicontinuity on compact windows after the Banach-bundle identification; lower semicontinuity is defined analogously. The order is strong enough to distinguish observable sections modulo the canonical zero-measure singular set because stable-leaf product boxes with smooth positive densities form a differentiation basis in each homogeneous component.

Lemma 7.18 (Density class is sufficient for comparison). *If $U \not\leq_{\#} V$, then there is a compact window K , a finite constant M , a regular homogeneous product box, and a density $\rho \in \text{Adm}_{M,K}$ supported in that box for which the paired difference is positive. Conversely, if all such localized admissible pairings are non-positive, then $U \leq_{\#} V$ and $U \leq V$ at Liouville-a.e. regular microscopic state whenever both sections satisfy the branchwise oscillation bound of Lemma 7.27.*

Proof. The positive cone of admissible densities contains normalized smooth densities on every stable-unstable product box inside a homogeneous component, with uniform anisotropic norm after the strip-tail truncation. If a paired inequality fails, weak compactness of the admissible family and the Hahn–Banach separation of the ordered dual cone produce a positive branchwise test density; regularizing it along stable leaves and cutting off the high-strip tail gives an element of some $\text{Adm}_{M,K}$ with the same sign. The converse is immediate for pairings. For pointwise separation, let the product boxes shrink to a Liouville density point and use the branchwise oscillation estimate to pass from box averages to point values. The only excluded points are the iterated grazing and multiple-collision singularities, which have zero Liouville–Birkhoff measure by the geometric trace ledger. $\square$

Lemma 7.19 (Transport of one density in the doubled comparison). *In the doubled-variable comparison proof, one admissible density may be used in the upper and lower inequalities after transporting it through the moving Banach-bundle charts. The transport error is bounded by*

$$(252) \quad C_M(|x - y| + |p_x - p_y|) + o_{\text{strip}}(1),$$

where $o_{\text{strip}}(1)$ is the high-strip tail sent to zero after the ordinary homogeneity truncation. The specular collision faces contribute no boundary measure, and the grazing and branch-change faces contribute only weak trace terms controlled by C_{tr} after Lemma 2.13 has already been proved.

Proof. Pull both microscopic fibers to the same reference bundle. The finite-response trivialization is C^{r-4} in (x, p) , so the push-forward of a density in Adm_M changes by at most $C_M(|x - y| + |p_x - p_y|)$ in the weak anisotropic norm. On physical collision faces the incoming and outgoing traces cancel by the specular Green identity and preservation of Liouville flux. On artificial homogeneity boundaries, grazing curves, and branch-change curves, the trace is a stable-leaf functional whose norm is bounded by the derived trace constant and whose high-strip tail is summable. Thus using the transported copy of the same density in both viscosity inequalities changes the doubled comparison only by a quantity that vanishes in the comparison limit. $\square$

Lemma 7.20 (Full-gradient Taylor ledger at a viscosity contact). *At a normalized contact with frozen macroscopic jet (p, X) and first corrector χ_p , the complete cotangent field generated by the perturbed test is*

$$(253) \quad P^\varepsilon(z) = p + \varepsilon \left(D_x^{\text{fr}} \chi_p(z) + X^T \chi(z) \right) + O_{\widehat{B}^\#}(\varepsilon^2).$$

The order-one expansion of the singular prelimit operator is

$$(254) \quad \begin{aligned} & -\varepsilon^{-2} \mathcal{A}_{x,P^\varepsilon}^{\text{fr}}(\varepsilon \chi_p + \varepsilon^2 \chi^{(2)}) - \varepsilon^{-1} b(x, P^\varepsilon, z) \cdot P^\varepsilon \\ & = -\varepsilon^{-1} (\mathcal{A}_\theta \chi_p + b(x, p, z) \cdot p) - \mathcal{A}_\theta \chi^{(2)} - \mathfrak{W}_{x,p}(D_x^{\text{fr}} \chi_p + X^T \chi) + O_{\widehat{\mathcal{B}}^\#, w}(\varepsilon). \end{aligned}$$

Thus no singular term of the form $\varepsilon^{-1} D_p \chi X$ and no uncentered connection mean remains. The $D_p \Theta$, $D_p b$, moving-boundary, and horizontal-work contributions are exactly the components of the complete work observable $\mathfrak{W}_{x,p}$.

Proof. The contact-frozen corrector is differentiated with p held fixed in the horizontal Banach-bundle chart. The slow Hessian of the macroscopic test acts only on the macroscopic factor multiplying the first corrector and therefore produces $X^T \chi$. Since the coefficients of the prelimit equation are evaluated at the complete gradient, the variation of the table, drift, and moving singularities must be expanded at this complete field. The universal bundle interpretation in Proposition 3.16 and the moving-boundary response formula give the weak Taylor expansion. The first parenthesis in (254) vanishes by the first cell equation. The remaining order-one term is the complete covariant work response; its invariant projection is kept in H and D , and its zero-mean part is inverted by the second deterministic cell problem. $\square$

Proposition 7.21 (Exact covariant full-gradient closure of the branchwise equation). *The viscosity–duality operator in (249) is the anisotropic closure of the branchwise deterministic Hamilton–Jacobi equation (70); it is not a different prelimit problem. The microscopic corrector contributes to the complete test gradient through (248). Coefficients depending on the cotangent port are evaluated by the finite-response branch operator on the universal billiard bundle, and this operator has an explicit first variation at every frozen contact jet.*

More precisely, fix (x, p) , write $\theta = \Theta(x, p)$, and let q be a branchwise finite-response cotangent field. If $P_\eta(z) = p + \eta q(z)$, then in the trivialization of Definition 2.21

$$(255) \quad \mathcal{A}_{x,P_\eta}^{\text{fr}} = \mathcal{A}_\theta + \eta(\nabla_\theta \mathcal{A}_\theta)[D_p \Theta(x, p)q] + O_{\widehat{\mathcal{B}}^\#, w}(\eta^2),$$

uniformly for q in bounded finite-response module families. Define the raw coefficient-response part $\mathcal{R}_{x,p}(q)$, the complete work response $\mathfrak{W}_{x,p}(q)$, and its invariant projection $\Lambda_{x,p}(q)$ as in Definition 7.1. Then

$$(256) \quad \mathfrak{W}_{x,p}(q) = \Lambda_{x,p}(q) + \mathfrak{W}_{x,p}^\circ(q), \quad \int \mathfrak{W}_{x,p}^\circ(q) d\mu_\theta = 0.$$

Consequently the reduced deterministic observable resolvent gives the unique zero-mean tangent corrector

$$(257) \quad \mathcal{T}_{x,p}(q) = \chi_\theta[\mathfrak{W}_{x,p}^\circ(q)], \quad \mathcal{A}_\theta \mathcal{T}_{x,p}(q) = -\mathfrak{W}_{x,p}^\circ(q), \quad \int \mathcal{T}_{x,p}(q) d\mu_\theta = 0.$$

Thus no centering is asserted for the raw connection term alone. The invariant part of the complete z -dependent cotangent response is kept as the coefficient $\Lambda_{x,p}(q)$; only the projected zero-mean part is inverted by the cell resolvent.

Proof. On a homogeneous flow box of the universal bundle, the fast vector field is the ordinary billiard vector field for the table whose parameter is read by the finite-response port. For a branchwise smooth observable F ,

$$(258) \quad \mathcal{A}_{x,P_\eta}^{\text{fr}} F(z) = F_{\Theta(x, P_\eta(z))}(z) \cdot D_z^{\text{br}} F(z)$$

with the specular reflection jump relation imposed at the moved boundary. Pulling the branch to the fixed fiber over θ by Lemma 2.20 and Taylor expanding the vector field, the roof, the reflection map, and the weak trace terms gives (255). The expansion is strong-to-weak: interior terms are branchwise C^{r-4} , while derivatives of moving singular indicators are normal-velocity trace functionals; the latter are summable over homogeneity strips by Lemma 4.7. Hence a non-constant $P(z)$ defines one deterministic first-order operator on the universal bundle, not a family of unrelated fixed billiards.

Insert the complete cotangent field $P_\eta = p + \eta q$ into the singular part of the branchwise Hamilton–Jacobi operator and expand at $\eta = 0$:

$$(259) \quad \begin{aligned} & -\eta^{-2} \mathcal{A}_{x, P_\eta}^{\text{fr}}(\eta \chi_p + \eta^2 F) - \eta^{-1} b(x, P_\eta, \cdot) \cdot P_\eta \\ & = -\eta^{-1} (\mathcal{A}_\theta \chi_p + b(x, p, \cdot) \cdot p) - \mathcal{A}_\theta F - \mathfrak{W}_{x, p}(q) + O_{\widehat{\mathcal{B}}^{\#}, w}(\eta). \end{aligned}$$

The first parenthesis vanishes by the first cell equation. The order-one observable is the complete work response $\mathfrak{W}_{x, p}(q)$: its physical part is $b \cdot q$, and its finite-response connection part is the derivative of the table, the centered drift, and the moving Banach-bundle trivialization.

The split (256) is the deterministic projection onto constants and the zero-mean observable subspace. The zero-mean component is in the domain of the reduced resolvent by Theorem 6.2, which gives (257). Taking $F = \mathcal{T}_{x, p}(q)$ in (259) cancels the entire zero-mean full-gradient response and leaves exactly the scalar coefficient $\Lambda_{x, p}(q)$.

At a viscosity contact the true increment is $q = Q_x \chi_p + X^T \chi$. By the definitions of H_{raw} and D and by the Hessian-sector reciprocity (192),

$$(260) \quad \int V(x, p, z) d\mu_\theta + \Lambda_{x, p}(q) = H_{\text{raw}}(x, p) + \text{Tr}(D(x, p)X).$$

After subtracting the zero-energy gauge this is $H(x, p) + \text{Tr}(D(x, p)X)$. Therefore the prelimit branch equation closes at the exact full gradient: all singular powers are cancelled by deterministic cell resolvents, while all invariant parts are retained in the displayed effective coefficients. The proof uses only deterministic transfer-resolvent and viscosity-stability estimates; no probabilistic limiting-law argument or hidden centering assumption enters. $\square$

Proposition 7.22 (Prelimit well-posedness from deterministic Hamilton–Jacobi theory). *For every $\varepsilon > 0$ and every $\phi \in C(\mathcal{M})$, (237) has a unique bounded viscosity–duality solution. It is the anisotropic completion of the branchwise deterministic Hamilton principal function in Definition 3.11.*

Proof. We give the deterministic comparison and existence argument because it is the point where possible circularity must be excluded.

Branchwise comparison. In a homogeneous flow box and after a local Banach-bundle trivialization, the prelimit operator is a continuous first-order Hamilton–Jacobi operator in the variables (t, x, z) with Hamiltonian evaluated on the test jet (p, Ψ_0) . The finite-response bounds make the Hamiltonian locally Lipschitz in the jet variables and bounded in the microscopic coefficients. The usual doubling argument for deterministic viscosity solutions applies in that box. The cotangent variable used in the coefficient is the cotangent component of the test jet; no value of a macroscopic limiting solution appears.

Pasting across singular hypersurfaces. If a doubled maximum approaches a singular curve, the singular curve is a codimension-one flux-null set. Cutting the box by the homogeneity partition produces boundary terms supported on moving singular curves. When these terms are paired with $\rho \in \text{Adm}_M(\theta)$ they are bounded by the weak anisotropic norm, and the

high-strip tail is summable by Lemma 4.7. Hence no positive boundary mass is created by grazing or multiple-collision curves. The branchwise comparison inequalities therefore paste to a comparison inequality in anisotropic duality.

Existence. The deterministic graph transform of Lemma 3.10 gives branchwise classical action solutions for smooth terminal payoffs and regularized data. Stability of viscosity solutions under monotone upper and lower envelopes gives Perron solutions for continuous terminal payoffs. The comparison just proved makes the Perron envelope unique. Completing the branchwise solutions in the observable Banach bundle gives exactly the normalized dual inequalities (249). Only the deterministic graph transform and viscosity stability are used. $\square$

Lemma 7.23 (Synchronized admissible density in doubled variables). *In the doubled-variable comparison for the prelimit viscosity–duality equation, one may use a single admissible microscopic density after transporting it through the Banach-bundle charts of the two contact fibers. More precisely, if (x, p, Ψ_U) and (y, p, Ψ_V) are the two normalized contact jets produced by the doubling argument and if $\rho_x \in \text{Adm}_M(\Theta(x, p + D_x^{\text{hor}}\Psi_U))$, then its transported density*

$$(261) \quad \rho_y = \mathfrak{T}_{\Theta(y, p + D_y^{\text{hor}}\Psi_V) \leftarrow \Theta(x, p + D_x^{\text{hor}}\Psi_U)}^{\text{den}} \rho_x$$

belongs to $\text{Adm}_{M_1}(\Theta(y, p + D_y^{\text{hor}}\Psi_V))$, with M_1 depending only on M , the compact jet window, and the finite-response bounds. Moreover all generator, drift, potential, and boundary-trace pairings differ by the structural modulus

$$(262) \quad C(|x - y| + |D_x^{\text{hor}}\Psi_U - D_y^{\text{hor}}\Psi_V|_{\mathcal{B}^{\#}, w})$$

after the common macroscopic Ishii penalty has been fixed.

Proof. The density transport is the positive cone preserving map of Definition 2.21. Its norm is uniformly bounded on compact finite-response parameter windows by Proposition 2.22, so unit mass and positivity are preserved and the admissible constant changes only from M to a deterministic M_1 . Interior generator terms are compared by (40); drift and potential terms are compared by the finite-response Lipschitz bounds. If a singular boundary is crossed in passing from one fiber to the other, the discrepancy is exactly a moving-boundary trace term of the form in Lemma 6.5, and the sum of such terms is bounded in the weak norm by (156). This proves (262). Hence the use of one synchronized density in the paired comparison is an anisotropic transport identity, not an unproved order assumption. $\square$

Theorem 7.24 (Full comparison and Perron theorem for viscosity–duality solutions). *For every fixed $\varepsilon > 0$ the prelimit equation (237), interpreted by Definition 7.16, satisfies the following deterministic well-posedness package.*

- (a) *If U is a bounded upper semicontinuous subsolution and V is a bounded lower semicontinuous supersolution with $U(T, \cdot, \cdot) \leq V(T, \cdot, \cdot)$ after pairing against every admissible density, then $U \leq V$ in the paired anisotropic order.*
- (b) *The Perron envelope of bounded subsolutions with terminal datum $\phi \in C(\mathcal{M})$ is the unique viscosity–duality solution.*
- (c) *The solution operator is monotone, constant preserving after the zero-energy gauge, and locally Lipschitz in the terminal datum in the paired sup norm.*

The proof is a deterministic viscosity comparison argument on the macroscopic variables coupled to the closed microscopic branch generator of Proposition 3.13. It does not use a stochastic representation or any limiting-law input.

Proof. We record the comparison proof in the paired order because this is the form used in the homogenization argument. Suppose that the paired maximum of $U - V$ is positive. For $\alpha, \delta > 0$ consider

$$(263) \quad \mathcal{M}_{\alpha, \delta} = \sup_{t, s, x, y, \rho} \left\{ \langle U(t, x, \cdot) - V(s, y, \cdot), \rho \rangle - \frac{|x - y|^2}{2\alpha} - \frac{|t - s|^2}{2\alpha} - \delta(|x|^2 + |y|^2) \right\},$$

where ρ ranges over a fixed bounded admissible family transported through the Banach-bundle charts. Compactness of the torus and weak compactness of bounded admissible families give maximizing triples after the standard upper/lower semicontinuous relaxation. The terminal inequality and the time penalty force the contact to occur at interior times for small α if the positive maximum exists.

At such a contact, the Crandall–Ishii matrix inequality in the macroscopic variables gives jets (a, p, X) for U and (b, p, Y) for V with

$$(264) \quad a - b \geq o_\alpha(1), \quad -\frac{3}{\alpha} \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} \leq \begin{pmatrix} X & 0 \\ 0 & -Y \end{pmatrix} \leq \frac{3}{\alpha} \begin{pmatrix} I & -I \\ -I & I \end{pmatrix}.$$

The microscopic component of the test is normalized as in Definition 7.15; its value at the contact is subtracted but its horizontal derivative remains in the complete cotangent field. Because $\mathcal{A}_{x, P}^{\text{fr}}$ has a closed graph and satisfies the Green identity (66), the microscopic density in the upper inequality is synchronized with the one in the lower inequality by Lemma 7.23. Thus the two pairings are compared in a common transported fiber with a deterministic modulus. The specular boundary terms cancel, while the singular boundary terms are the explicit moving-boundary functionals of Lemma 6.5; their high-strip tails vanish in the comparison limit.

Subtracting the subsolution and supersolution inequalities gives

$$(265) \quad \begin{aligned} 0 \leq & -(a - b) + C(|x - y| + |t - s|) + C|P_U - P_V| \\ & + \omega \left(|x - y| + \frac{|x - y|^2}{\alpha} \right) + o_\alpha(1) + o_\delta(1), \end{aligned}$$

where P_U and P_V are the complete cotangent fields of the two normalized tests and ω is the modulus obtained from the finite-response bounds and from the generator covariance estimate (40). At the doubled contact the normalized microscopic gauges have the same horizontal derivative up to $o_\alpha(1)$ in the weak norm, hence $|P_U - P_V| \rightarrow 0$. Letting $\alpha \downarrow 0$ and then $\delta \downarrow 0$ contradicts the assumed positive maximum. This proves comparison.

For Perron’s method, the deterministic graph-transform solutions with smooth terminal data give nonempty sub- and supersolution classes. The supremum of subsolutions has an upper semicontinuous envelope that is a subsolution by the standard viscosity stability argument applied to the normalized dual jets; the lower envelope is a supersolution by the same bumping argument. Comparison identifies the two envelopes. Monotonicity and Lipschitz dependence on terminal data follow by applying comparison to $U \pm \|\phi - \psi\|_\infty$ after the zero-energy gauge, which makes constants exact solutions. Every operation in this proof is deterministic: doubling of variables, Banach-bundle transport, Green’s identity, and Perron stability. $\square$

Lemma 7.25 (Deterministic residual identity for the contact-frozen perturbed test). *Let $\varphi \in C^4$ be localized near a strict contact point (t_0, x_0) of a half-relaxed limit. Let $(t_\varepsilon, x_\varepsilon)$ be*

the corresponding prelimit contact point and set

$$(266) \quad p_\varepsilon = D_x \varphi(t_\varepsilon, x_\varepsilon), \quad X_\varepsilon = D_x^2 \varphi(t_\varepsilon, x_\varepsilon), \quad \theta_\varepsilon = \Theta(x_\varepsilon, p_\varepsilon).$$

Write

$$(267) \quad \chi_{p_\varepsilon} = \sum_i p_{\varepsilon,i} \chi_i(x_\varepsilon, p_\varepsilon, \cdot), \quad \mathcal{A}_{\theta_\varepsilon} \chi_{p_\varepsilon} = -b(x_\varepsilon, p_\varepsilon, \cdot) \cdot p_\varepsilon.$$

The complete gradient of the normalized first corrector is the actual chain-rule quantity obtained from (206). With

$$(268) \quad \chi_\varepsilon(z) = (\chi_1(x_\varepsilon, p_\varepsilon, z), \dots, \chi_m(x_\varepsilon, p_\varepsilon, z))^T,$$

it is

$$(269) \quad q_\varepsilon(z) = D_x^{\text{fr}} \chi_{p_\varepsilon}(x_\varepsilon, p_\varepsilon, z) + X_\varepsilon^T \chi_\varepsilon(z).$$

Equivalently, the j th component is

$$(270) \quad (q_\varepsilon)_j = \partial_{x_j}^{\text{fr}} \chi_{p_\varepsilon} + \sum_{i=1}^m X_{\varepsilon,ij} \chi_i.$$

No derivative of the coefficient family with respect to p is part of this chain rule. The p -dependence of the table and of the centered drift is combined with the physical Euler work $b \cdot q_\varepsilon$ into the complete covariant work response

$$(271) \quad \mathfrak{W}_\varepsilon(z) = \mathfrak{W}_{x_\varepsilon, p_\varepsilon}(q_\varepsilon)(z)$$

of Definition 7.1. Define the full-gradient order-one observable by

$$(272) \quad \mathcal{G}_\varepsilon^\varphi(z) = V^\circ(x_\varepsilon, p_\varepsilon, z) + \mathfrak{W}_\varepsilon(z).$$

The zero-mean part of $\mathfrak{W}_\varepsilon$ is included in the second cell datum. Its invariant part is retained. Since $q_\varepsilon = Q_x \chi_{p_\varepsilon} + X_\varepsilon^T \chi_\varepsilon$, (201) and (192) give the exact effective average. Therefore

$$(273) \quad \overline{\mathcal{G}}_\varepsilon^\varphi = \int \mathcal{G}_\varepsilon^\varphi d\mu_{\theta_\varepsilon} = \text{Tr}(D(x_\varepsilon, p_\varepsilon) X_\varepsilon) + H(x_\varepsilon, p_\varepsilon).$$

Let

$$(274) \quad \tilde{\mathcal{G}}_\varepsilon^\varphi = \mathcal{G}_\varepsilon^\varphi - \overline{\mathcal{G}}_\varepsilon^\varphi, \quad \chi_\varepsilon^{(2)} = \chi_{\theta_\varepsilon}[\tilde{\mathcal{G}}_\varepsilon^\varphi],$$

so that

$$(275) \quad \mathcal{A}_{\theta_\varepsilon} \chi_\varepsilon^{(2)} = -\tilde{\mathcal{G}}_\varepsilon^\varphi, \quad \int \chi_\varepsilon^{(2)} d\mu_{\theta_\varepsilon} = 0.$$

The normalized microscopic gauge is

$$(276) \quad \Psi^\varepsilon(t, x, z) = \varepsilon \Xi_\varepsilon[\varphi](t, x, z) + \varepsilon^2 \chi_\varepsilon^{(2)}(z), \quad \Psi^{\varepsilon, \circ}(t, x, z) = \Psi^\varepsilon(t, x, z) - \Psi^\varepsilon(t_\varepsilon, x_\varepsilon, z),$$

where Ξ_ε is the contact-jet-frozen first corrector from (206). For every bounded admissible family $\rho \in \text{Adm}_M(\theta_\varepsilon)$, inserting $\varphi + \Psi^{\varepsilon, \circ}$ into the prelimit dual operator gives the identity

$$(277) \quad -\partial_t \varphi(t_\varepsilon, x_\varepsilon) - \text{Tr}(D(x_\varepsilon, p_\varepsilon) X_\varepsilon) - H(x_\varepsilon, p_\varepsilon) + \langle R^\varepsilon[\varphi], \rho \rangle$$

for the value of the left-hand side of (249). The residual satisfies

$$(278) \quad |\langle R^\varepsilon[\varphi], \rho \rangle| \leq C_M \left(\varepsilon + |t_\varepsilon - t_0| + |x_\varepsilon - x_0| + |p_\varepsilon - p_0| + |X_\varepsilon - X_0| \right),$$

where $p_0 = D_x \varphi(t_0, x_0)$ and $X_0 = D_x^2 \varphi(t_0, x_0)$. In particular, neither an $\varepsilon^{-1} |p_\varepsilon - p_0|$ term nor an unaccounted chain-rule contribution $D_p \chi D_x^2 \varphi$ appears. The only Hessian dependence

in the physical gradient is the controlled vector $X_\varepsilon^T \chi$ displayed in (269); any $D_p \chi_p[X_\varepsilon^T \chi]$ term is an exact response primitive absorbed by (274).

Proof. All expansions are made at the actual contact jet $(x_\varepsilon, p_\varepsilon, X_\varepsilon)$ after transporting nearby fibers to the Banach fiber over θ_ε . This order is essential: the singular terms are cancelled before one lets the contact point converge.

The first singular term is

$$(279) \quad -\varepsilon^{-1} b(x_\varepsilon, p_\varepsilon, z) \cdot p_\varepsilon - \varepsilon^{-2} \mathcal{A}_{\theta_\varepsilon}(\varepsilon \chi_{p_\varepsilon})(z),$$

which is identically zero by the first cell equation. Thus no $\varepsilon^{-1}(p_\varepsilon - p_0)$ term can be produced.

The complete-gradient dependence of the coefficient fields is handled by Proposition 7.21. The expansion is made with the true increment

$$(280) \quad q_\varepsilon = D_x \Xi_\varepsilon(t_\varepsilon, x_\varepsilon, \cdot),$$

identified in (269). Thus the order-one response is the complete work observable $\mathfrak{W}_\varepsilon$ in (271). Its zero-mean projection is cancelled by the tangent corrector (257); its invariant projection is retained. No step asserts that the raw coefficient connection or the physical work term is separately centered for a z -dependent increment.

After the first cancellation and the response split, the order-one part is (272). Its average is (273): the horizontal component of $\Lambda_{x,p}(q_\varepsilon)$ gives the normalized Hamiltonian, and the symmetric Hessian component gives the Green–Kubo contraction through (192). The second corrector solves (274), so the centered order-one observable is cancelled by the deterministic observable resolvent.

It remains to bound the terms not displayed in the principal identity. They are of five deterministic types:

source	analytic form	bound
smooth test Taylor remainder	third and fourth order remainders of φ near $(t_\varepsilon, x_\varepsilon)$	$C(t_\varepsilon - t_0 + x_\varepsilon - x_0 + p_\varepsilon - p_0 + X_\varepsilon - X_0)$
contact-frozen first corrector	$\varepsilon \partial_t \Xi_\varepsilon$ and horizontal products with finite-response observables	$C_M \varepsilon$
second corrector after cancellation	$\varepsilon b \cdot D_x^{\text{hor}} \chi_\varepsilon^{(2)}$ and $\varepsilon^2 \partial_t \chi_\varepsilon^{(2)}$	$C_M \varepsilon$
moving singular curves	weak boundary distributions from changing homogeneous charts	$C_M \varepsilon$ after strip-tail summation
parameter continuity	replacing the actual contact jet by the limiting jet only after cancellation	$C(x_\varepsilon - x_0 + p_\varepsilon - p_0 + X_\varepsilon - X_0)$.

The product and horizontal-derivative estimates are Lemma 5.5; the moving-boundary estimates are Proposition 2.22 and Lemma 4.7; and the resolvent bound for $\chi_\varepsilon^{(2)}$ is Theorem 6.2. Summing the displayed bounds proves (278). The proof is a deterministic branchwise expansion plus transfer-resolvent estimates; no stochastic limiting-law input is used. $\square$

Proposition 7.26 (Expanded deterministic scale ledger for the perturbed test). *In the contact-frozen perturbed-test expansion, every term of order ε^{-2} , ε^{-1} , 1, and ε is accounted*

for by the following deterministic ledger:

$$\begin{aligned}
 (281) \quad & \varepsilon^{-2} : \mathcal{A}\varphi = 0 \quad (\varphi \text{ is independent of } z), \\
 & \varepsilon^{-1} : \mathcal{A}\chi_p + b \cdot p = 0 \quad \text{by the first cell equation,} \\
 & 1 : \mathfrak{W} + V^\circ \quad \text{has mean } \text{Tr}(DX) + H, \\
 & \quad \text{and its centered part is solved by the second cell equation,} \\
 & \varepsilon : \text{Taylor, horizontal, roof, and trace remainders,} \\
 & \quad \text{bounded by product estimates, response bounds, and strip summability.}
 \end{aligned}$$

Here the order-one line uses the full slow gradient $P = p + \varepsilon q + O(\varepsilon^2)$ with $q = D_x^{\text{hor}} \chi_p + X^T \chi$. Therefore the expansion contains no uncontrolled ε^{-1} coefficient mismatch and no omitted $D_p \chi X$ chain-rule contribution.

Proof. The prelimit branch generator is evaluated at the physical covector of the perturbed test, not at a later limiting covector. Since the macroscopic test has no fast variable, the fast generator annihilates it at scale ε^{-2} . The first corrector is frozen at the contact jet and solves $\mathcal{A}\chi_p = -b \cdot p$, which cancels every scale- ε^{-1} term exactly. The derivative of the first corrector with respect to the slow variable is the horizontal derivative plus the Hessian contribution $X^T \chi$; the dependence of the table, drift, potential, and projector on p is differentiated by the closed response chain rule (176). Grouping those terms gives the complete covariant work observable. Its invariant projection is computed by single-action reciprocity and the deterministic correlation identity, yielding $\text{Tr}(DX) + H$. Its centered projection is in the range of the generator by Proposition 6.12, hence is cancelled by the second cell corrector. The terms one scale lower contain at least one explicit factor of ε after the two cancellations. Their pairings with admissible densities are bounded by the product estimate, response estimate, and moving-trace estimate already proved in the anisotropic calculus. This gives the residual bound in (278). $\square$

Lemma 7.27 (Uniform deterministic oscillation modulus on homogeneous rectangles). *Let u^ε be the prelimit viscosity–duality solution with continuous terminal datum. After excluding the canonical iterated singular set, every homogeneous standard rectangle R in the suspension has a deterministic modulus ω_R and a number $r_\varepsilon \rightarrow 0$ such that for all subrectangles $R' \subset R$ and all compact macroscopic cylinders,*

$$(282) \quad \text{osc}_{R'}(u^\varepsilon(t, x, \cdot) - u(t, x)) \leq \omega_R(\text{diam } R') + r_\varepsilon,$$

locally uniformly in (t, x) . The modulus is obtained from deterministic branchwise Hamilton–Jacobi comparison and the strip-adapted distortion bounds; it is not an additional assumption in the pointwise upgrade.

Proof. On a fixed homogeneous flow box the graph-transform construction gives branchwise C^1 bounds for the action representative in the rescaled billiard coordinates. The constants depend on the finite-response bounds and on the finite-time branch distortion, and the latter is controlled by Lemma 4.6 and Theorem 4.9. When a subrectangle crosses a later singular boundary it is subdivided; the descendants in high strips have total terminal contribution bounded by (95). Choosing the strip cutoff first and then the rectangle diameter gives the modulus ω_R . The viscosity completion is obtained as the monotone stability envelope of smooth branchwise action solutions, so the same modulus passes to u^ε by deterministic comparison. Since the macroscopic limit u is continuous, subtracting $u(t, x)$ does not change

the microscopic oscillation. The error r_ε is the vanishing tail from the terminal children and from the regularized branchwise approximation. All steps are measure-theoretic or deterministic viscosity estimates on the billiard phase space. $\square$

Lemma 7.28 (Standard-family upgrade from densities to Liouville-a.e. states). *Let $F_\varepsilon(z)$ be a uniformly bounded family of branchwise anisotropic observables on the billiard suspension. Suppose that for every finite admissible constant M ,*

$$(283) \quad \sup_{\rho \in \text{Adm}_M(\theta)} |\langle F_\varepsilon, \rho \rangle| \longrightarrow 0,$$

and that on each homogeneous standard rectangle the oscillation of F_ε on nested subrectangles is controlled by a deterministic modulus which tends to zero with the rectangle diameter, uniformly in ε up to an error $r_\varepsilon \rightarrow 0$. Then, after discarding the canonical zero-measure set of singular orbits, $F_\varepsilon(z) \rightarrow 0$ for Liouville-a.e. microscopic state. The convergence is sequence-free; it is the ordinary limit as $\varepsilon \downarrow 0$.

Proof. Disintegrate the suspension Liouville measure into countably many homogeneous standard rectangles subordinate to unstable leaves and roof intervals. For each fixed rectangle R choose a C^1 stable-smoothed density ρ_R which is equal to the normalized indicator of R except in a boundary layer of arbitrarily small conditional measure. Its anisotropic norm is finite; it may be large, but it belongs to Adm_M for some finite $M = M(R)$. The hypothesis therefore gives convergence of the averages of F_ε over every member of a countable determining algebra of rectangles.

If $\limsup_{\varepsilon \downarrow 0} |F_\varepsilon| > \eta$ on a set of positive conditional measure in some standard rectangle, the leafwise differentiation basis gives a smaller algebra rectangle on which the average of either the positive or the negative part is at least $\eta/2$, after excluding a boundary layer. The smoothed density of that fixed rectangle has finite admissible norm, contradicting the convergence of pairings for its value of M . Taking a countable intersection over rational $\eta > 0$ gives a.e. convergence. The iterated grazing and multiple-collision sets have zero Liouville measure by the finite-horizon singularity stratification. This is a deterministic density basis argument; no sampling argument or external limiting law is involved. $\square$

Lemma 7.29 (Deterministic leafwise extraction from admissible densities). *Let $G^\varepsilon(t, x, z)$ be a family of microscopic observables obtained from solutions of the prelimit Hamilton–Jacobi equation after subtracting a macroscopic continuous limit. Suppose that, for every $M < \infty$,*

$$(284) \quad \sup_{\rho \in \text{Adm}_M(\theta)} |\langle G^\varepsilon(t, x, \cdot), \rho \rangle| \longrightarrow 0$$

locally uniformly in (t, x) , and that on every homogeneous standard rectangle R the branchwise oscillation satisfies

$$(285) \quad \text{osc}_{R'} G^\varepsilon(t, x, \cdot) \leq C \text{diam}(R')^\alpha + r_\varepsilon$$

for all subrectangles $R' \subset R$, with $r_\varepsilon \rightarrow 0$ outside the iterated singular set. Then $G^\varepsilon(t, x, z) \rightarrow 0$ for Liouville-a.e. microscopic state z , locally uniformly along every countable dense family of macroscopic points (t, x) , and hence for the viscosity limit after the usual modulus-of-continuity interpolation.

Proof. For each point of the dense macroscopic family, apply Lemma 7.28 to $F_\varepsilon(z) = G^\varepsilon(t, x, z)$. The hypotheses of that lemma are exactly (284) and (285). A countable intersection of full-measure microscopic sets gives convergence for the whole dense family. The deterministic

equicontinuity of the viscosity envelopes in (t, x) , obtained from the branchwise comparison estimate in Proposition 7.22, extends the convergence to compact subsets of the macroscopic cylinder. The argument is disintegration, density-basis differentiation, and deterministic oscillation control. $\square$

Lemma 7.30 (Contact selection for normalized frozen gauges). *Let $\varphi \in C^3$ touch an upper half-relaxed limit strictly at (t_0, x_0) . For each sufficiently small ε there exist a prelimit contact point $(t_\varepsilon, x_\varepsilon) \rightarrow (t_0, x_0)$ and an admissible density ρ_ε such that the normalized contact-jet-frozen perturbed test*

$$(286) \quad \varphi(t, x) + \Psi^{\varepsilon, \circ}(t, x, z), \quad \Psi^{\varepsilon, \circ}(t, x, z) = \Psi^\varepsilon(t, x, z) - \Psi^\varepsilon(t_\varepsilon, x_\varepsilon, z),$$

with Ψ^ε defined in (276), belongs to the upper normalized dual jet of u^ε at $(t_\varepsilon, x_\varepsilon; \rho_\varepsilon)$. The analogous statement holds for lower contacts.

Proof. Make the macroscopic contact strict by adding $\eta(|t - t_0|^2 + |x - x_0|^4)$ and later letting $\eta \downarrow 0$. On a compact cylinder the half-relaxed definition gives maximizing triples $(t_\varepsilon, x_\varepsilon, \rho_\varepsilon)$ for the paired quantity $\langle u^\varepsilon - \varphi, \rho \rangle$, up to $o(1)$; strictness forces $(t_\varepsilon, x_\varepsilon) \rightarrow (t_0, x_0)$.

Define the frozen parameters from the actual maximizing point:

$$(287) \quad p_\varepsilon = D_x \varphi(t_\varepsilon, x_\varepsilon), \quad X_\varepsilon = D_x^2 \varphi(t_\varepsilon, x_\varepsilon).$$

Solve the first and second cell equations with these parameters. The unnormalized microscopic gauge Ψ^ε has uniformly bounded anisotropic norm after multiplication by ε and ε^2 . Subtracting its value at the contact point leaves the generator and horizontal derivative used in the PDE unchanged, but makes the contact equality automatic for every admissible density. Hence the earlier obstruction $\langle \Psi^\varepsilon(t_\varepsilon, x_\varepsilon), \rho \rangle = 0$ is not present.

The addition of the normalized gauge changes the paired maximum by $o(1)$ uniformly over bounded admissible families. A deterministic bump supported in the macroscopic cylinder converts the approximate maximum into an exact upper normalized dual contact. Singular hypersurfaces do not create boundary contacts: admissible densities have summable high-strip tails and the singular sets have zero flux measure. Reversing inequalities gives the lower-contact statement. $\square$

Lemma 7.31 (Order of quantifiers in density convergence and pointwise extraction). *In Theorem 7.33 the constants are chosen in the following order:*

$$(288) \quad (\text{geometric class, finite-response bounds}) \longrightarrow (K, M, \text{macroscopic cylinder}) \\ \longrightarrow (\text{contact jet and density}) \longrightarrow \varepsilon \downarrow 0.$$

For each fixed finite M and compact jet window K , the convergence of paired envelopes is uniform over $\text{Adm}_{M, K}$. For the Liouville-a.e. state statement, the rectangle used in the differentiation-basis argument is fixed first; its admissible norm $M(R)$ is finite but may be large, and the already proved convergence is applied with that value of M . Thus no diagonal choice of M depending on ε is used.

Proof. The residual estimate (278) is uniform on bounded admissible density balls and compact jet windows because the transfer-resolvent, product, trace, and response bounds are uniform there. This proves paired convergence for each fixed (M, K) . In the pointwise upgrade, choose a countable algebra of homogeneous rectangles. The smoothed density of a rectangle R has a definite finite anisotropic norm, so it belongs to $\text{Adm}_{M(R)}$. Applying the paired convergence with $M = M(R)$ gives the vanishing of averages on that rectangle. If

pointwise convergence failed on a positive-measure subset, the leafwise differentiation basis and the oscillation modulus would produce a fixed smaller rectangle whose average has a fixed sign, contradicting the paired limit for its own finite norm. Since the countable algebra is fixed before ε varies, the resulting a.e. convergence is the ordinary sequence-free limit. $\square$

Proposition 7.32 (Subsolution and supersolution residuals in full branch form). *Let φ touch an upper relaxed paired envelope from above and ψ touch a lower relaxed paired envelope from below. The normalized contact-frozen gauges constructed from φ and ψ satisfy, respectively,*

$$(289) \quad -\partial_t \varphi - \text{Tr}(D(x, D\varphi)D^2\varphi) - H(x, D\varphi) \leq o(1),$$

$$(290) \quad -\partial_t \psi - \text{Tr}(D(x, D\psi)D^2\psi) - H(x, D\psi) \geq -o(1),$$

where the $o(1)$ is uniform over bounded admissible density balls and compact contact-jet windows. The sign reversal is the only difference between the two halves of the perturbed-test argument; the cell equations, moving-boundary traces, and full-gradient response terms are identical.

Proof. At an upper contact, the viscosity-duality subsolution inequality is applied to $\varphi + \Psi^{\varepsilon, \circ}$. Lemma 7.25 reduces the left-hand side exactly to the effective operator plus a residual whose bound is uniform in the admissible density. Letting the contact point converge gives the displayed upper inequality. At a lower contact, the same normalized gauge is inserted into the supersolution inequality; the branchwise Hamilton–Jacobi identity and the two cell equations do not change. Only the viscosity inequality is reversed, yielding the lower inequality. The singular sets are handled in both cases by the same weak trace estimates, and no pointwise claim is made on those zero-measure hypersurfaces. $\square$

Theorem 7.33 (Coefficient identification under paired closure assumptions). *Assume the geometric inputs in assumption 2.4, the full analytic admissibility package assumption 2.5, and a specified finite-response port compatible with the single action in (A4). Let $\phi \in C(\mathcal{M})$. Then the unique viscosity–duality solution u^ε of the well-posed prelimit equation (237) converges to the unique viscosity solution u of*

$$(291) \quad -\partial_t u(t, x) - \text{Tr}(D(x, \nabla u)\nabla^2 u) - H(x, \nabla u) = 0, \quad u(T, x) = \phi(x).$$

The coefficients D and H are the deterministic coefficients in Definitions 7.3 and 7.5. The convergence is locally uniform in (t, x) after pairing against microscopic densities, in the quantitative compact-jet form

$$(292) \quad \sup_{(p, \rho) \in \text{Adm}_{M, K}(x)} \left| \langle u_{\Theta(x, p)}^\varepsilon(t, x, \cdot), \rho \rangle - u(t, x) \right| \longrightarrow 0$$

for every finite admissible constant M , every compact cotangent window $K \Subset T^*\mathcal{M}$ containing the contact gradients under consideration, and every compact macroscopic cylinder. At an actual viscosity contact the parameter p in (292) is the contact gradient; the supremum over K records uniformity of the Banach-bundle transport and does not introduce a free macroscopic coefficient. No Liouville-a.e., sequence-free pointwise convergence in the microscopic state is claimed. Such an upgrade requires an additional ε -uniform microscopic oscillation or maximal estimate. Conditional on the package, the response formula, Green–Kubo tensor, Hamiltonian, and full-gradient second cell equation are used in the forward order

$$(293) \quad \begin{aligned} &\text{geometry} \rightarrow \text{singular hyperbolicity} \rightarrow \text{anisotropic transfer} \\ &\rightarrow \text{resolvent} \rightarrow \text{cell equations} \rightarrow \text{viscosity limit}, \end{aligned}$$

and no implication in this chain invokes the macroscopic HJB equation or an external probabilistic limiting theorem. The spectral, response, coherent prelimit, and uniform closure properties themselves remain hypotheses. In particular, this theorem packages the consequences of (A5); it does not prove contact selection or residual vanishing from weaker billiard data.

Proof. The argument is a deterministic perturbed-test proof in anisotropic duality. It uses only the branchwise action equation, the transfer-resolvent cell problems, and viscosity stability.

Step 1: invoke the compactness clause of (A5). Fix M and a compact cotangent window $K \subseteq T^*\mathcal{M}$. The density bundle $\text{Adm}_{M,K}(x)$ is compact after transport to a local Banach-bundle chart by Lemma 7.12. Define the relaxed paired envelopes by

$$(294) \quad \begin{aligned} \bar{u}_K(t, x) &= \limsup_{\substack{\varepsilon \downarrow 0 \\ (s, y) \rightarrow (t, x)}} \sup_{(p, \rho) \in \text{Adm}_{M,K}(y)} \langle u_{\Theta(y,p)}^\varepsilon(s, y, \cdot), \rho \rangle_{\Theta(y,p)}, \\ \underline{u}_K(t, x) &= \liminf_{\substack{\varepsilon \downarrow 0 \\ (s, y) \rightarrow (t, x)}} \inf_{(p, \rho) \in \text{Adm}_{M,K}(y)} \langle u_{\Theta(y,p)}^\varepsilon(s, y, \cdot), \rho \rangle_{\Theta(y,p)}. \end{aligned}$$

The deterministic comparison estimate for the prelimit problem gives bounded macroscopic equicontinuity of these envelopes. Compactness of the transported density bundle prevents loss of mass or escape to grazing strips in the relaxed-limit passage. The terminal condition follows from $u^\varepsilon(T, x, z) = \phi(x)$ and is therefore independent of the microscopic density and of the window parameter.

Step 2: identify the upper effective operator. Let $\varphi \in C^4$ touch $\bar{u}_K$ strictly from above at (t_0, x_0) . By Lemma 7.30 there are prelimit contact points $(t_\varepsilon, x_\varepsilon) \rightarrow (t_0, x_0)$, admissible densities ρ_ε , and physical contact jets

$$(295) \quad p_\varepsilon = D_x \varphi(t_\varepsilon, x_\varepsilon), \quad X_\varepsilon = D_x^2 \varphi(t_\varepsilon, x_\varepsilon),$$

such that the normalized contact-frozen perturbed test is an upper dual test for u^ε . The full-gradient dual closure Proposition 7.21 ensures that this test is applied to the exact prelimit action equation, not to a modified equation.

The first corrector satisfies

$$(296) \quad \mathcal{A}_{\Theta(x_\varepsilon, p_\varepsilon)} \chi_{p_\varepsilon} = -b(x_\varepsilon, p_\varepsilon, \cdot) \cdot p_\varepsilon,$$

and cancels the ε^{-1} work term at the actual prelimit contact jet. The second corrector is the deterministic resolvent solution of (274). Therefore Lemma 7.25 gives, after insertion into the subsolution inequality,

$$(297) \quad -\partial_t \varphi(t_\varepsilon, x_\varepsilon) - \text{Tr}(D(x_\varepsilon, p_\varepsilon) X_\varepsilon) - H(x_\varepsilon, p_\varepsilon) + \langle R^\varepsilon[\varphi], \rho_\varepsilon \rangle \leq 0.$$

The residual tends to zero by (278). Passing to the limit and using Proposition 7.9 gives

$$(298) \quad -\partial_t \varphi(t_0, x_0) - \text{Tr}(D(x_0, D\varphi(t_0, x_0)) D^2 \varphi(t_0, x_0)) - H(x_0, D\varphi(t_0, x_0)) \leq 0.$$

Thus $\bar{u}_K$ is a viscosity subsolution of (291).

Step 3: identify the lower effective operator. If $\psi \in C^4$ touches $\underline{u}_K$ strictly from below, the lower-contact part of Lemma 7.30 gives lower dual contacts with the same physical frozen jets. The first and second cell equations are unchanged and the residual estimate is identical. Reversing the viscosity inequality in (297) yields

$$(299) \quad -\partial_t \psi(t_0, x_0) - \text{Tr}(D(x_0, D\psi(t_0, x_0)) D^2 \psi(t_0, x_0)) - H(x_0, D\psi(t_0, x_0)) \geq 0.$$

Thus $\underline{u}_K$ is a viscosity supersolution.

Step 4: invoke comparison and conclude paired convergence. The coefficient structure in Proposition 7.9 gives the Crandall–Ishii modulus and degenerate ellipticity, while the prelimit paired compactness and contact selection are justified by Lemma 7.12 and Theorem 7.24. The comparison theorem in Appendix A.1 therefore yields $\bar{u}_K \leq \underline{u}_K$. The opposite inequality is built into the definitions of the relaxed limits, so the two envelopes coincide with the unique viscosity solution u of (291). Since the residual constants in Lemma 7.25 are uniform over bounded admissible families and over compact contact-jet windows, the convergence is uniform in $(p, \rho) \in \mathbf{Adm}_{M,K}(x)$ and gives (292).

Step 5: scope of the conclusion. Steps 1–4 deduce only the paired convergence (292). Convergence of averages over a restricted anisotropic density class does not entail pointwise convergence, and the graph-transform bounds displayed earlier depend on ε . We therefore do not invoke the later density-basis lemmas to obtain a statewise upgrade. All compactness, contact, residual, and comparison inputs used in Steps 1–4 are precisely the assumptions collected in (A5). Thus the argument identifies the effective operator selected by those inputs; it does not derive convergence from weaker billiard data. $\square$

Theorem 7.34 (End-to-end conditional dependency theorem). *Assume (G1)–(G6), (A1)–(A5), and a specified compatible finite-response port. Then the a.e. billiard and Liouville flux are defined from geometry; (A1)–(A3) yield the resolvents, correctors, and response coefficients; and (A4)–(A5) yield the paired viscosity limit in Theorem 7.33. No statewise upgrade is included.*

Proof. This is the dependency concatenation recorded in Proposition 1.2. The optical, strip, and branch calculations give verification obligations for the analytic clauses, not proofs of map mixing, UNI, response closure, or homogenization closure. $\square$

7.3. Verification status of the analytic package. The branch and strip formulae above are a verification ledger for a concrete family. They do not prove the uniform constants, compatible operator domains, endpoint realization, contact selection, or residual bounds in (A1)–(A5). The logical content used below is only this: after those properties have been independently established, the reduced resolvents define the displayed correctors and the perturbed-test algebra identifies the coefficients in (291). No rate of convergence and no statewise microscopic limit follows from the package as stated.

7.4. Scale hierarchy.

Theorem 7.35 (Scale I: fixed-fiber homogenization). *For fixed θ and fixed (x, p) , the terminal singular perturbation generated by*

$$(300) \quad -\varepsilon^{-2} \mathcal{A}_\theta - \varepsilon^{-1} b_\theta(z) \cdot \nabla_x$$

converges to the deterministic operator

$$(301) \quad -\mathrm{Tr}(D_\theta \nabla_x^2).$$

The convergence is obtained by the first and second observable cell problems on $\widehat{\mathcal{B}}_\theta^\#$.

Proof. This is the preceding two-scale expansion with Θ , V , and all x, p dependence frozen. Since the observable resolvent estimates are uniform and the centered drift has zero invariant average, the order ε^{-1} term is an exact coboundary in the anisotropic observable sense. Averaging the order-one term gives the Green–Kubo tensor. $\square$

Theorem 7.36 (Scale II: local moving-fiber homogenization). *When $\theta = \Theta(x, p)$ varies on a compact parameter set, the fixed-fiber convergence is locally uniform in (x, p) . The limiting terminal operator is*

$$(302) \quad -\operatorname{Tr}(D(x, p)\nabla_x^2) - H(x, p).$$

Proof. Uniform hyperbolicity with singularities gives constants independent of θ . Theorem 6.4 supplies differentiability of the invariant density, correlation tensor, and corrector. Therefore the residual estimate in the two-scale expansion is uniform on compact subsets of (x, p) . $\square$

Theorem 7.37 (Scale III: global nonlinear HJB limit). *The global feedback problem with $\theta = \Theta(x, \nabla u)$ converges to (291). The Hamiltonian has the rigid form*

$$(303) \quad H_\theta(x, p, X) = \operatorname{Tr}(D(x, p)X) + H(x, p), \quad H(x, 0) = 0.$$

Proof. Combine Scale II with viscosity stability. At a contact point the parameter is evaluated at the test gradient $p = \nabla\varphi$, not at a nonexistent classical gradient of u^ε . The prelimit feedback is therefore non-circular, and the limiting equation follows from the subsolution and supersolution inequalities proved in Theorem 7.33. $\square$

7.5. Scale IV: deterministic fluctuation equation. Let

$$(304) \quad u^\varepsilon = u + \varepsilon w^\varepsilon.$$

Expanding the already derived effective Hamiltonian around $(x, \nabla u, \nabla^2 u)$ gives the deterministic linearized operator

$$(305) \quad \mathfrak{L}_u w = -\partial_t w - \operatorname{Tr}(D(x, \nabla u)\nabla^2 w) - \partial_p H(x, \nabla u) \cdot \nabla w - \sum_{i,j} \left(\partial_p D_{ij}(x, \nabla u) \cdot \nabla w \right) \partial_{ij} u.$$

At the next order, the quadratic part of the Hamiltonian produces the deterministic viscous Burgers hierarchy

$$(306) \quad \mathfrak{L}_u w - \sum_{i,j} \left(\partial_p D_{ij}(x, \nabla u) \cdot \nabla w \right) \partial_{ij} w$$

$$(307) \quad - \frac{1}{2} \sum_{i,j} \left(\nabla w^T \partial_{pp}^2 D_{ij}(x, \nabla u) \nabla w \right) \partial_{ij} u - \frac{1}{2} \nabla w^T \partial_{pp}^2 H(x, \nabla u) \nabla w = 0.$$

This equation is deterministic and is obtained by Taylor expansion of the derived HJB operator.

8. THE MICROSCOPIC ORIGIN OF MACROSCOPIC NON-CONVEXITY AND NON-SUBADDITIVITY

This section separates two algebraic facts that are often conflated. A negative Hessian direction proves local non-convexity of $p \mapsto H(x, p)$. Failure of subadditivity of the induced nonlinear expectation is instead detected by the direct Hamiltonian inequality

$$(308) \quad H(x, p + q) > H(x, p) + H(x, q)$$

for some x, p, q . The deterministic billiard construction supplies both effects by explicit microscopic potentials, but the second is not inferred merely from the first.

8.1. Perturbation of the invariant density. Let $p \mapsto \theta(p)$ be a smooth feedback curve and let the normalized potential contribution be defined by Definition 7.5. By Theorem 6.4, H is twice differentiable whenever the table and V are sufficiently smooth.

Lemma 8.1 (Second derivative formula). *Assume near $p = p_*$ that*

$$(309) \quad V(p, z) = V_0(z) + \ell(z) \cdot (p - p_*) + \frac{1}{2}(p - p_*)^T A(z)(p - p_*) + O(|p - p_*|^3),$$

and

$$(310) \quad \theta(p) = \theta_* + \Theta_1(p - p_*) + \frac{1}{2}\Theta_2[p - p_*, p - p_*] + O(|p - p_*|^3).$$

Then

$$(311) \quad \nabla_p^2 H(p_*) = \langle A \rangle_* + \mathcal{R}_\mu(V_0, \ell, \Theta_1, \Theta_2) + \nabla_p^2 H_{\text{work}}(p_*),$$

where $\mathcal{R}_\mu$ is the deterministic invariant-density response obtained by differentiating the reduced transfer resolvent, and $H_{\text{work}}(x, p) := \Lambda_{x,p}(Q_x \chi_p)$ is the complete covariant horizontal work response in Definition 7.5.

Proof. Differentiate the normalized expression for H twice. The explicit second derivative of V gives $\langle A \rangle_*$. Differentiating the invariant density gives the two response terms: one linear in the second parameter derivative Θ_2 acting on V_0 , and one quadratic in the first parameter derivative Θ_1 acting on the first and second reduced-resolvent derivatives. These are precisely the terms collected in $\mathcal{R}_\mu$. The corrector contribution is differentiated by the identity

$$(312) \quad \partial_p(-\mathcal{A}_{\Theta(x,p)})^{-1} = (-\mathcal{A})^{-1}(\partial_p \mathcal{A})(-\mathcal{A})^{-1}$$

on the moving zero-mean bundle, together with the derivatives of the centered projection. The normalization subtracts $H_{\text{raw}}(x, 0)$ and therefore contributes no Hessian at p_* unless $p_* = 0$ and the chosen gauge is moved; in the fixed gauge used here it is a deterministic additive reference. $\square$

Theorem 8.2 (Directional non-convexity criterion). *If there exist x, p_* and $v \in \mathbb{R}^m$ such that*

$$(313) \quad v^T \nabla_p^2 H(x, p_*) v < 0,$$

then $p \mapsto H(x, p)$ is not convex on any neighborhood of p_* containing the segment $p_* + [-\delta, \delta]v$ for sufficiently small $\delta > 0$. In terms of (311), condition (313) is

$$(314) \quad v^T \langle A \rangle_* v + v^T \mathcal{R}_\mu(V_0, \ell, \Theta_1, \Theta_2) v + v^T \nabla_p^2 H_{\text{work}}(p_*) v < 0.$$

Proof. For s small, Taylor's formula gives

$$(315) \quad H(x, p_* + sv) + H(x, p_* - sv) - 2H(x, p_*) = s^2 v^T \nabla_p^2 H(x, p_*) v + o(s^2).$$

If the displayed quadratic coefficient is negative, Jensen convexity along this line segment fails for all sufficiently small nonzero s . This proves local non-convexity. The formula in terms of microscopic data follows from Lemma 8.1. $\square$

Assumption 8.3 (Verified non-trivial response direction for the example). For the particular admissible billiard/port family used in the example, assume there is a C^r one-parameter

normal deformation θ_s remaining in the admissible parameter window and a branchwise C^{r-4} finite-response observable W such that

$$(316) \quad L(W) := \left. \frac{d}{ds} \right|_{s=0} \int W d\mu_{\theta_s} \neq 0.$$

Assume also that this derivative is the normalized invariant-suspension response supplied by (A3), including roof, dynamics, normalization, and moving-singularity contributions. This is not asserted for every table.

Proposition 8.4 (Conditional response-generated non-convexity and non-subadditivity). *Assume (A1)–(A5), the verified response direction in assumption 8.3, and a mechanically compatible port that realizes the bounded feedback curve used below. For any unit vector $e \in \mathbb{R}^m$, if that port admits the cubic–quartic scalar profile, then the normalized Hamiltonian is non-convex and satisfies the direct defect (308). More precisely, along $p = re$ one obtains*

$$(317) \quad H(x, re) = \alpha r^3 - \kappa r^4 + O(r^6), \quad \alpha > 0, \quad \kappa > 0,$$

where both coefficients are deterministic invariant-density response coefficients of the billiard table under a finite-response feedback.

Proof. Let θ_s and W be given by assumption 8.3, and replace W by $-W$ if necessary so that $L(W) > 0$. Choose functions $Q_3, Q_4 \in C_b^\infty(\mathbb{R})$ with bounded derivatives of all orders and with

$$(318) \quad Q_3(r) = r^3, \quad Q_4(r) = r^4 \quad (|r| \leq R), \quad Q_3(0) = Q_4(0) = 0.$$

For $r = p \cdot e$ define the table feedback by

$$(319) \quad \Theta(x, p) = \theta_{\eta_3 Q_3(r) - \eta_4 Q_4(r)},$$

where $\eta_3, \eta_4 > 0$ are chosen small enough that the entire curve of tables stays in the admissible parameter window. This feedback is finite-response because the Q_j are bounded with bounded derivatives and because its realization is part of the specified compatible-port hypothesis; no arbitrary finite-jet realization theorem is invoked. Choose a centered drift $b_0(z)$ independent of (x, p) in the working chart and with no x -dependence, so $Q_x \chi_p = 0$ and the complete horizontal work response in H vanishes along the line $p = re$. The Green–Kubo tensor is irrelevant for the direct Hamiltonian defect because the terminal test functions used in Corollary 8.6 have zero Hessian at the chosen point.

Take the microscopic potential to be

$$(320) \quad V(x, p, z) = W(z)$$

on the working chart, with the harmless global finite-response saturation of Definition 3.1. Thus the potential has no explicit cotangent dependence near the jet interval; all displayed powers of r below come from the feedback of the deterministic billiard table.

By (A3) applied to the independently verified geometric deformation,

$$(321) \quad \int W d\mu_{\theta_s} = \int W d\mu_{\theta_0} + L(W)s + O(s^2),$$

with the remainder controlled by the differentiated transfer resolvent. Using (319) and the normalization $H(x, 0) = 0$, for $|r| \leq R$ we obtain

$$(322) \quad H(x, re) = L(W)\eta_3 r^3 - L(W)\eta_4 r^4 + O((\eta_3 r^3 - \eta_4 r^4)^2).$$

Put $\alpha = L(W)\eta_3$ and choose η_4 so that $\kappa = L(W)\eta_4$ is positive and the deformation remains geometric. Then choose $0 < \eta_3 \ll \eta_4$ so that $r_* := 3\alpha/(4\kappa)$ lies in $(0, R/2)$ and the $O(r^6)$ remainder in (322), together with its second derivative, is dominated on $[0, R]$ by the displayed cubic-quartic terms. At $r = r_*$,

$$(323) \quad \frac{d^2}{dr^2}(\alpha r^3 - \kappa r^4) \Big|_{r=r_*} = -\frac{9}{4} \frac{\alpha^2}{\kappa} < 0,$$

and therefore Theorem 8.2 gives local non-convexity.

For the direct Hamiltonian defect, take $a > 0$ with $2a \leq R$ and $a < \alpha/(4\kappa)$, after reducing a further so that the $O(a^6)$ term is smaller than αa^3 . Then

$$(324) \quad H(x, 2ae) - 2H(x, ae) = 6\alpha a^3 - 14\kappa a^4 + O(a^6) > 0.$$

Thus (308) holds. The leading defect is the response-generated cubic term, not an explicit quadratic term in the microscopic potential and not an effective Hamiltonian coefficient inserted by hand. The construction uses bounded finite-response microscopic data fixed before the transfer operator, the cell problem, and the effective HJB equation. $\square$

Proposition 8.5 (Algebraic status of the conditional example). *Under the realization and response hypotheses of Proposition 8.4, the coefficients α and κ in (317) are not chosen as effective Hamiltonian data; they are the first response coefficients*

$$(325) \quad \alpha = L(W)\eta_3, \quad \kappa = L(W)\eta_4$$

produced by the verified normalized invariant response. This proposition does not assert that every billiard admits such a port.

Proof. For the specified port, map the scalar feedback $s = \eta_3 Q_3(p \cdot e) - \eta_4 Q_4(p \cdot e)$ to this normal deformation. This is a mechanically compatible finite-dimensional prescription by hypothesis. The observable W is supported in the same rectangle and is fixed before the invariant density is differentiated. The value $L(W)$ is then computed by the moving-singularity response formula; it is not part of the input. The triangular choice of Q_3, Q_4 only separates the odd and even powers in the cotangent feedback so that the Taylor expansion has a cubic term creating the subadditivity defect and a quartic term creating a negative second derivative at an interior point. Thus the algebra does not insert the effective coefficients by hand, while existence of the response direction and compatible realization remains an explicit hypothesis. $\square$

Corollary 8.6 (Non-subadditivity of the induced expectation). *If the derived Hamiltonian satisfies (308) at some x, p, q , then the nonlinear semigroup associated with (291) is not subadditive. Hence the resulting expectation lies outside the sublinear G -expectation class. Under the additional hypotheses of Proposition 8.4, its algebraic profile satisfies this condition.*

Proof. Assume, for contradiction, that the semigroup is subadditive for all smooth terminal data. Work in a coordinate chart around x and choose smooth periodic terminal functions ϕ_p, ϕ_q whose gradients at x are p and q and whose Hessians vanish at x . The short-time expansion of the viscosity semigroup, obtained directly from the deterministic generator, gives

$$(326) \quad \mathcal{E}_{T-\tau, T}^\theta[\phi_p](x) = \phi_p(x) + \tau H(x, p) + o(\tau),$$

$$(327) \quad \mathcal{E}_{T-\tau, T}^\theta[\phi_q](x) = \phi_q(x) + \tau H(x, q) + o(\tau),$$

$$(328) \quad \mathcal{E}_{T-\tau, T}^\theta[\phi_p + \phi_q](x) = \phi_p(x) + \phi_q(x) + \tau H(x, p + q) + o(\tau).$$

Subadditivity would imply $H(x, p + q) \leq H(x, p) + H(x, q)$ after division by τ and letting $\tau \downarrow 0$, contradicting (308). Therefore the semigroup is not subadditive. $\square$

9. ACYCLIC DETERMINISTIC DERIVATION AND DEPENDENCY CLOSURE

This section records the final logical closure of the first part as part of the proof. The chain is forward directed:

$$(329) \quad \begin{array}{c} \text{geometry} \implies \text{singular hyperbolicity} \\ \xrightarrow{(A1)-(A2)} \text{anisotropic transfer gap} \implies \text{observable resolvent} \\ \xrightarrow{(A3)-(A5)} \text{cell correctors} \implies \text{viscosity homogenization.} \end{array}$$

The macroscopic equation appears only at the final arrow. It is not used in the construction of the prelimit graph, the spectral gap, the cell problem, or the coefficient response.

Theorem 9.1 (No circular dependence in the deterministic proof). *Every object used in Theorem 7.33 is obtained before the homogenized solution u is introduced. More precisely:*

- (i) the billiard flow, singular sets, homogeneity strips, invariant measure, and Banach bundle are constructed from assumption 2.4;
- (ii) the prelimit solution u^ε is the viscosity-duality completion of the deterministic jet graph of Lemma 3.10;
- (iii) the spectral gap and flow resolvent are supplied by (A1)–(A2) in the forms recorded in Theorem 5.9 and Lemmas 5.13 and 5.16;
- (iv) the correctors χ_i and $\chi^{(2)}$ are obtained by applying the reduced observable resolvent to centered deterministic observables;
- (v) the effective coefficients D and H are invariant-measure pairings of those deterministic correctors;
- (vi) the homogenized function u is introduced only as the unique viscosity solution selected by comparison after the subsolution and supersolution inequalities have been proved.

No step in this list assumes the conclusion of a later step.

Proof. Items (i) and (ii) are Sections 2 and 3. Their inputs are scatterer geometry, specular reflection, and finite-response mechanical coupling. Items (iii) and (iv) are the transfer and cell-problem sections: the hypotheses act on densities and observables in the anisotropic bundle and are independent of the value function. Item (v) is the algebraic averaging in Definitions 7.3 and 7.5. Item (vi) follows from Proposition 7.9 and the perturbed-test proof of Theorem 7.33. The only occurrence of a gradient of a solution before the final HJB equation is the prelimit cotangent jet in the branchwise deterministic graph transform and, in viscosity form, the cotangent component of a test jet. Neither object is the gradient of the unknown homogenized solution. The dependency graph (329) is therefore acyclic. $\square$

Theorem 9.2 (Conditional support of the main theorem). *The convergence theorem Theorem 7.33 is supported by the geometric construction together with (A1)–(A5). The following display is a dependency chain, not a claim that every arrow follows from ordinary curvature*

and finite horizon:

(330)

ordinary dispersing billiard geometry $\implies$ a.e. singular collision map with branch geometry

$\xrightarrow{(A1)}$ anisotropic spectral gap

$\xrightarrow{(A2)}$ Dolgopyat suspension resolvent $\xrightarrow{(A3)}$ moving-singularity response

$\implies$ reduced observable correctors $\xrightarrow{(A4)-(A5)}$ full-gradient residual identity

$\implies$ viscosity–duality comparison $\implies$ effective HJB and nonlinear semigroup.

Each arrow is either a geometric construction or an explicitly identified analytic hypothesis. The finite-response port supplies only compact deterministic microscopic action data and bounded cotangent read-out; it does not supply an effective coefficient, a macroscopic value function, or a representation of the limiting semigroup. The uncertainty tensor D , the nonlinear Hamiltonian H , and the failure of subadditivity are therefore conditional outputs of the billiard transfer-resolvent calculation.

Proof. Geometry constructs the a.e. branch map. The successive analytic arrows are, respectively, (A1), (A2), (A3), and (A4)–(A5). Once those inputs hold, the reduced-resolvent definitions and coefficient pairings occur in the displayed forward order.

The compact finite-response port appears only in the primitive input layer: it moves the table and supplies branchwise action read-outs through a fixed finite-dimensional Hamiltonian mechanism. All its Taylor coefficients are fixed before transfer operators are defined. Since no object produced after the transfer-resolvent stage is used to define the port, and since the limiting semigroup is introduced only after the HJB comparison input, the dependency chain is acyclic. This acyclicity does not prove any clause of (A1)–(A5). $\square$

Corollary 9.3 (Closure of the conditional deterministic implication). *The derivation of Equation (291) uses only deterministic microscopic trajectories, anisotropic transfer estimates, and viscosity stability. The positive tensor D is a deterministic Green–Kubo bilinear form represented by the transfer resolvent. The conclusion is paired convergence; no Liouville-a.e. microscopic statement is included.*

Proof. The limiting operations used in the conclusion are Banach-space limits of transfer resolvents, contour-resolvent limits for the suspension semigroup, and viscosity half-relaxed limits, under (A1)–(A5). The representation language introduced in Part II is a post-derivation representation of the already derived nonlinear semigroup and is not used in proving Theorem 7.33. $\square$

Part 2. The Probabilistic Representation

10. THE θ -EXPECTATION AS A NONLINEAR SEMIGROUP

The probabilistic representation begins only after the deterministic derivation is complete. The primitive object in this part is the normalized nonlinear viscosity semigroup generated by the derived HJB equation. The normalization $H(x, 0) = 0$ was obtained in Definition 7.5; it is the deterministic zero-energy gauge that makes constants genuine constants.

Definition 10.1 (The θ -expectation semigroup). For terminal payoff $\phi \in C(\mathcal{M})$, let u be the unique viscosity solution of

$$(331) \quad -\partial_t u - \text{Tr}(D(x, \nabla u) \nabla^2 u) - H(x, \nabla u) = 0, \quad u(T, x) = \phi(x).$$

For $0 \leq t \leq T$, define

$$(332) \quad \mathcal{E}_{t,T}^\theta[\phi](x) = u(t, x).$$

Theorem 10.2 (Foundational properties). *The family $\mathcal{E}_{t,T}^\theta$ is constant preserving, cash additive, monotone, and time consistent:*

$$(333) \quad \mathcal{E}_{t,T}^\theta[c] = c,$$

$$(334) \quad \mathcal{E}_{t,T}^\theta[\phi + c] = \mathcal{E}_{t,T}^\theta[\phi] + c,$$

$$(335) \quad \phi \leq \psi \implies \mathcal{E}_{t,T}^\theta[\phi] \leq \mathcal{E}_{t,T}^\theta[\psi],$$

$$(336) \quad \mathcal{E}_{s,T}^\theta[\phi] = \mathcal{E}_{s,t}^\theta[\mathcal{E}_{t,T}^\theta[\phi]].$$

It is generally nonlinear and, whenever the direct Hamiltonian defect in Corollary 8.6 holds, not subadditive.

Proof. If $u \equiv c$, then $\nabla u = 0$, $\nabla^2 u = 0$, and the PDE reduces to $-H(x, 0) = 0$, which holds by the deterministic normalization in Definition 7.5. The same observation gives cash additivity, because adding a constant to a solution changes neither its gradient nor its Hessian. Monotonicity and time consistency are consequences of the comparison and uniqueness theorem proved in Proposition 7.9 and Appendix A.1. Nonlinearity and failure of subadditivity follow from the Hamiltonian's dependence on ∇u and from Corollary 8.6. $\square$

11. WHY A NONLINEAR MARTINGALE PROBLEM IS NOT THE PRIMITIVE OBJECT

Theorem 11.1 (Linearity obstruction). *Let $\mathcal{G}$ be an operator on smooth test functions. Suppose there exists a single probability law under which*

$$(337) \quad f(t, X_t) - f(0, X_0) - \int_0^t \mathcal{G}f(s, X_s) ds$$

is a martingale for every smooth f . Then $\mathcal{G}$ is additive on this class of tests. In particular, a genuinely nonlinear Hamiltonian

$$(338) \quad \mathcal{G}f = \partial_t f + \text{Tr}(D(x, \nabla f) \nabla^2 f) + H(x, \nabla f)$$

cannot be the generator of a single classical martingale problem.

Proof. If the displayed process is a martingale for f , g , and $f + g$, then the difference

$$(339) \quad \int_0^t (\mathcal{G}(f + g) - \mathcal{G}f - \mathcal{G}g)(s, X_s) ds$$

is a finite-variation martingale. It is therefore constant. Differentiating in t gives additivity along the support of the law. Varying the initial point gives additivity on the test class. A nonlinear dependence on ∇f or $\nabla^2 f$ violates this unless the Hamiltonian is accidentally linear. $\square$

Remark 11.2. This theorem does not enter the deterministic derivation. It identifies the correct representation layer after the HJB equation has already been obtained: the nonlinear semigroup is primitive, while classical probability laws appear only after fixing and linearizing a payoff.

12. LINEARIZATION, CARRÉ DU CHAMP, AND ITÔ CALCULUS

Let u solve the effective HJB equation for a fixed terminal payoff. Define

$$(340) \quad a^u(t, x) = 2D(x, \nabla u(t, x)),$$

and, when the coefficients are differentiable,

$$(341) \quad b^u(t, x) = \partial_p H(x, \nabla u(t, x)) + \sum_{i,j} \partial_p D_{ij}(x, \nabla u(t, x)) \partial_{ij} u(t, x).$$

The linearized generator acting on a smooth test f is

$$(342) \quad L_t^u f(x) = \frac{1}{2} \text{Tr}(a^u(t, x) \nabla^2 f(x)) + b^u(t, x) \cdot \nabla f(x).$$

The carré du champ is

$$(343) \quad \Gamma_t^u(f, g) = \frac{1}{2} (L_t^u(fg) - fL_t^u g - gL_t^u f) = \nabla f^T a^u \nabla g / 2.$$

Theorem 12.1 (Linearized representation for a fixed payoff). *For a fixed payoff ϕ and its decoupling field u , the increment of the nonlinear semigroup along perturbations of the payoff is represented by the linear parabolic equation generated by L^u . Equivalently, any classical diffusion representation uses the generator L^u , which depends on u and therefore on ϕ .*

Proof. Perturb the terminal condition to $\phi + \delta\eta$ and write $u^\delta = u + \delta v + o(\delta)$. Expanding the HJB equation to first order in δ gives

$$(344) \quad -\partial_t v - L_t^u v = 0, \quad v(T) = \eta.$$

Thus the derivative of the nonlinear semigroup is represented by the linearized generator. Since u depends on ϕ , the representing law is payoff-dependent. $\square$

Theorem 12.2 (Degenerate calibrated diffusion representation by deterministic viscosity regularization). *After the deterministic HJB equation has been derived, fix a terminal payoff whose viscosity solution is the locally uniform limit of smooth solutions u^δ to the uniformly parabolic regularizations*

$$(345) \quad -\partial_t u^\delta - \text{Tr}((D(x, \nabla u^\delta) + \delta I) \nabla^2 u^\delta) - H(x, \nabla u^\delta) = 0, \quad u^\delta(T, \cdot) = \phi^\delta.$$

For each $\delta > 0$, the payoff-calibrated linearization has an ordinary Markov diffusion representation. As $\delta \downarrow 0$, the corresponding linear semigroups converge locally uniformly to the nonlinear viscosity semigroup calibrated by the derived solution u . This representation is a post-derivation analytic identity and is never used in Part 1.

Proof. For fixed $\delta > 0$, the matrix $D + \delta I$ is uniformly positive and the smooth solution u^δ freezes the coefficients

$$(346) \quad a^\delta(x, t) = D(x, \nabla u^\delta(t, x)) + \delta I, \quad b^\delta(x, t) = \partial_p H(x, \nabla u^\delta) + \sum_{i,j} \partial_p D_{ij}(x, \nabla u^\delta) \partial_{ij} u^\delta.$$

The resulting operator is linear in the test function, so the standard parabolic representation applies. Stability of viscosity solutions and the Crandall–Ishii modulus already derived in Proposition 7.9 identify the locally uniform limit with the original degenerate HJB semigroup. The construction starts only after D and H have been obtained from the billiard transfer calculus; it is therefore not a microscopic probabilistic proof of homogenization. $\square$

Theorem 12.3 (Downstream character of all probabilistic representations). *Every probabilistic object in Part 2 is determined by the deterministic semigroup of Theorem 7.33. Fixing a terminal payoff and a regular decoupling field u determines the linearized coefficients a^u and b^u ; any diffusion, FBSDE, or Girsanov formula below represents this already-derived linearized parabolic problem. If two deterministic microscopic billiard models produce the same coefficients (D, H) on the relevant jet range, then all representation-layer objects in this part coincide on that range. Changing the payoff changes u and therefore changes the representing linear law, while the underlying nonlinear semigroup remains the same deterministic HJB semigroup.*

Proof. The nonlinear semigroup is the solution operator of the effective HJB equation whose coefficients were constructed in Part 1. Linearizing this semigroup at a payoff is ordinary differentiation of a deterministic parabolic equation and gives the operator L^u in Theorem 12.1. The diffusion representation, when the coefficient factorization is nondegenerate on the visited region, is a representation theorem for this linear operator. The BSDE identity is Itô's formula applied to the same smooth decoupling field, and the Girsanov identity is an algebraic Hamiltonian shift of the same deterministic generator. None of these constructions is used in the proof of Theorem 7.33; they depend only on the derived pair (D, H) and on the selected payoff. $\square$

13. THE θ -BSDE REPRESENTATION

Assume in this representation section that the derived matrix D is uniformly positive definite on the region visited by the decoupling field, and write

$$(347) \quad \sigma(x, p)\sigma(x, p)^T = 2D(x, p).$$

The BSDE control variable is

$$(348) \quad Z_t = \sigma(X_t, \nabla u(t, X_t))^T \nabla u(t, X_t),$$

not the gradient itself unless $\sigma = I$.

Theorem 13.1 (Decoupled FBSDE for a smooth solution). *Let u be a smooth solution of the effective HJB equation and set $p(t, x) = \nabla u(t, x)$. Let X solve*

$$(349) \quad dX_t = b^u(t, X_t) dt + \sigma(X_t, p(t, X_t)) dW_t.$$

Define

$$(350) \quad Y_t = u(t, X_t), \quad Z_t = \sigma(X_t, p(t, X_t))^T p(t, X_t).$$

Then (X, Y, Z) satisfies

$$(351) \quad dY_t = -g(t, X_t, Z_t) dt + Z_t dW_t, \quad Y_T = \phi(X_T),$$

where, with $p = (\sigma^T)^{-1}Z$,

$$(352) \quad g(t, x, Z) = H(x, p) - b^u(t, x) \cdot p.$$

Conversely, a sufficiently regular Markovian solution of this FBSDE recovers the viscosity solution of the HJB equation.

Proof. Apply Itô's formula to $Y_t = u(t, X_t)$. The martingale coefficient is $Z_t = \sigma^T \nabla u$. The drift is

$$(353) \quad \partial_t u + b^u \cdot \nabla u + \frac{1}{2} \text{Tr}(\sigma \sigma^T \nabla^2 u) = \partial_t u + b^u \cdot p + \text{Tr}(D \nabla^2 u).$$

Using the HJB equation,

$$(354) \quad \partial_t u + \text{Tr}(D \nabla^2 u) = -H(x, p),$$

so the drift equals $b^u \cdot p - H(x, p) = -g$. The converse follows by reversing the calculation and using the Markovian identification $Y_t = u(t, X_t)$. $\square$

14. THE θ -GIRSANOV TRANSFORMATION

Girsanov transformations are classical transformations of the linearized law associated with a fixed payoff. At the nonlinear semigroup level they are encoded by a deterministic Hamiltonian shift and by the corresponding generator defect.

Definition 14.1 (Hamiltonian shift and deterministic defect). For a deterministic drift shift $h(t, x)$, define

$$(355) \quad H^h(x, p) = H(x, p + h) - H(x, h), \quad D^h(x, p) = D(x, p + h).$$

The shifted semigroup is generated by

$$(356) \quad -\partial_t u - \text{Tr}(D^h(x, \nabla u) \nabla^2 u) - H^h(x, \nabla u) = 0.$$

Relative to the unshifted generator, the deterministic defect is

$$(357) \quad \Lambda_h(t, x, p, X) = \text{Tr} \left((D(x, p + h(t, x)) - D(x, p)) X \right) + H(x, p + h(t, x)) - H(x, h(t, x)) - H(x, p).$$

Theorem 14.2 (Nonlinear Cameron–Martin formula at semigroup level). *Let u_h be the shifted solution. Then u_h is equivalently the solution of the unshifted HJB equation with deterministic source Λ_h evaluated along its own jet:*

$$(358) \quad -\partial_t u_h - \text{Tr}(D(x, \nabla u_h) \nabla^2 u_h) - H(x, \nabla u_h) - \Lambda_h(t, x, \nabla u_h, \nabla^2 u_h) = 0.$$

Consequently the shifted expectation is the unshifted nonlinear semigroup with this deterministic generator defect. When D is independent of p and H is linear-quadratic, (357) reduces after linearization to the classical Cameron–Martin–Girsanov drift term.

Proof. Substitute the definitions of D^h and H^h into the shifted PDE and add and subtract the unshifted terms $\text{Tr}(D(x, \nabla u_h) \nabla^2 u_h) + H(x, \nabla u_h)$. The remainder is exactly (357). If D is independent of p and $H(x, p) = \frac{1}{2} p^T a(x) p + \ell(x) \cdot p$, then

$$(359) \quad H(x, p + h) - H(x, h) - H(x, p) = p^T a(x) h,$$

which is the drift-tilt term appearing in the payoff-dependent linearized diffusion. The exponential density is therefore a representation of this already-derived linearized identity, not an input to the deterministic homogenization. $\square$

15. POST-DERIVATION REPRESENTATION HIERARCHY AND UNIVERSALITY

The four deterministic scales have corresponding representation layers.

- (i) Scale I is represented by a linear diffusion with deterministic Green–Kubo covariance.
- (ii) Scale II is represented by an inhomogeneous diffusion whose coefficients are deterministic functions of (x, p) .
- (iii) Scale III is represented by the nonlinear HJB semigroup and, after fixing a payoff, by the corresponding decoupled FBSDE.
- (iv) Scale IV is represented by the deterministic viscous Burgers hierarchy (306); probabilistic formulae are optional linearized representations, not inputs to the derivation.

16. COMPUTATIONAL FALSIFICATION AND REPRODUCIBILITY PROTOCOL

The results above are analytic. This section states a prospective numerical protocol for testing whether finite, explicitly simulated billiards display the four computable consequences on which the proposed interpretation relies. It reports no experiment and no fitted quantity. In particular, the thresholds below are fixed before simulation and are not additional assumptions in any proof. A failed threshold falsifies the corresponding *computational claim* for the preregistered configurations; it does not by itself identify which analytic hypothesis, discretization, or implementation component failed.

16.1. Claims, estimands, and decision rules. Let the collision observable be centered with respect to the numerically approximated invariant collision measure, and let D and H denote the coefficient functionals defined in Definitions 7.3 and 7.5. The tests are fixed as follows.

C1 Correlation and positivity. Compare exponential and power-law fits to the centered correlation tail by AIC; estimate D both by the truncated Green–Kubo sum and by block displacements. Pass only if the exponential fit has smaller AIC, the minimum eigenvalue is at least minus three bootstrap standard errors, and the two estimates differ by less than 10% in relative Frobenius norm.

C2 Coefficient response. For regular table and port perturbations, compare centered finite differences of D and H at $h, h/2, h/4$ with tangent/Ulam reduced-resolvent derivatives. Pass only if the discrepancy decreases at every refinement and the final relative error is below 10% (with an absolute-error report when the reference derivative is numerically zero).

C3 Billiard-to-HJB convergence. At $\varepsilon \in \{1/4, 1/8, 1/16, 1/32\}$, compare the deterministic prelimit value and the monotone-scheme HJB solution on the same space–time grid. Report L^∞ , relative L^2 , and the log–log slope. Pass only if both errors decrease over the last three levels and the finest relative L^2 error is below 5%.

C4 Nonconvexity and non-subadditivity. At points fixed in the run manifest, estimate the smallest eigenvalue of $D_p^2 H$ and $\Delta(p, q) = H(p + q) - H(p) - H(q)$. Pass only if a preregistered Hessian point has a bootstrap confidence interval wholly below zero and a preregistered pair (p, q) has a confidence interval for Δ wholly above zero.

All relative errors must state their denominator and the regularization used near zero. Confidence levels, block lengths, correlation-fit windows, Ulam partition sequences, HJB meshes, and the set of (p, q) points are part of the manifest and may not be chosen after inspecting outcomes.

16.2. Simulation design and numerical safeguards. The design uses three periodic finite-horizon tables: a symmetric geometry, an asymmetric geometry, and a near-grazing stress geometry. Each is evaluated at three regular port amplitudes. Every configuration receives 10^7 post-burn-in collisions for each of the fixed seeds 1103, 2207, 3301, 4409, 5501; seeds choose initial conditions and bootstrap resamples, not the deterministic dynamics. The event-driven integrator solves the next-impact equation with grazing-safe bracketing and applies the specular reflection map. Each run records energy drift, impact-equation residual, empirical lower and upper flight-time bounds, homogeneity-strip index, discarded singular events, burn-in, and effective sample size.

Computation stops after all five seeds reach the collision budget and the finest two discretizations, or earlier for numerical invalidity: energy drift above 10^{-8} , collision residual above 10^{-10} , effective sample size below 500, or Ulam mass-conservation error above 10^{-10} . Invalid runs are reported rather than replaced. One rerun is permitted only for a documented implementation bug, with both the original manifest and the patch retained. Time-shuffled observables, the frozen-table/no-port system, block displacement versus Green–Kubo estimation, the linear heat equation, and the HJB equation with $H = 0$ are baselines. The planned ablations remove, one at a time, moving-boundary response, full-gradient work terms, the second corrector, and high homogeneity strips; they diagnose sensitivity and never replace the primary model.

16.3. Artifact contract and independent rerun. Reproducibility requires more than a seed list. A conforming release contains (i) immutable geometry and port parameters with units and admissibility checks; (ii) source revision, compiler/interpreter and dependency lockfiles, platform metadata, and the complete command line; (iii) raw initial conditions, trajectories or deterministic checkpoints, collision logs, and derived arrays; (iv) the manifest fixing all choices listed above; and (v) a machine-readable report mapping each claim to its estimand, uncertainty interval, threshold, and pass/fail/invalid status. Every artifact is indexed by a cryptographic hash, and reports are generated only from hashed inputs. Raw and derived data remain separate; preprocessing is deterministic and records parent hashes.

Before long runs, unit tests must verify specular energy conservation, time-reversal away from singularities, observable centering, Ulam mass conservation, and convergence of the monotone solver on constant-coefficient heat and manufactured HJB solutions. An independent rerun consists of validating hashes and the environment, executing the unit tests, regenerating one short trajectory from its initial condition, reproducing one coefficient estimate from a stored checkpoint, and rebuilding all tables from the raw artifacts. This protocol makes numerical evidence auditable without making it a premise of the analytical argument.

17. SCOPE AND LIMITATIONS

The theory is conditional on both the geometric class in assumption 2.4 and the independent analytic package assumption 2.5. No claim is made here that every periodic finite-horizon dispersing billiard satisfies that package. Infinite horizon, cusps, focusing or flat components, noncompact energy shells, topology-changing table perturbations, and perturbations that destroy the uniform response constants are outside the theorem. Near-grazing trajectories are retained through homogeneity strips, but the estimates are not uniform under degeneration of the primitive curvature, horizon, or transversality margins.

The effective equation is a finite-horizon terminal-value statement on the compact slow manifold. It does not establish rates of homogenization unless a rate is stated separately,

interchange of infinite-time and homogenization limits, pathwise convergence for every initial condition, or robustness to measurement noise. Positive semidefiniteness of D allows degeneracy; the FBSDE and Girsanov representations therefore require the additional smoothness and nondegeneracy assumptions stated in Part 2. Nonconvexity and non-subadditivity are conditional on verifying the indicated response coefficients for a mechanically compatible port; they are not consequences of ordinary billiard geometry.

Finally, the numerical protocol has finite-sample and discretization error. Bootstrap intervals can be misleading under poorly resolved long correlations; Ulam derivatives can be biased by a partition that does not resolve moving singularities; and a monotone HJB solver can converge too slowly for the chosen grid hierarchy. The invalidity rules, estimator comparisons, stress geometry, and ablations expose some of these failures but cannot prove the analytic theorems. Conversely, passing all four tests corroborates only the registered finite configurations and scales, not the entire admissible class.

18. CONCLUSION

We treat the billiard map as a global a.e. collision map with singularities. Grazing collisions form the singular set around which homogeneous charts are organized. The geometric construction defines this dynamics; the spectral gap, suspension UNI, moving-singularity response scale, coherent prelimit mechanics, and uniform homogenization closure are assumed in (A1)–(A5).

The resulting Hamiltonian is affine in the Hessian and may be non-convex in the gradient when the chosen admissible port has the required response coefficients. The probabilistic part is formulated through nonlinear semigroups and payoff-dependent linearizations. Within the revised scope in Section 17, the paper gives a conditional θ -expectation framework: the uncertainty tensor, nonlinear Hamiltonian, and downstream representation calculus follow once the independently stated spectral, response, coherent-prelimit, and homogenization hypotheses (A1)–(A5) have been verified for a concrete billiard/port family. Establishing those hypotheses from ordinary curvature and finite horizon remains outside the present theorem.

APPENDIX A.1. ANALYTIC VERIFICATION LEDGER AND EFFECTIVE COMPARISON

This appendix replaces the former repeated branch/strip proof templates with a single dependency ledger. For any proposed billiard/port family, the following items must be proved independently, with spaces, domains, and constants fixed uniformly on the parameter window:

- (A1) quasi-compact map transfer operators with simple invariant eigenvalue and a uniform gap on the zero-mean space;
- (A2) a quantitative non-lattice/UNI estimate and the resulting suspension resolvent strip;
- (A3) closable moving-boundary derivatives, summable singular traces, and all intermediate strong-to-weak bounds needed for differentiated resolvent words;
- (A4) one globally coherent prelimit action or generator satisfying the reflection, collision-face, reciprocity, and comparison requirements;
- (A5) uniform paired compactness, contact selection, corrector residual convergence, and effective comparison.

Optical identities, homogeneity coordinates, and formal resolvent identities are useful in such a verification, but none substitutes for these five items. In particular, positive curvature

and finite horizon alone are not used here to deduce mixing, UNI, response closure, or homogenization closure.

Theorem A.1.1 (Effective comparison under the closure hypothesis). *Let D, H be the coefficients in Definitions 7.3 and 7.5. Under (A5), if u is a bounded upper-semicontinuous subsolution and v a bounded lower-semicontinuous supersolution of (291), with $u(T, \cdot) \leq v(T, \cdot)$, then $u \leq v$ on $[0, T] \times \mathcal{M}$.*

Proof. This is the effective-comparison clause of (A5). The coefficient calculation identifies D and H , but does not derive the global Crandall–Ishii modulus from ordinary billiard geometry. A concrete application must verify that modulus (or another valid comparison criterion) before invoking the theorem. $\square$

Scope. The main theorem is therefore an assumption-to-coefficient-and-limit reduction. Its falsifiable content is the explicit port compatibility, corrector equations, Green–Kubo coefficient formula, and full-gradient residual identity to be checked against a proposed model. It is not a new unconditional spectral or homogenization theorem for finite-horizon billiards, and the nonconvexity example requires the additional verified response and rank hypotheses stated in assumption 8.3 and Proposition 8.4.

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PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: `qiqian@pku.edu.cn`